Conversión CC/CC

Convertidores con Transformador: Puente Completo o Full-bridge

Convertidores con Transformador

Los montajes de convertidores con transformador se subdividen en dos categorías:

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- **Asimétricos**

Convertidores con Transformador

Los montajes de convertidores con transformador se subdividen en dos categorías:

- **Asimétricos**

- **Simétricos**

Convertidores con Transformador

Los montajes de convertidores con transformador se subdividen en dos categorías:

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    → La corriente que alimenta al transformador es unidireccional;

- **Simétricos**

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Los montajes de convertidores con transformador se subdividen en dos categorías:

- **Asimétricos**
  - → La corriente que alimenta al transformador es unidireccional;
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- **Simétricos**

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  - → La corriente que alimenta al transformador es unidireccional;
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  - → Son circuitos sencillos;
- **Simétricos**

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Los montajes de convertidores con transformador se subdividen en dos categorías:

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  - → La corriente que alimenta al transformador es unidireccional;
  - → El transformador estará “mal utilizado”, producto de utilizar un sólo interruptor;
  - → Son circuitos sencillos;
  - → Se utilizan generalmente en baja potencia.

- **Simétricos**

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  Ejemplo: Convertidor Flyback

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  Ejemplo: Convertidor Flyback

- **Simétricos**
  - → La corriente que alimenta al transformador es bidireccional/alterna;

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  Ejemplo: Convertidor Flyback

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  - → La corriente que alimenta al transformador es bidireccional/alterna;
  - → Mejor uso de los circuitos magnéticos (transformador), utilizan al menos dos interruptores;

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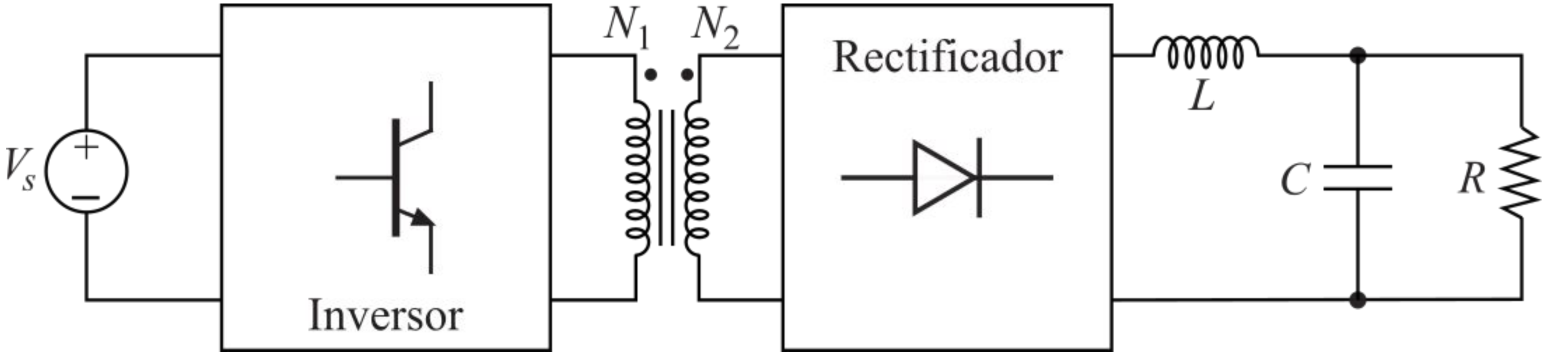
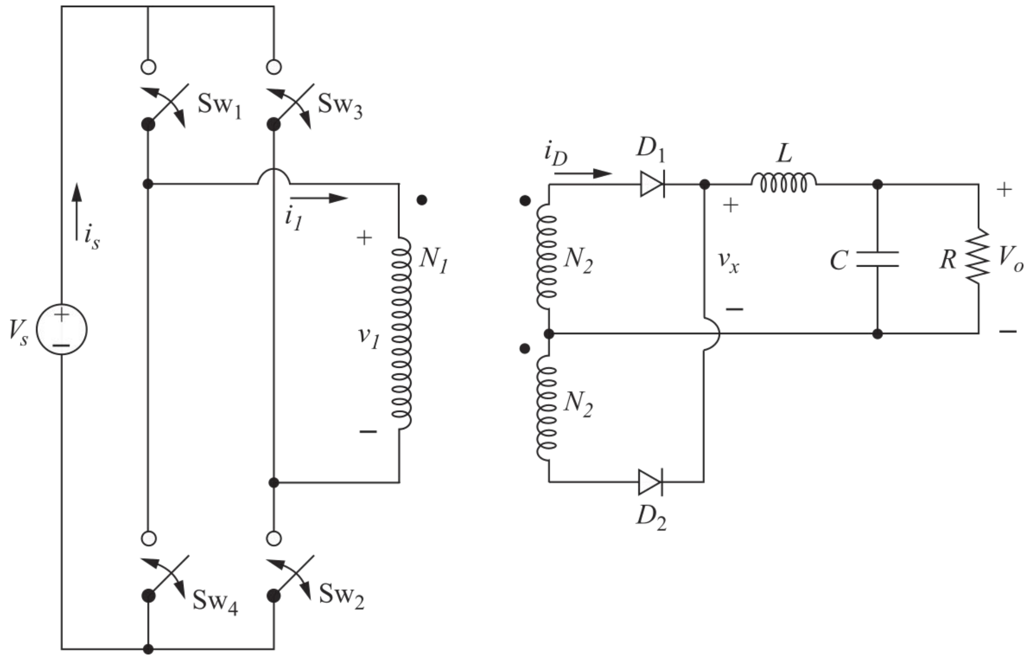
  Ejemplo: Convertidor Full-bridge

Convertidores Simétricos: concepto

Un convertidor simétrico se basa en el concepto de generar una tensión alterna en el primario del transformador y luego rectificar y filtrar a la salida.



Convertidor Full-bridge: esquema



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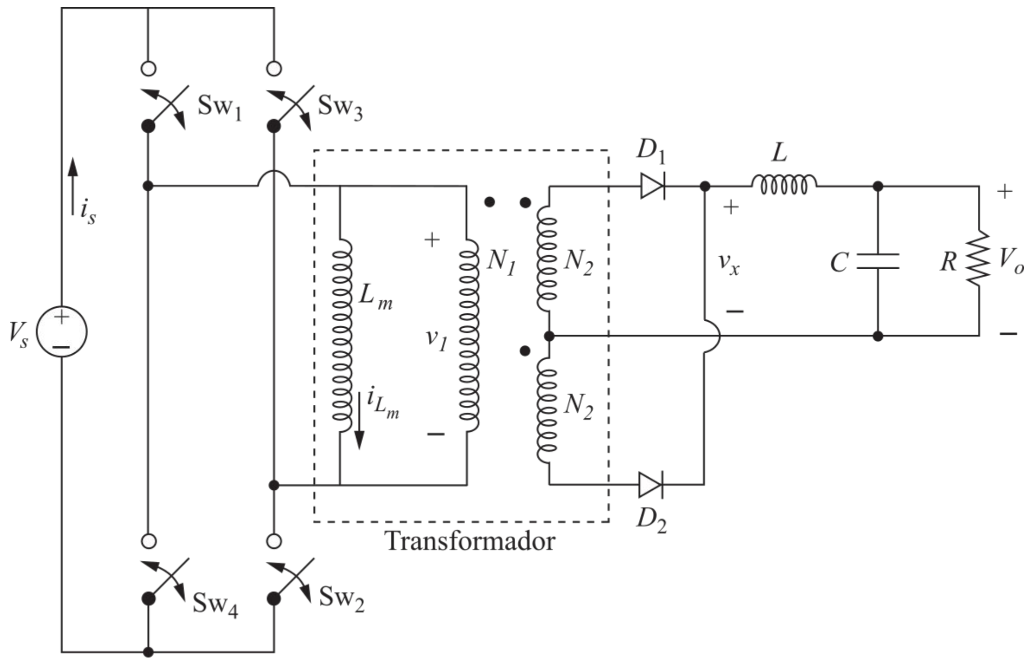
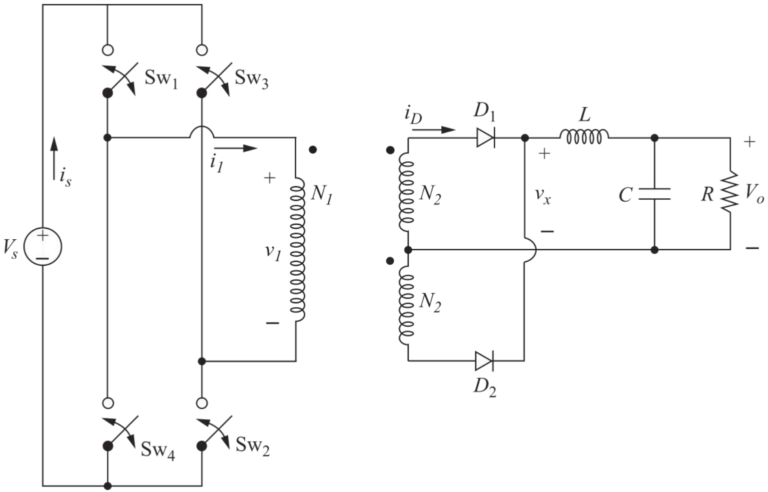


Convertidor Full-bridge: funcionamiento



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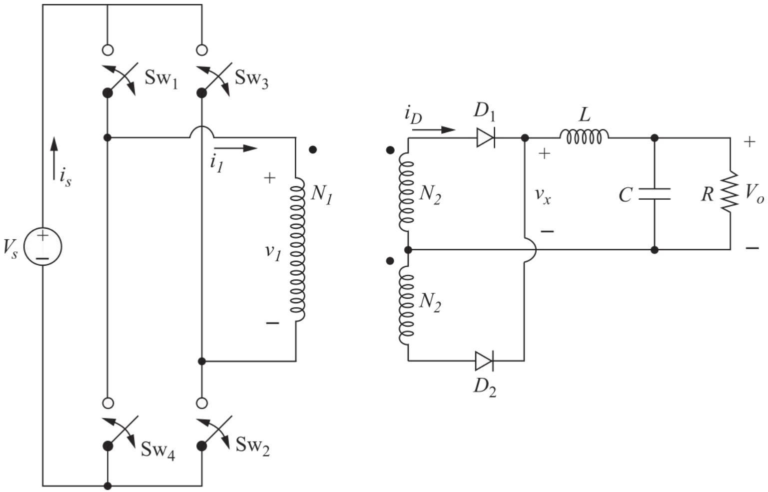

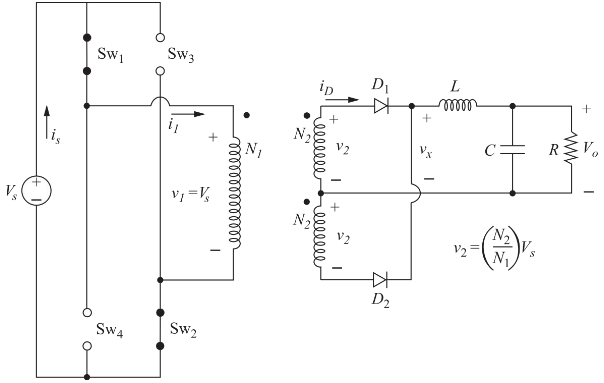

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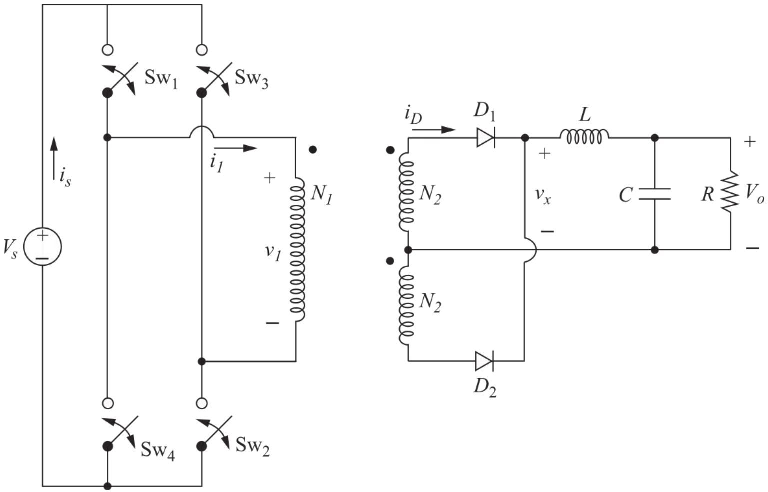

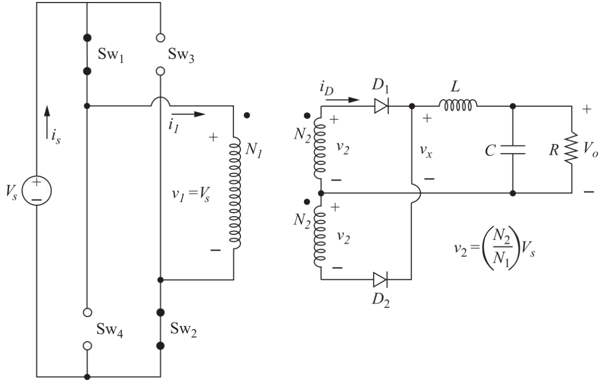

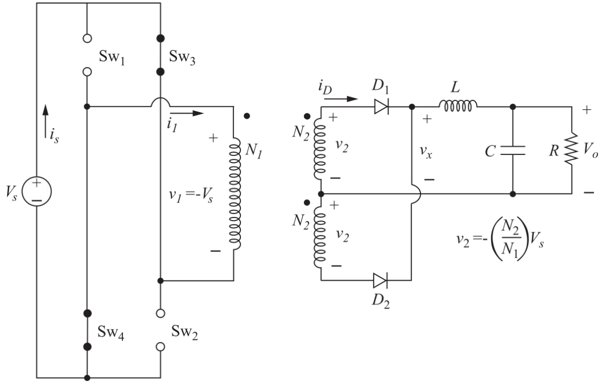

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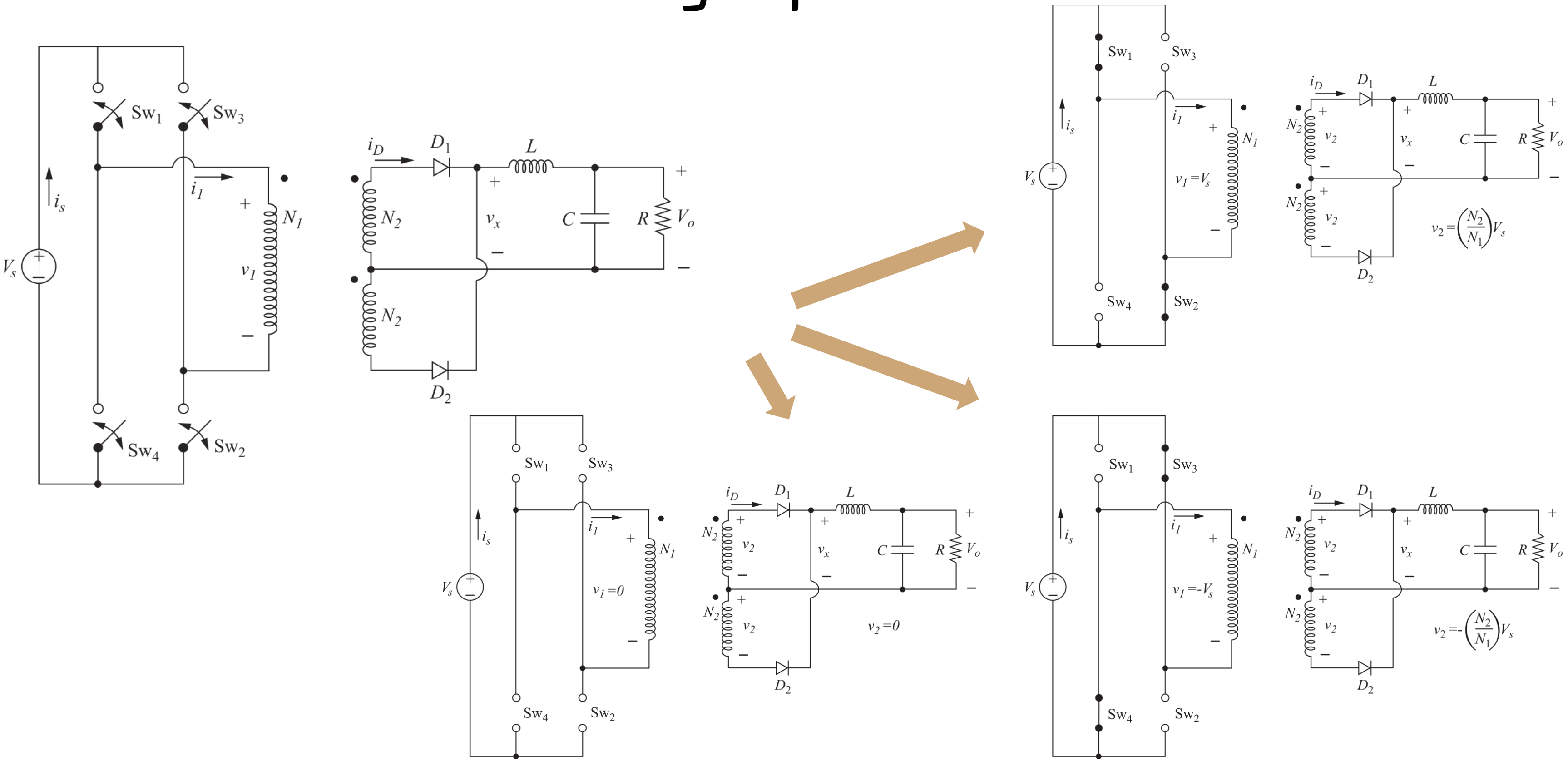
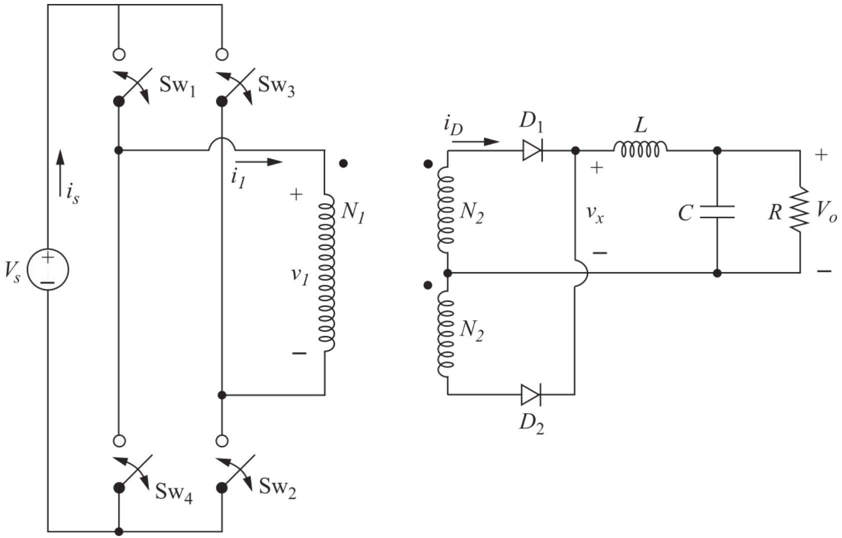
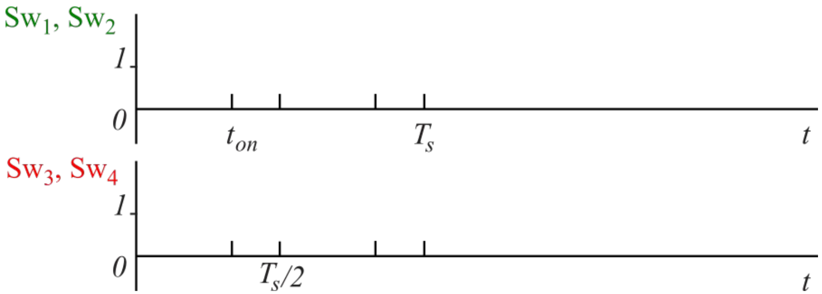


Convertidor Full-bridge: formas de onda





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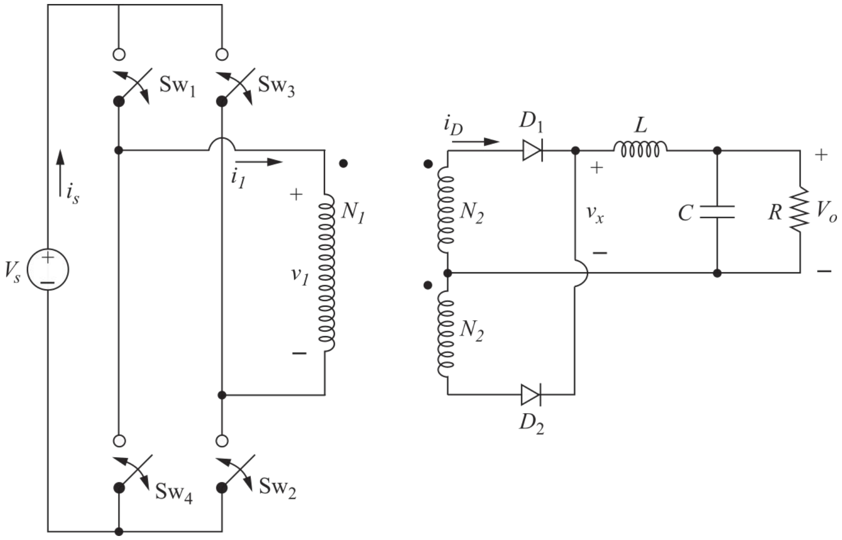

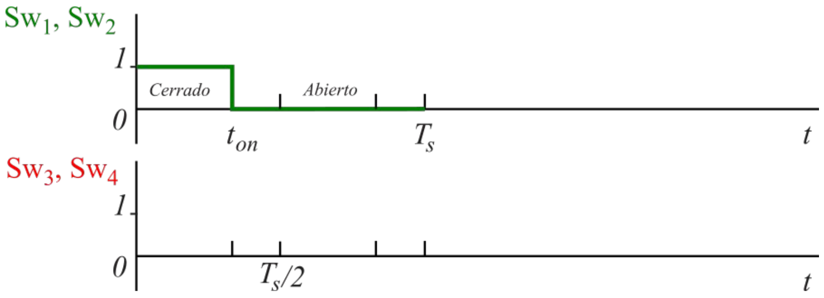

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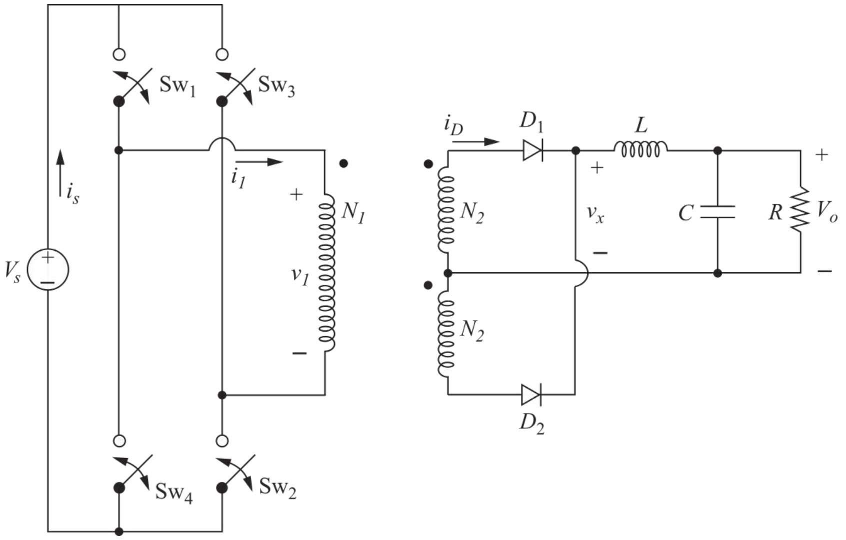

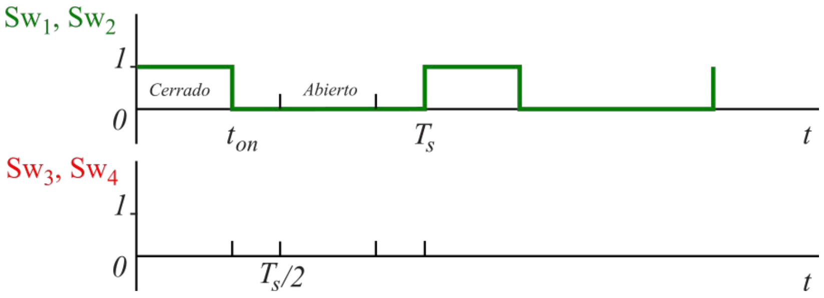

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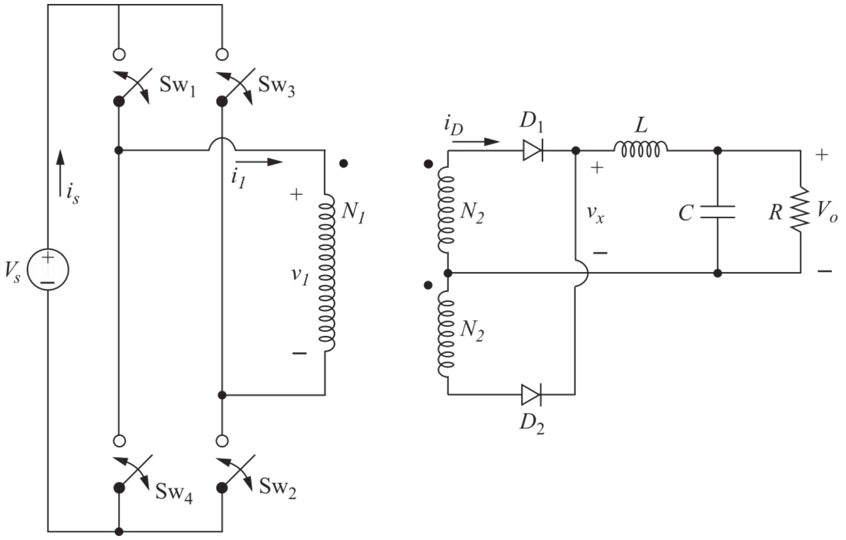

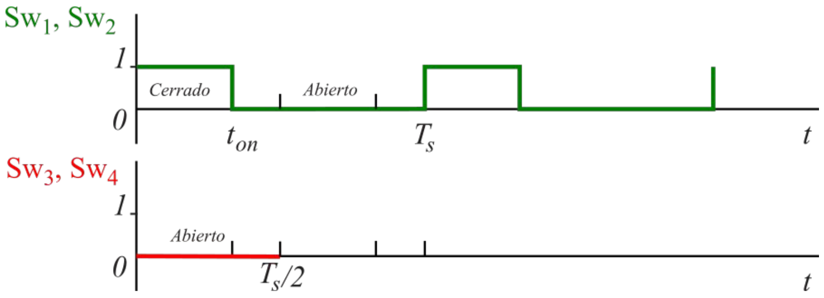

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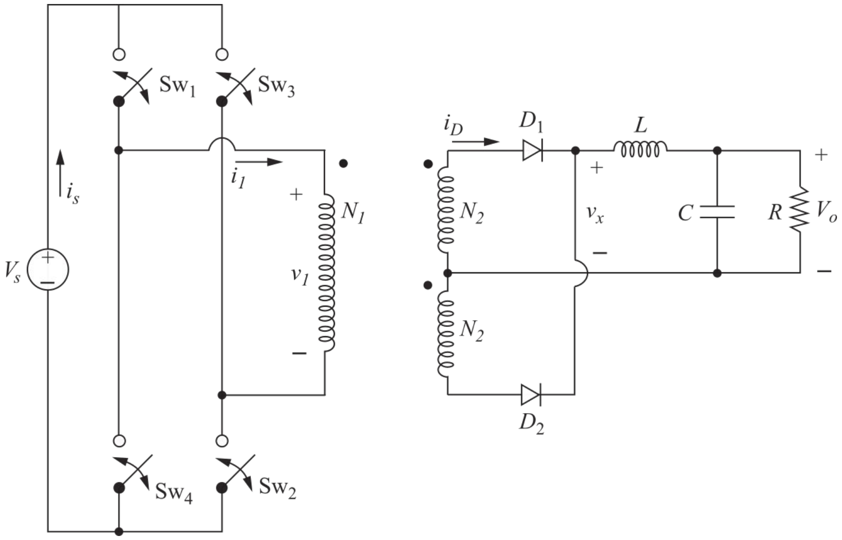

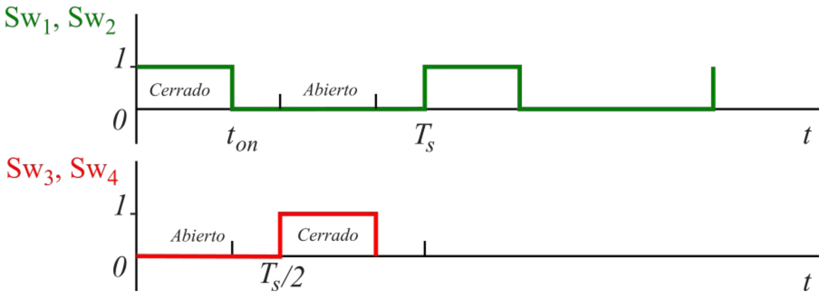

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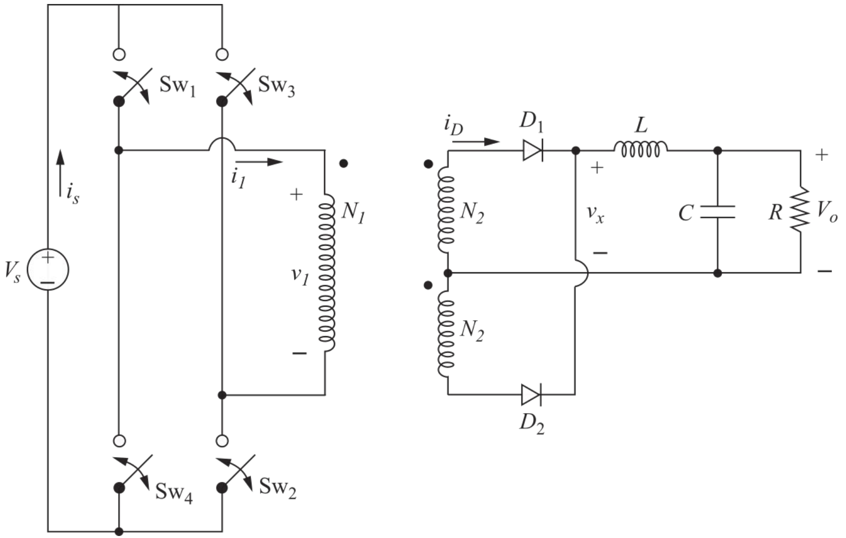

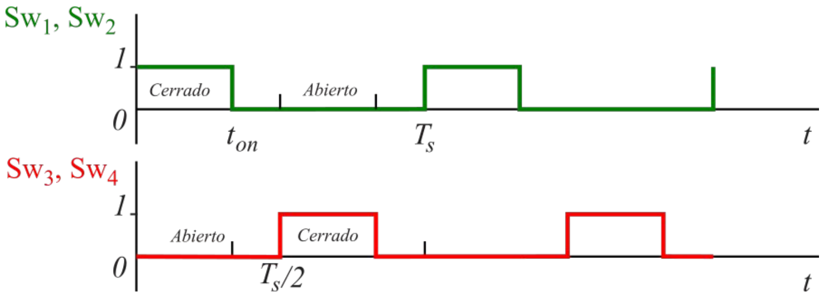

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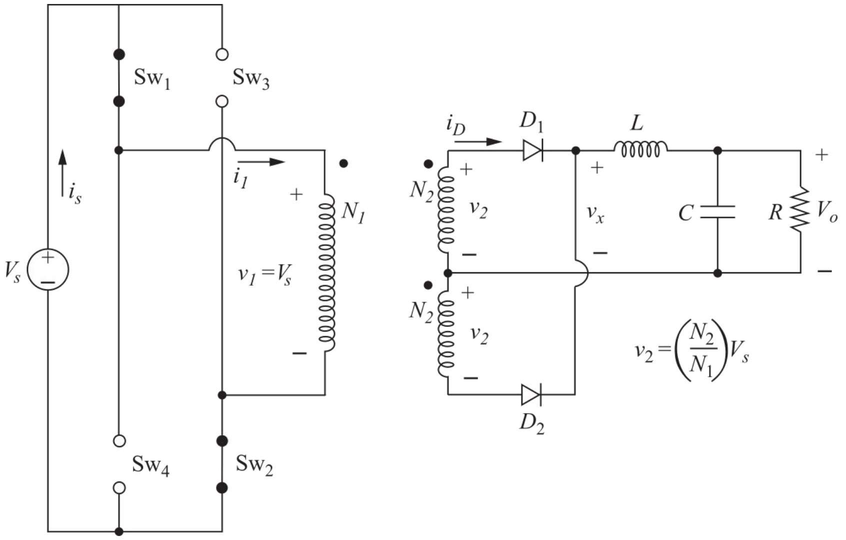

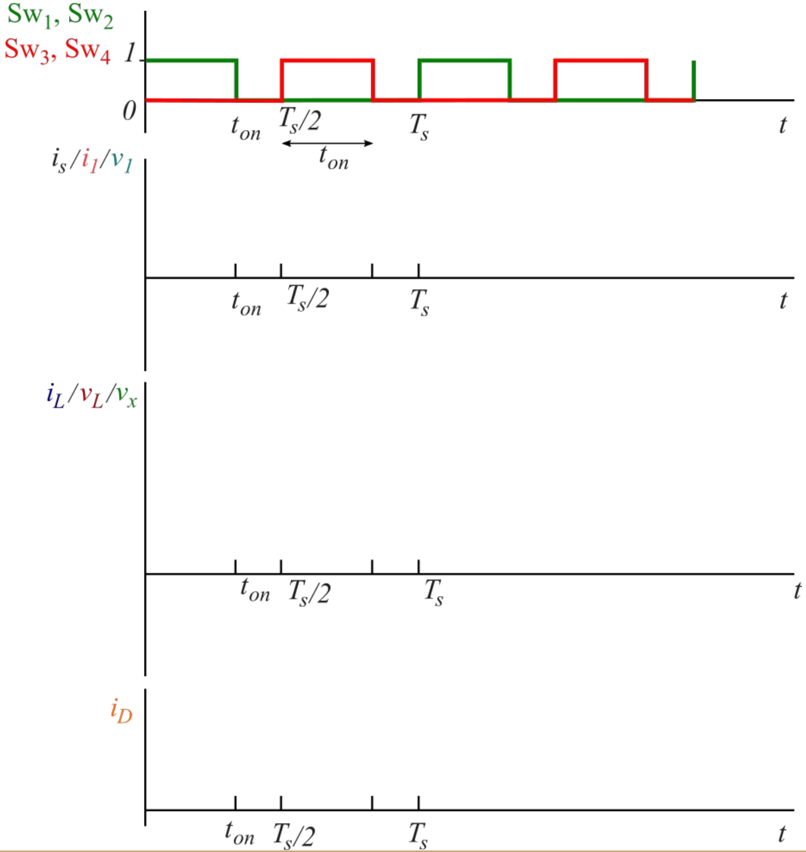

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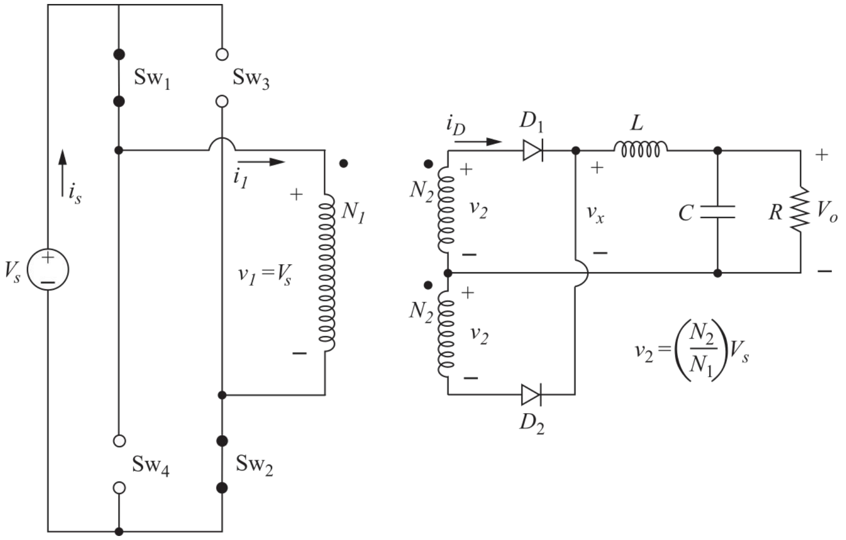

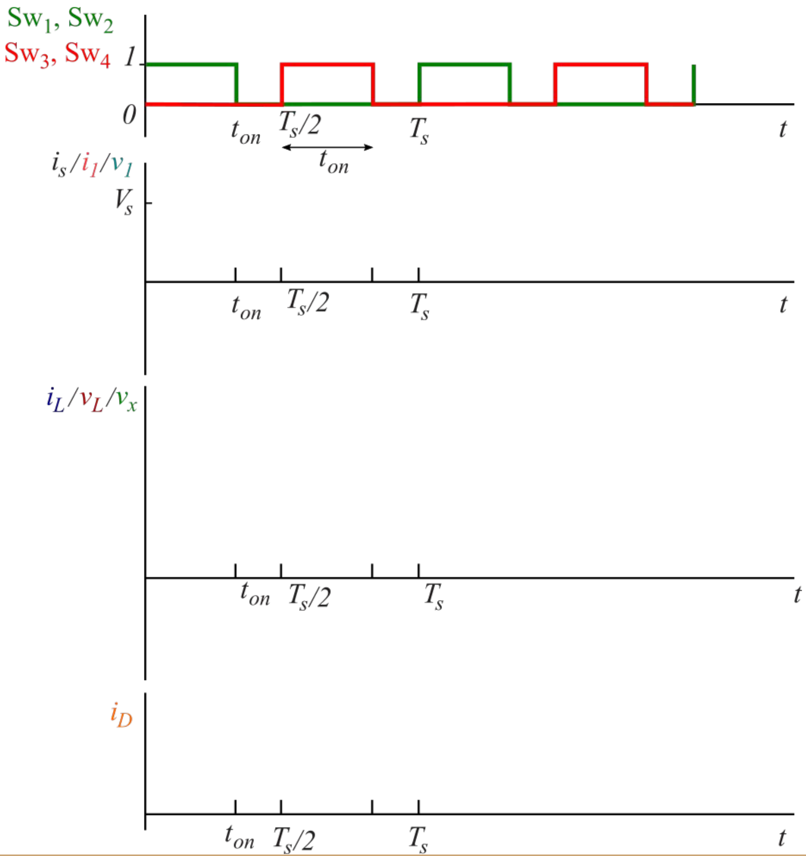

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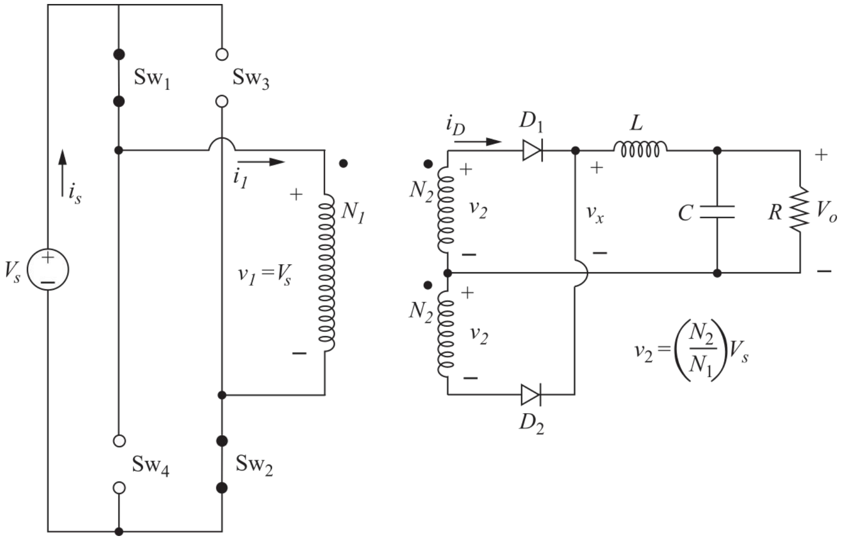

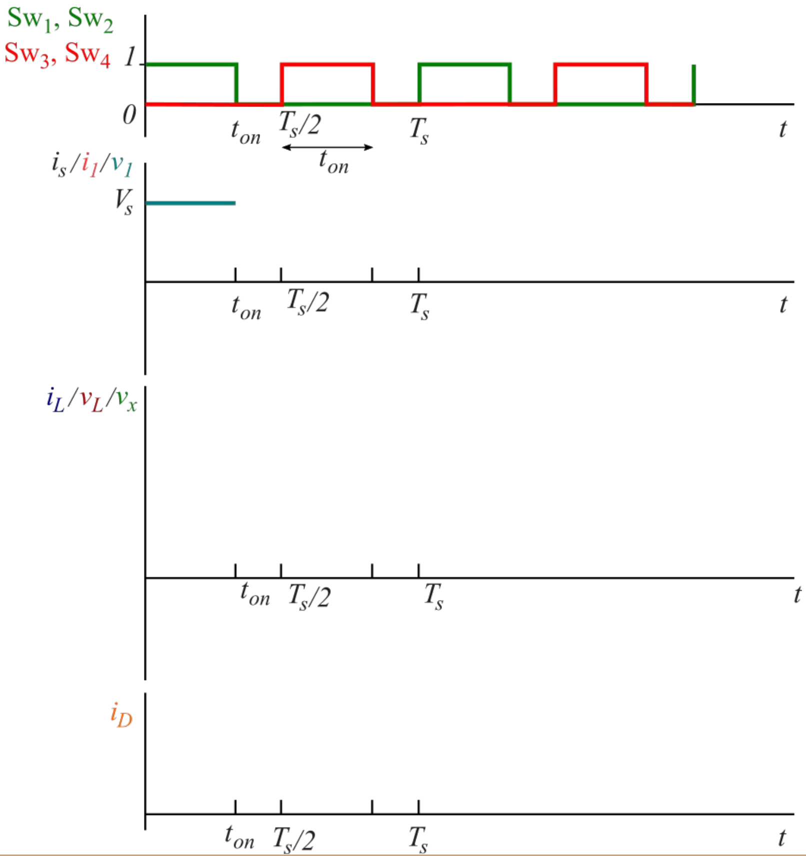

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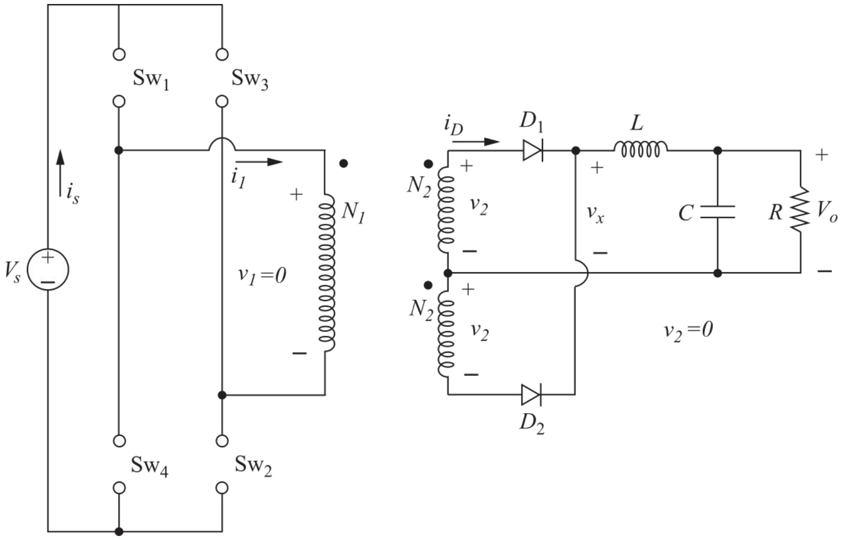

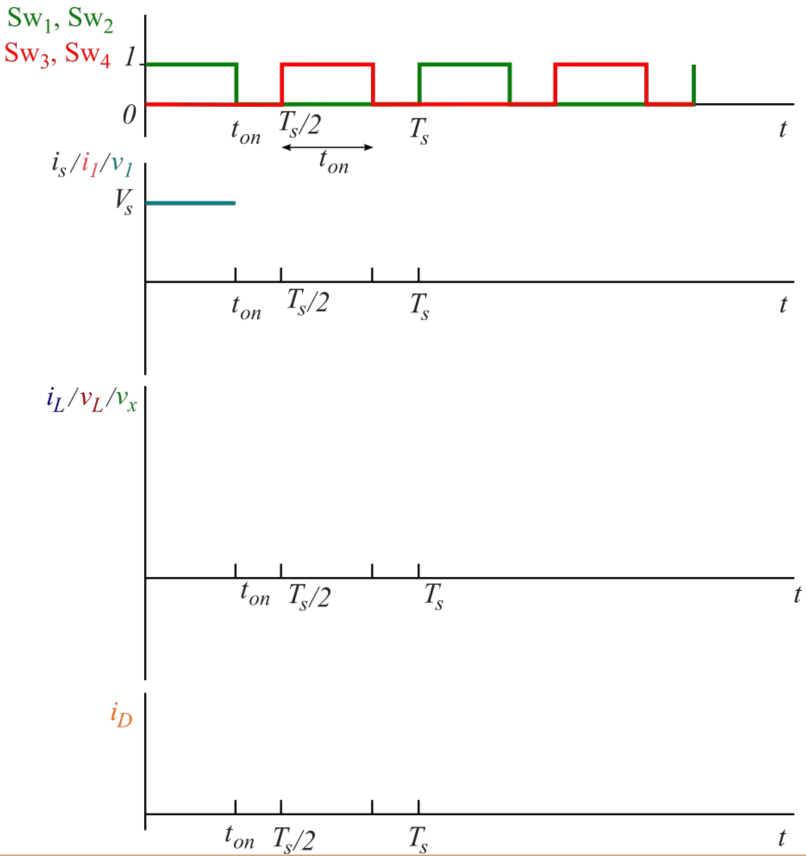

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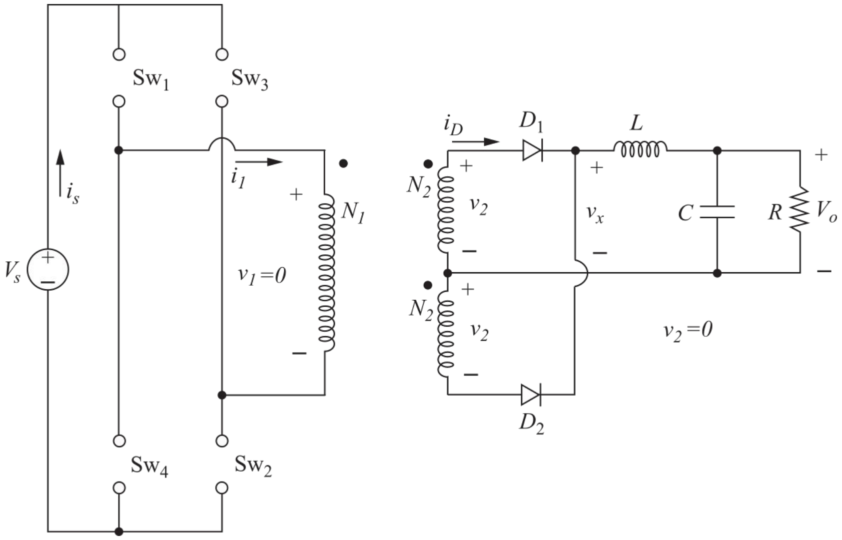

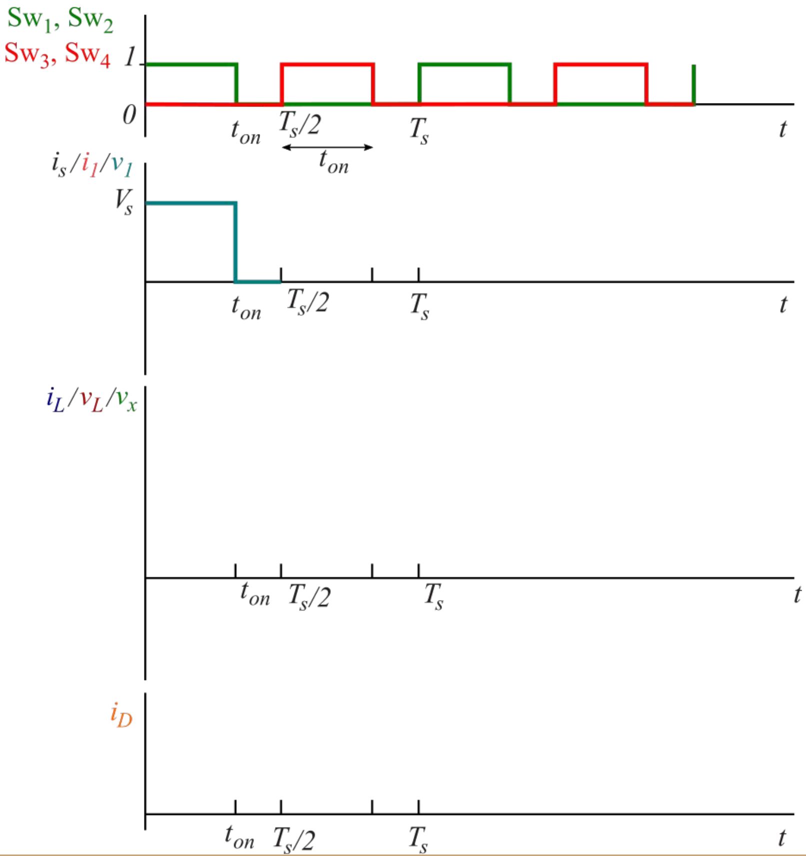

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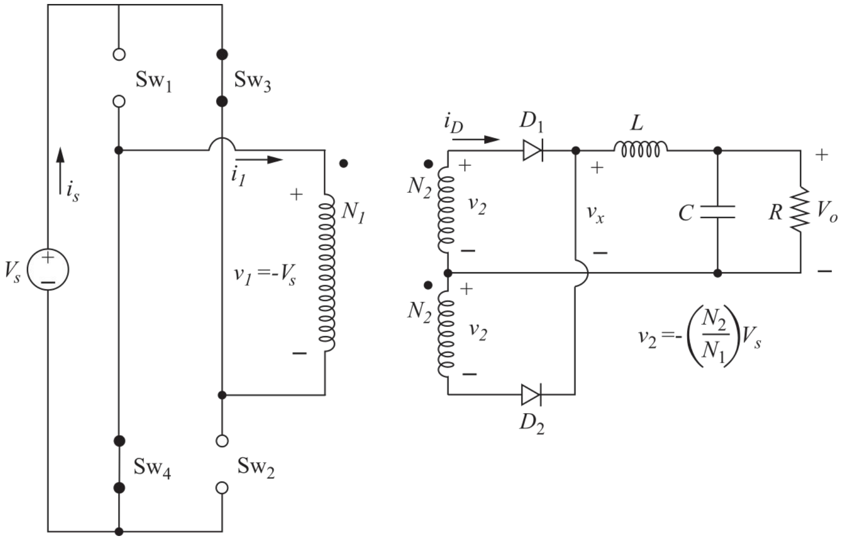

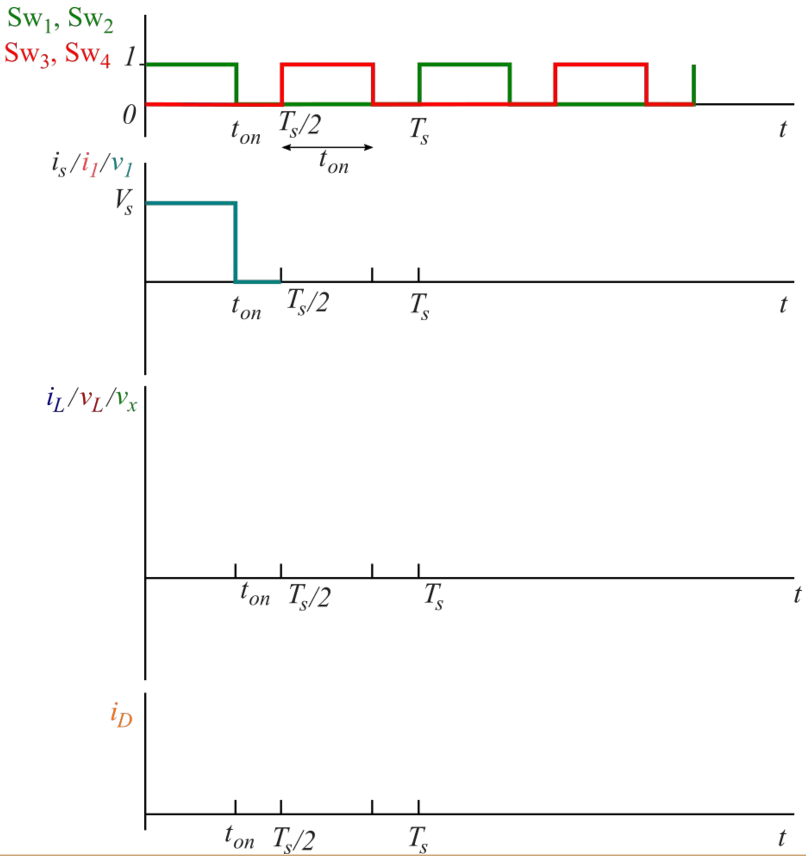

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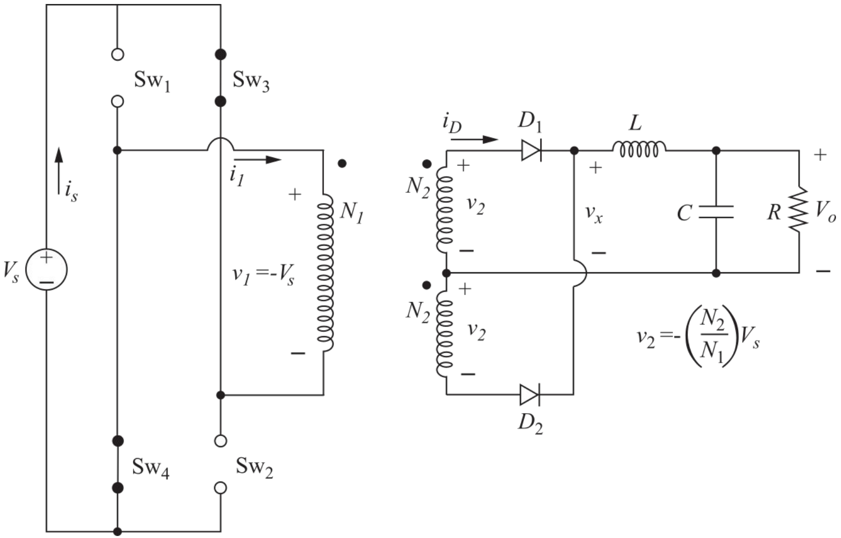

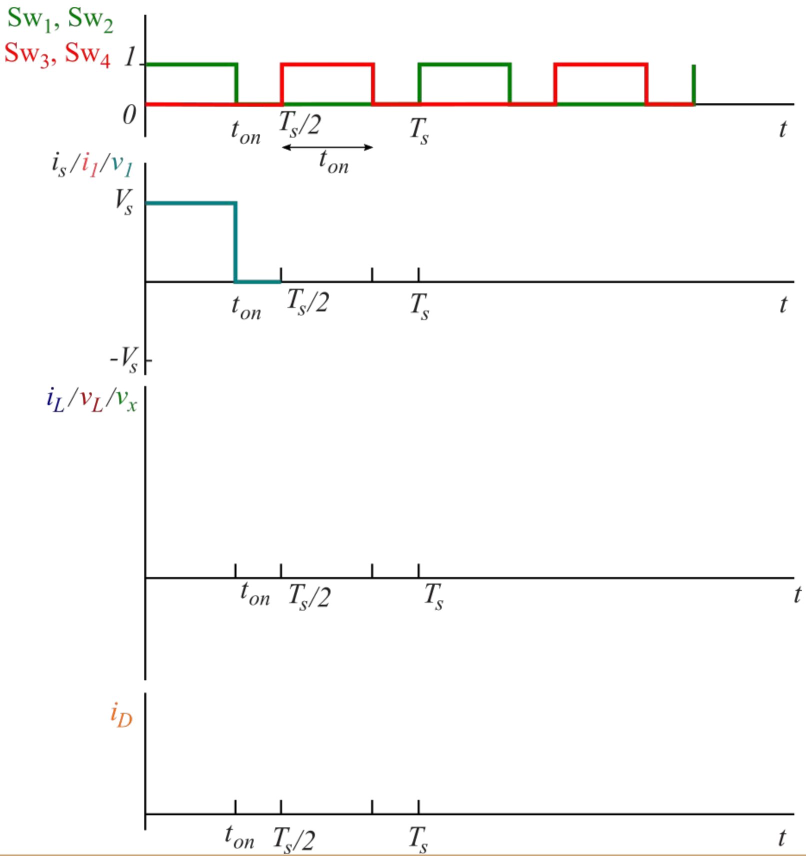

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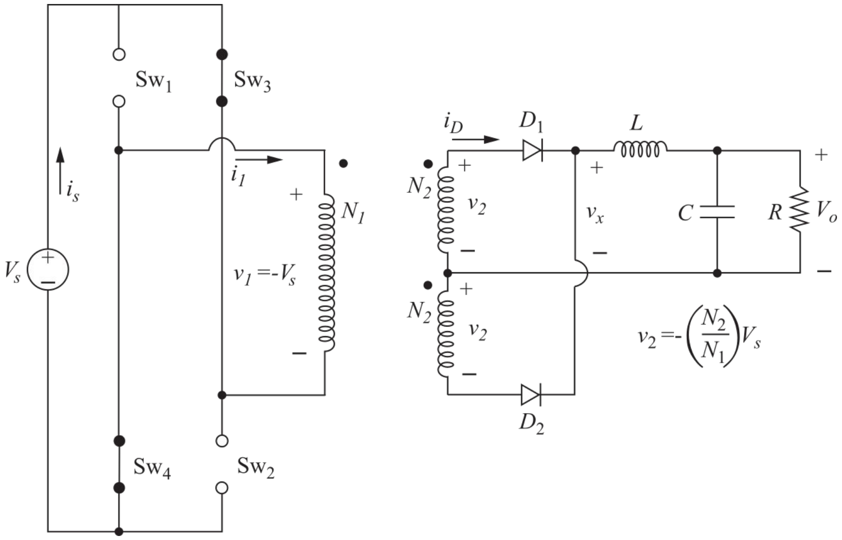

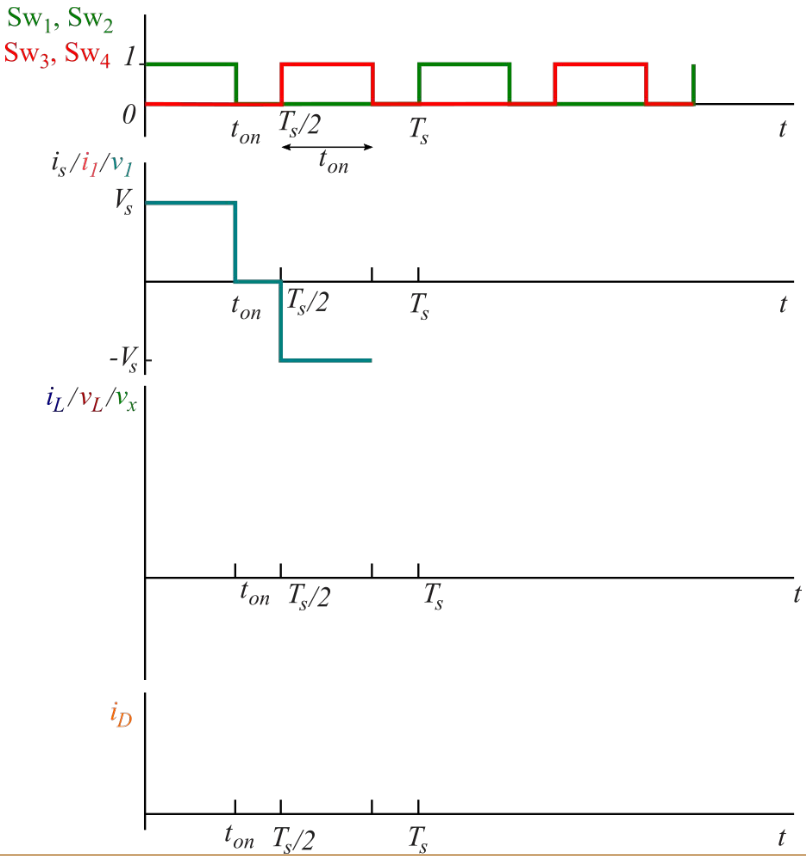

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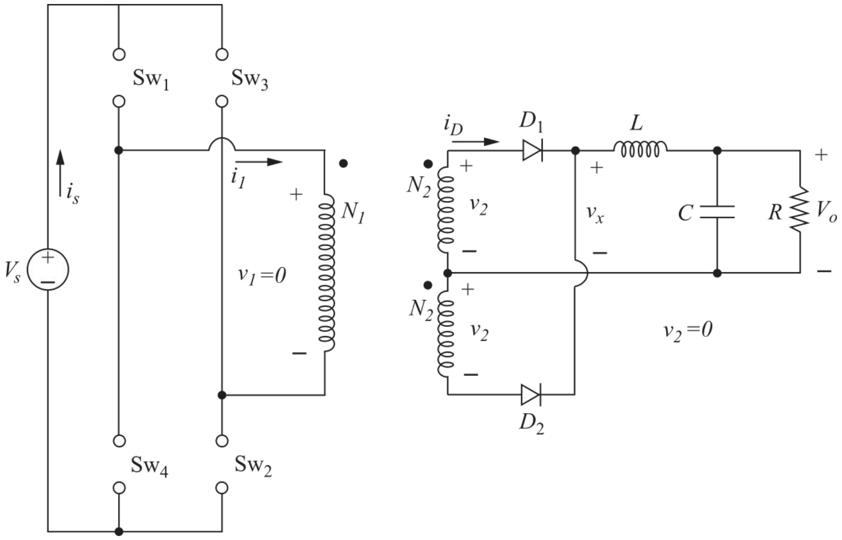

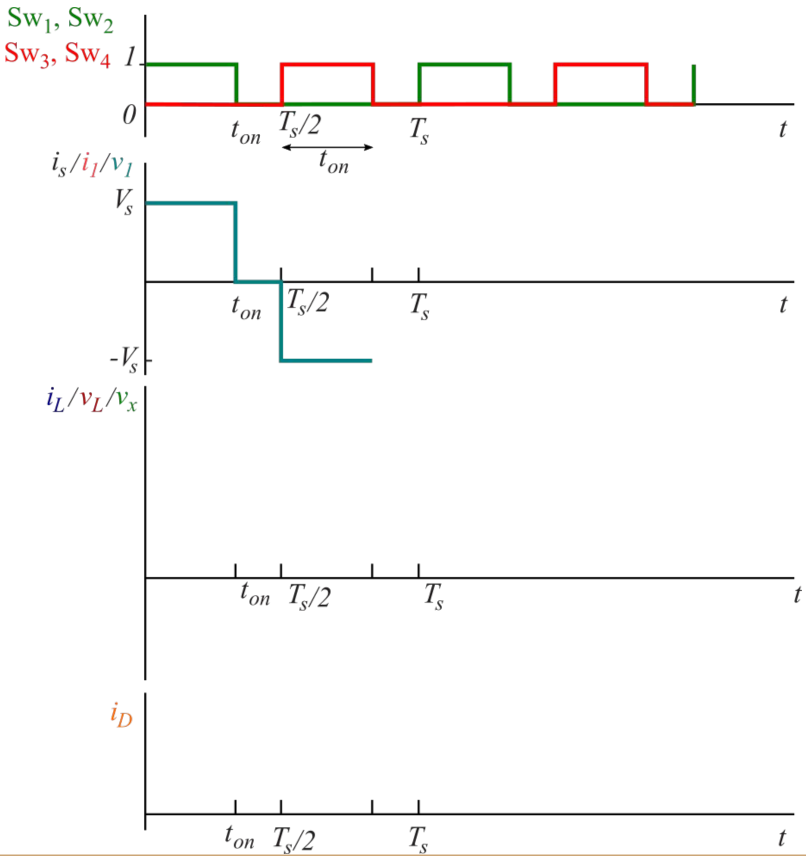

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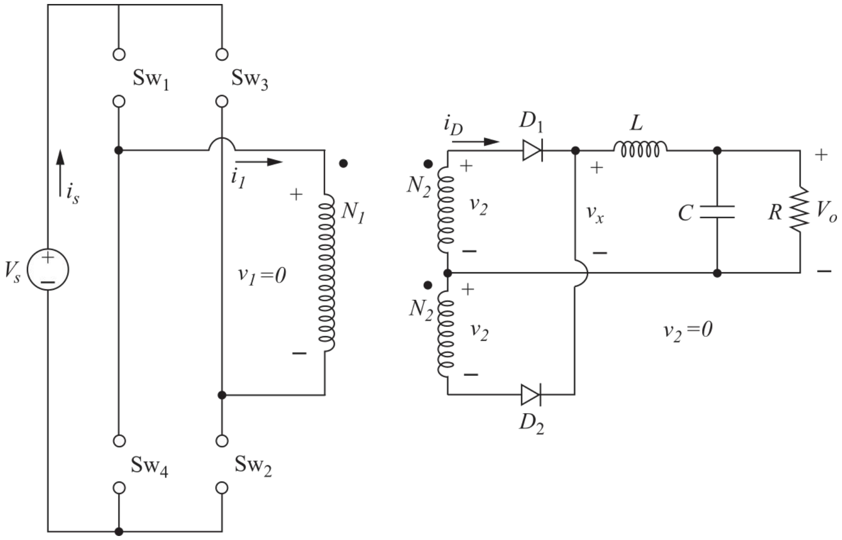

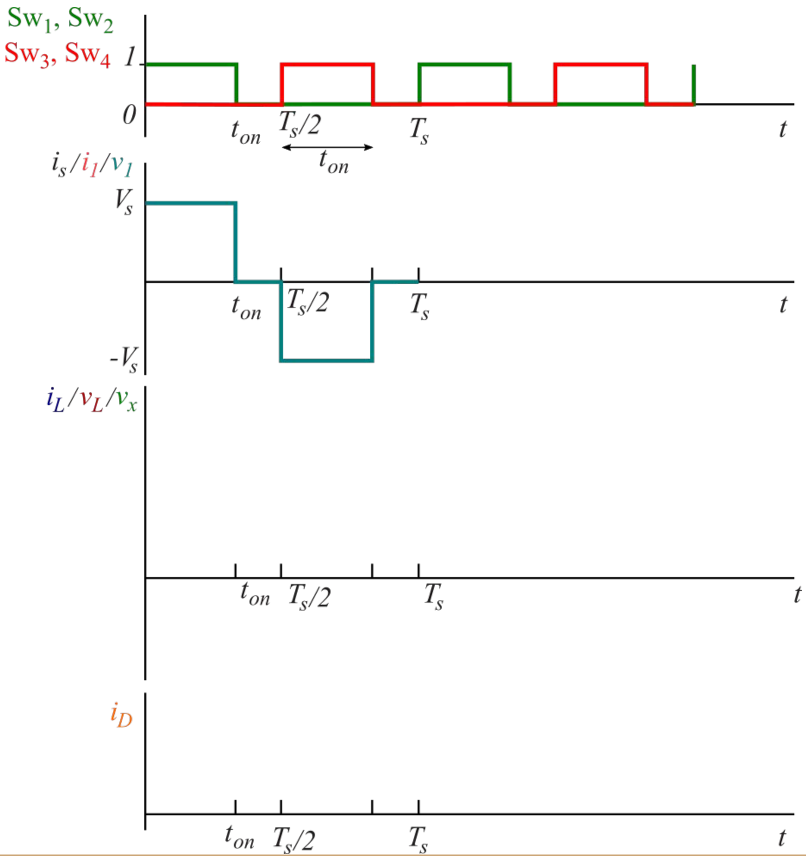

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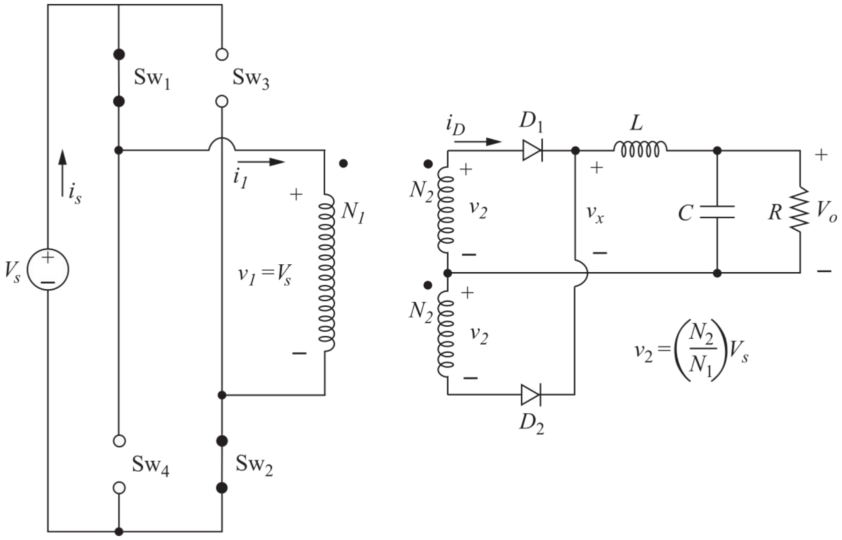

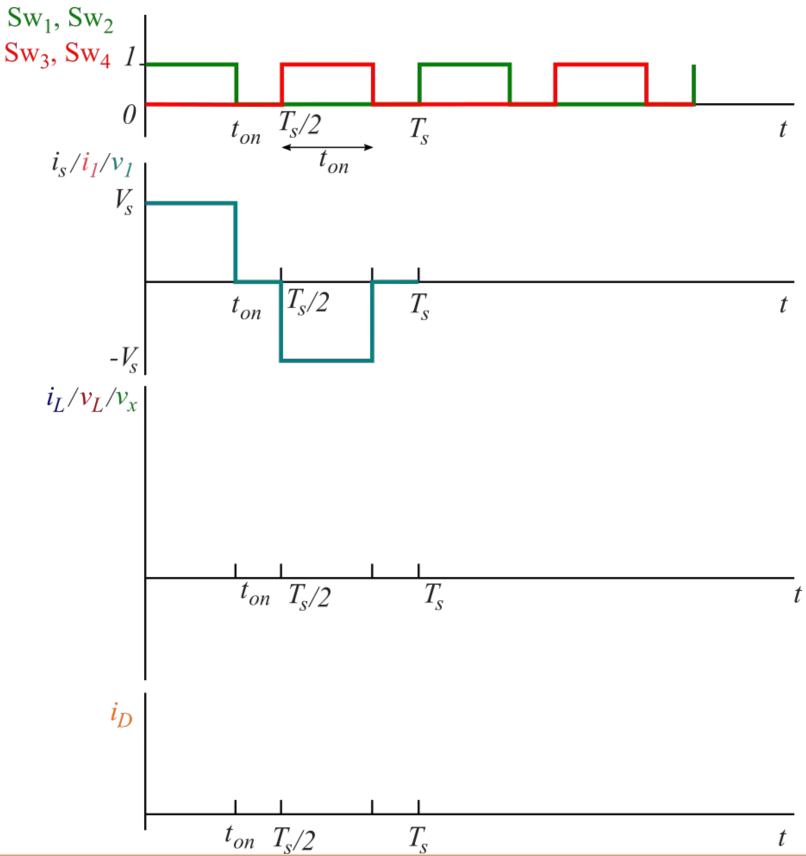

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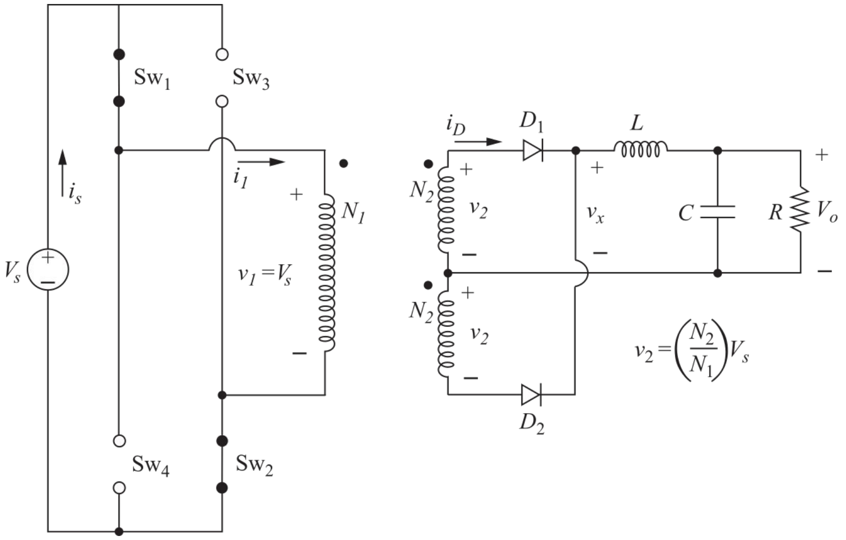

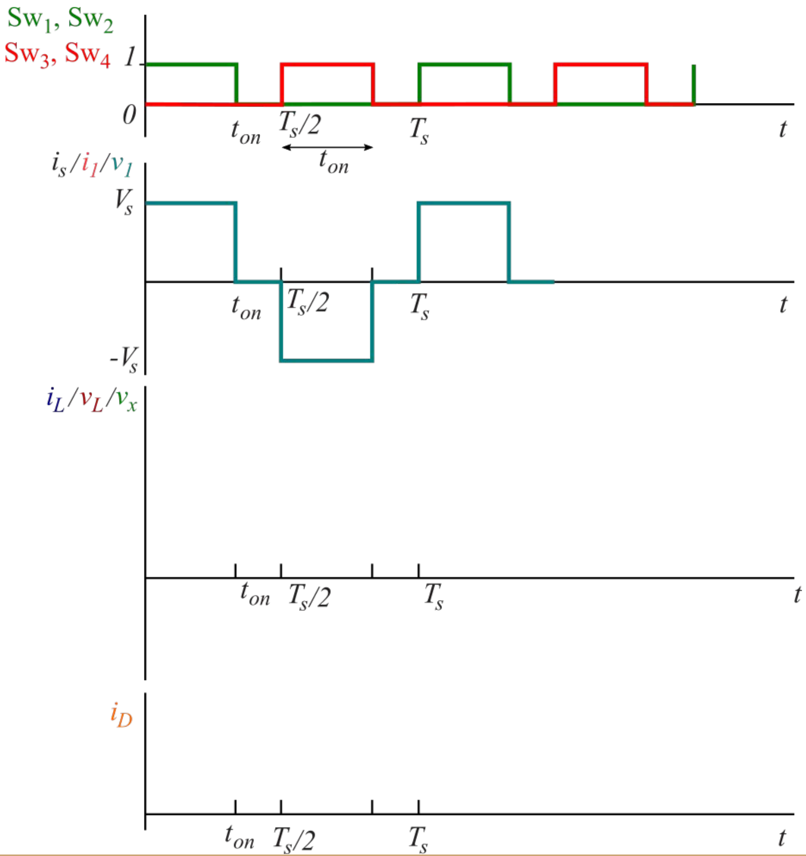

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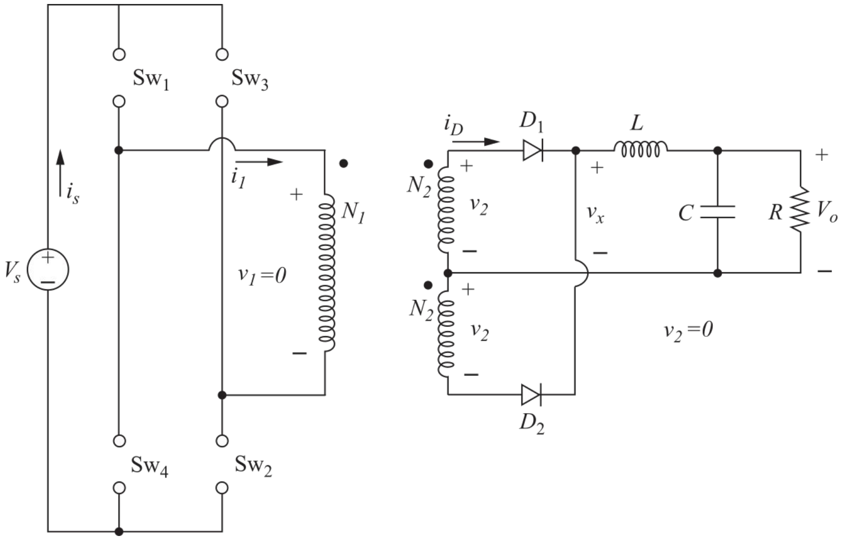

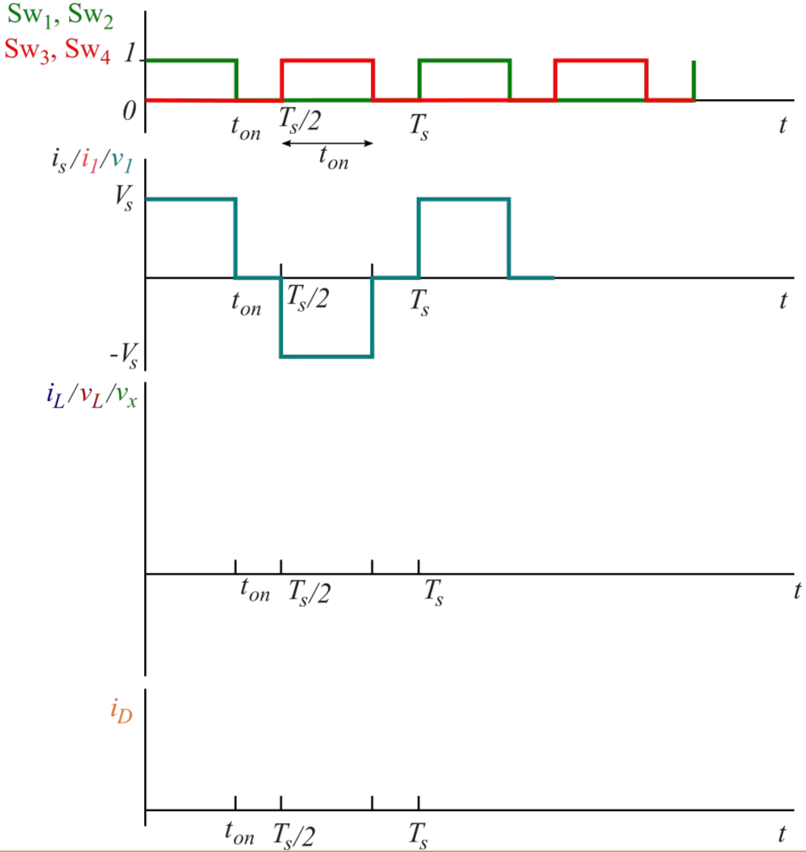

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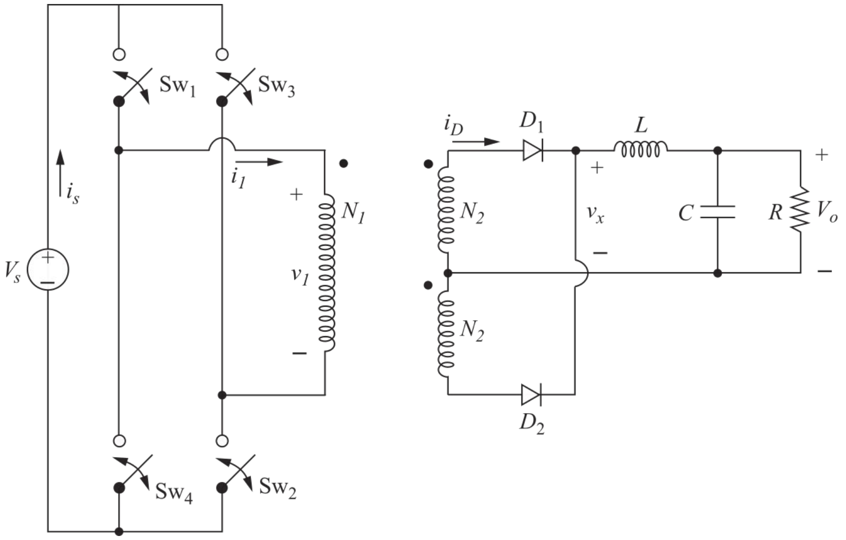

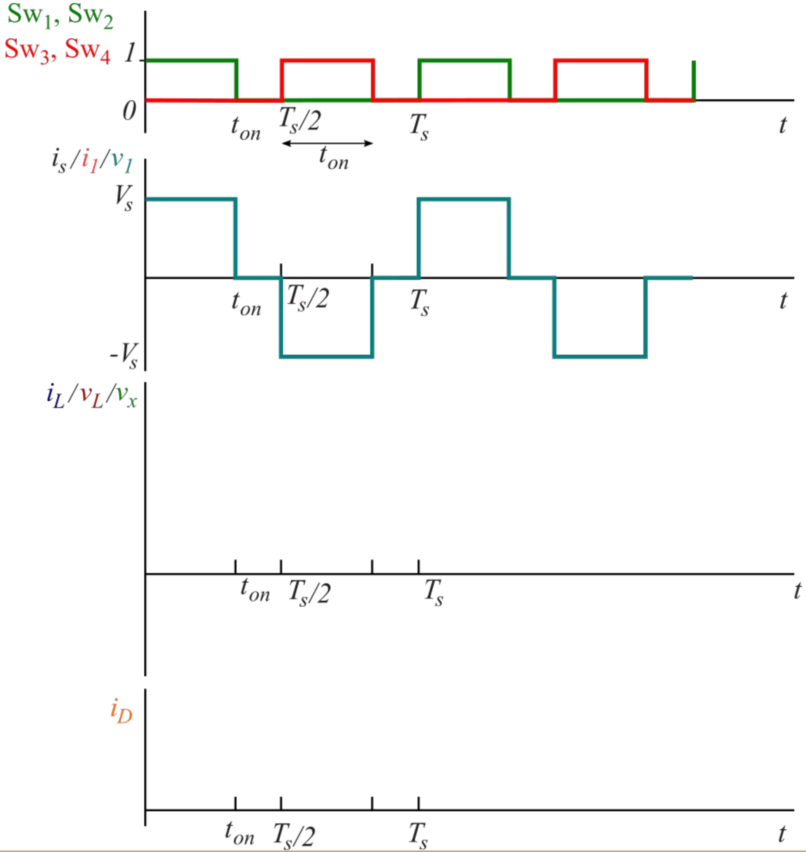

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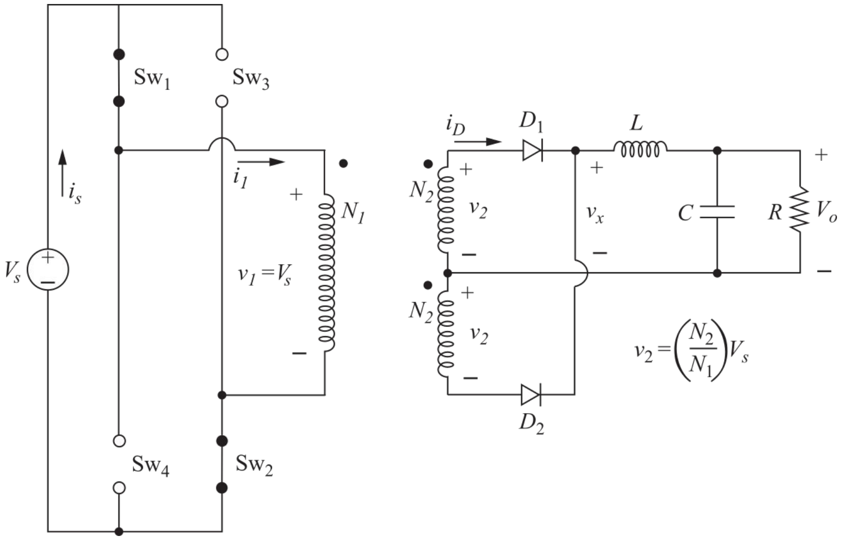

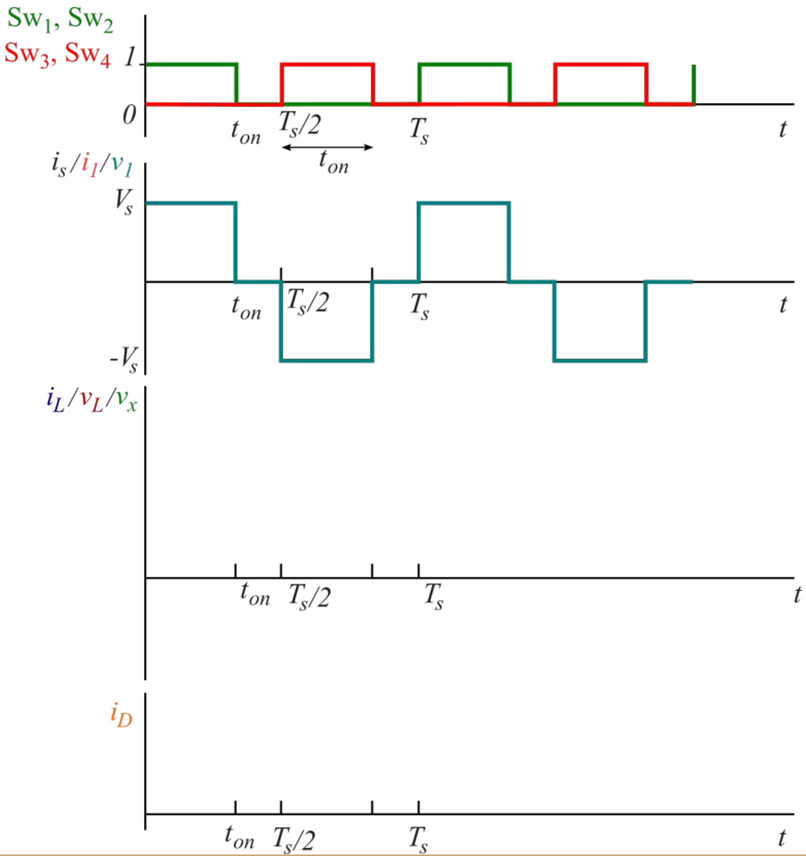

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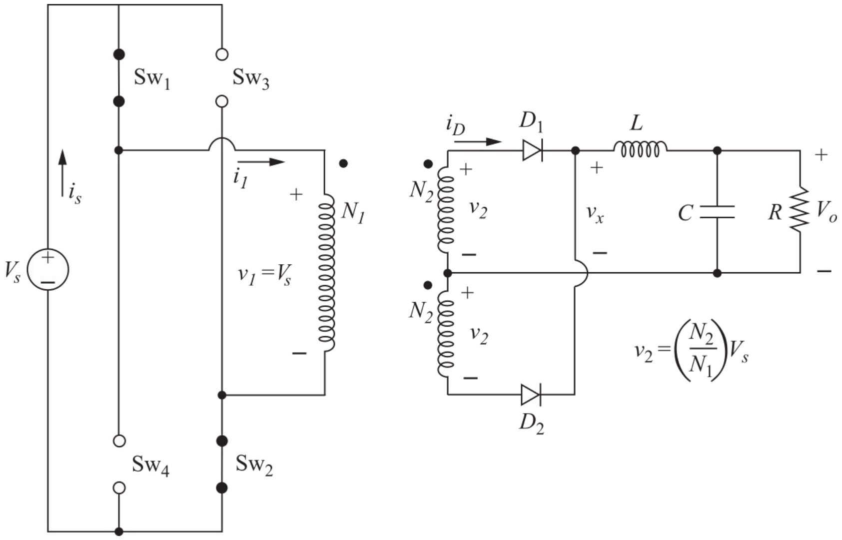

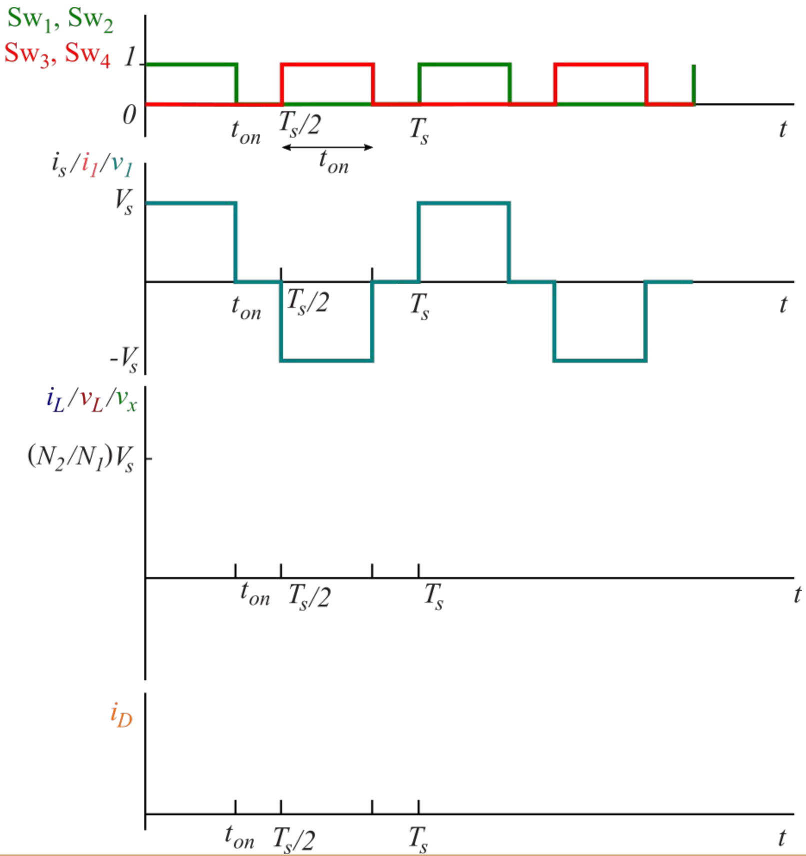

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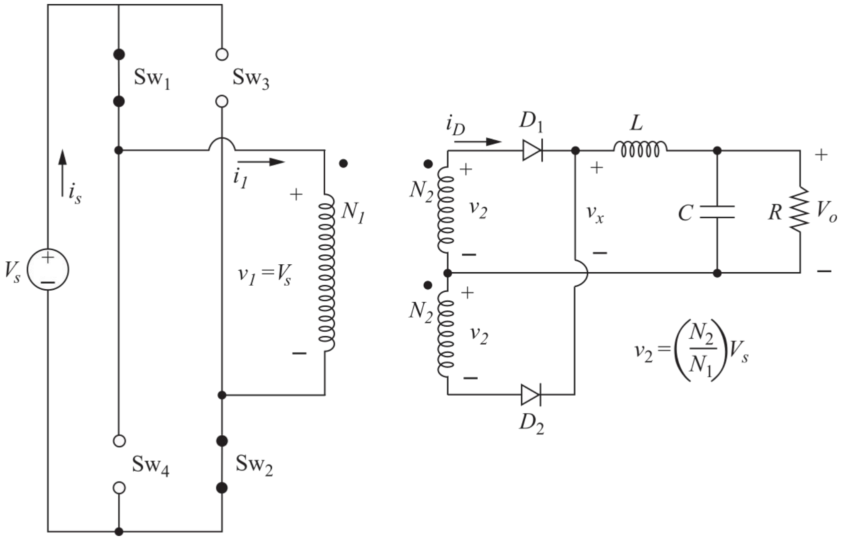

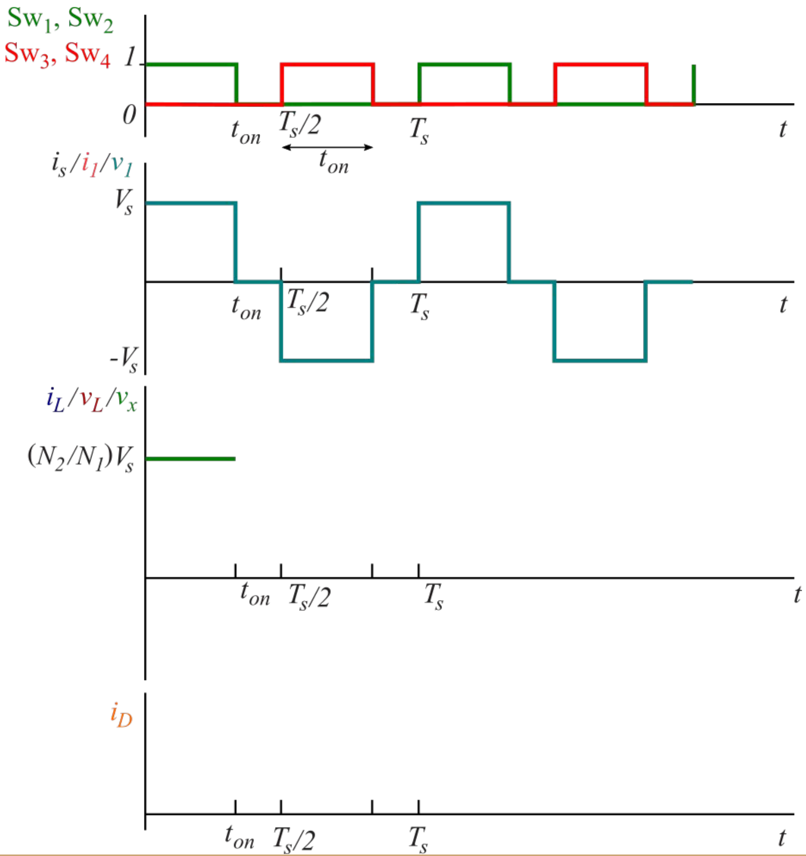

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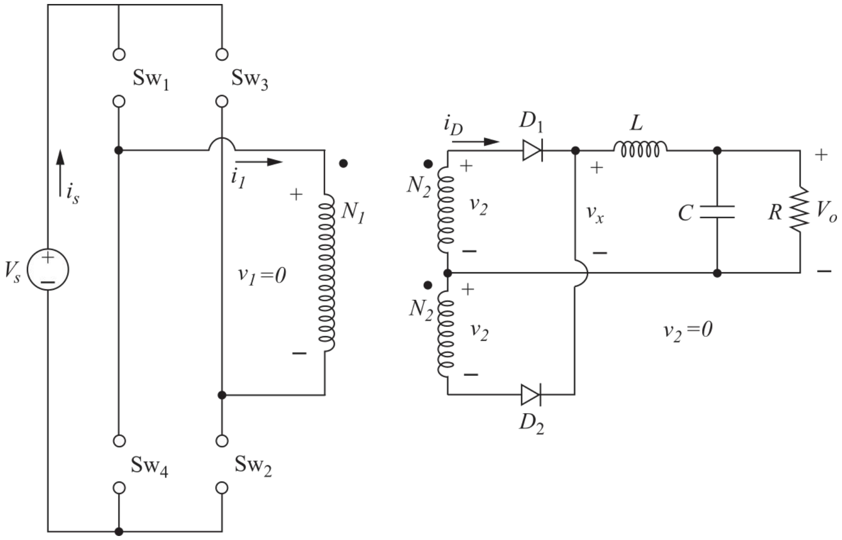

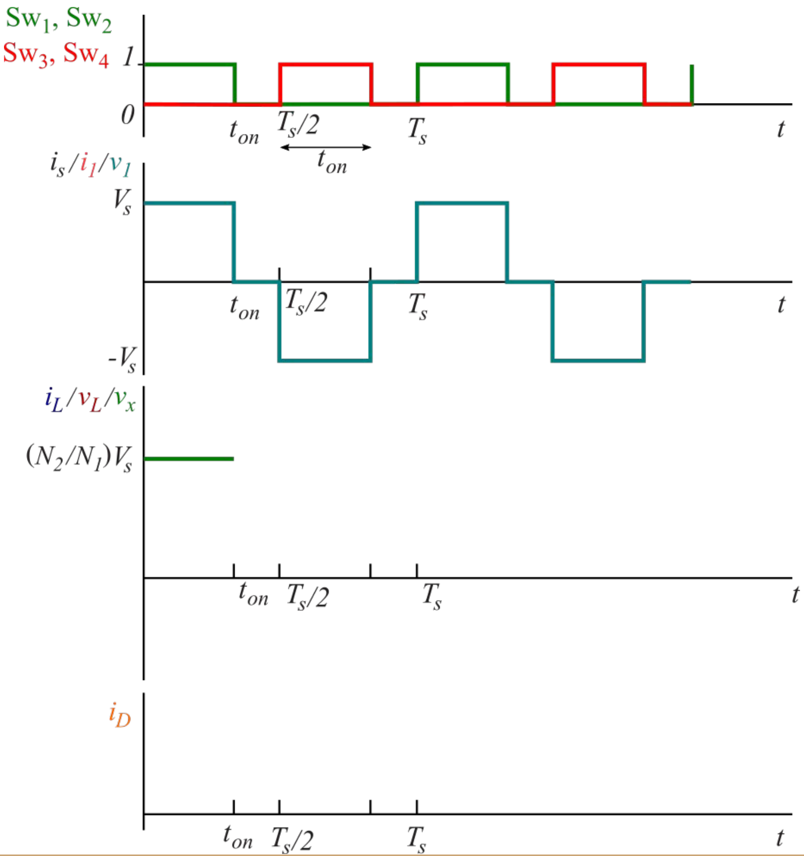

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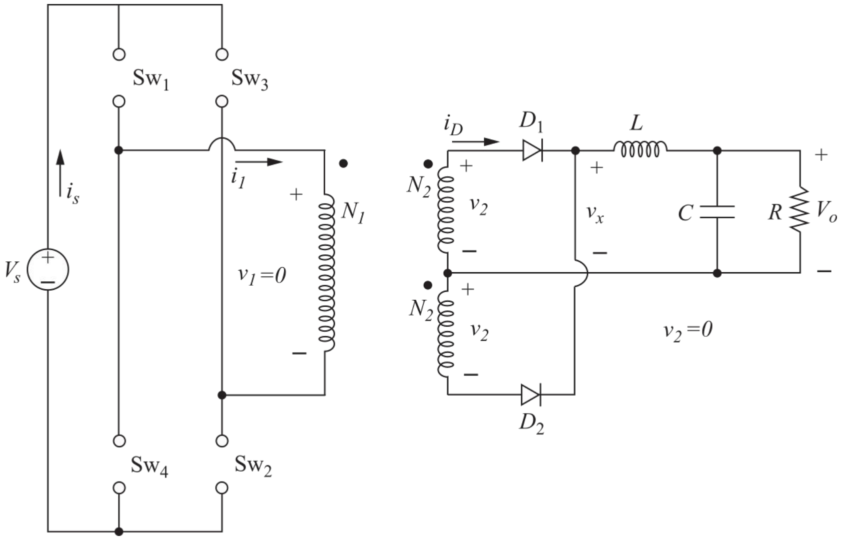
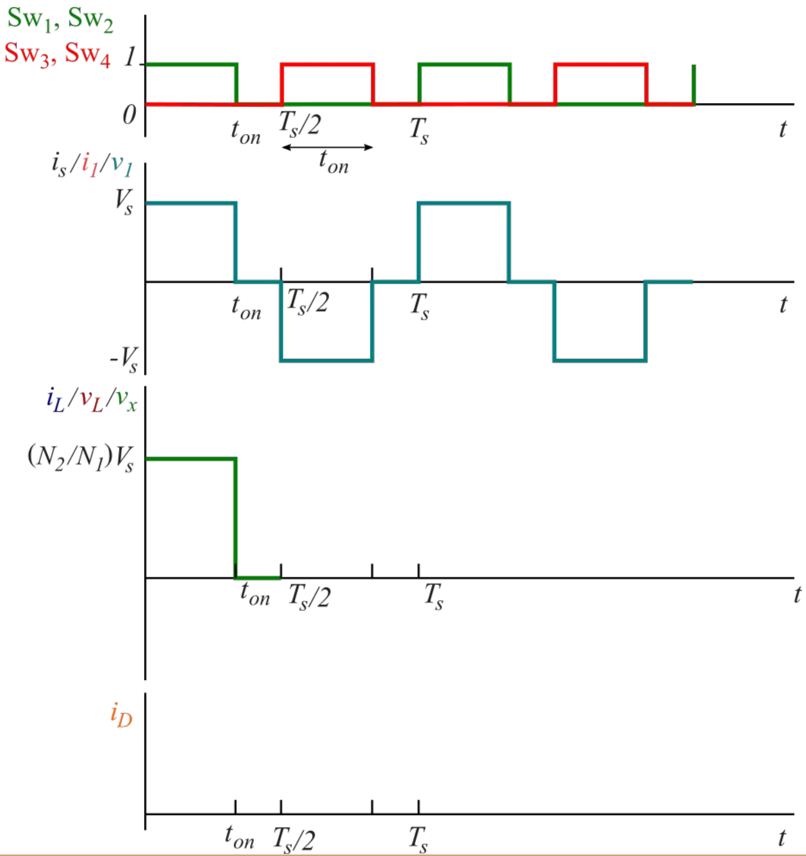
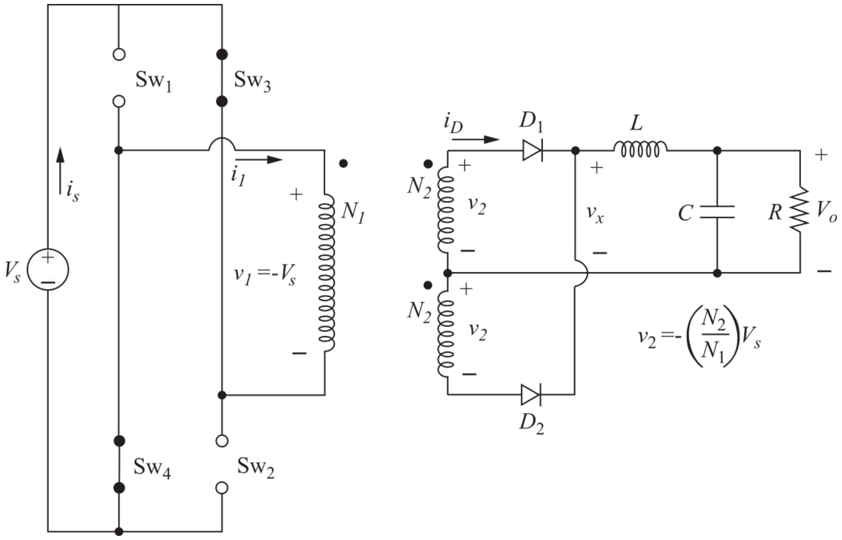
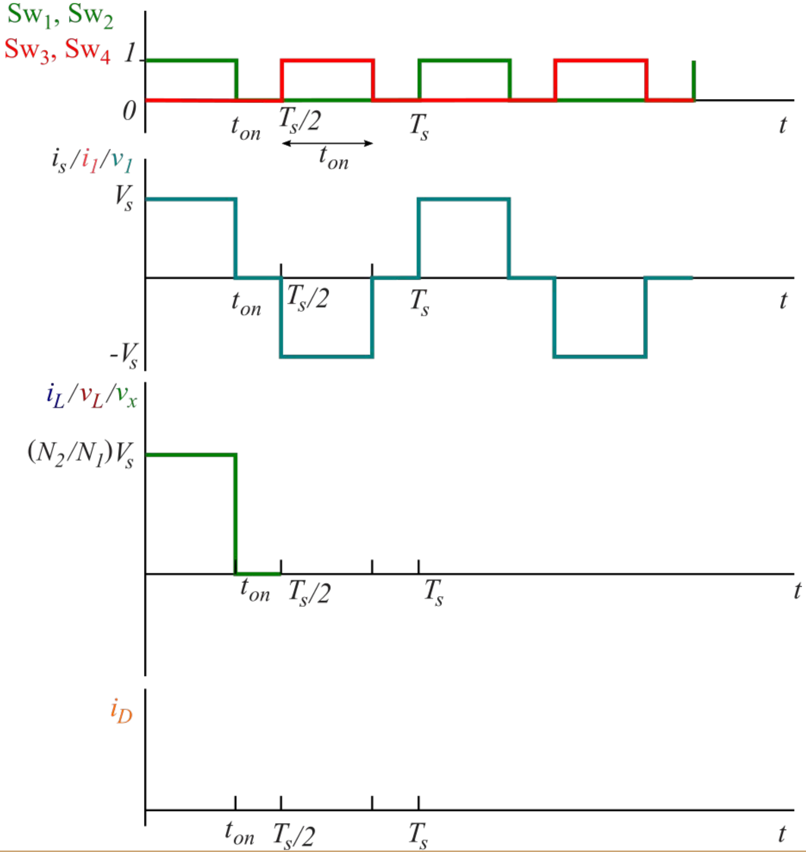




Convertidor Full-bridge: formas de onda





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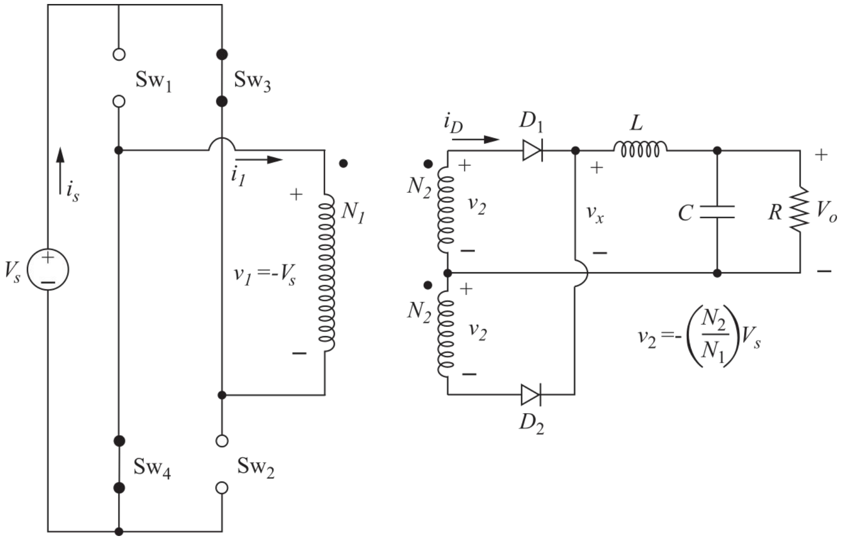

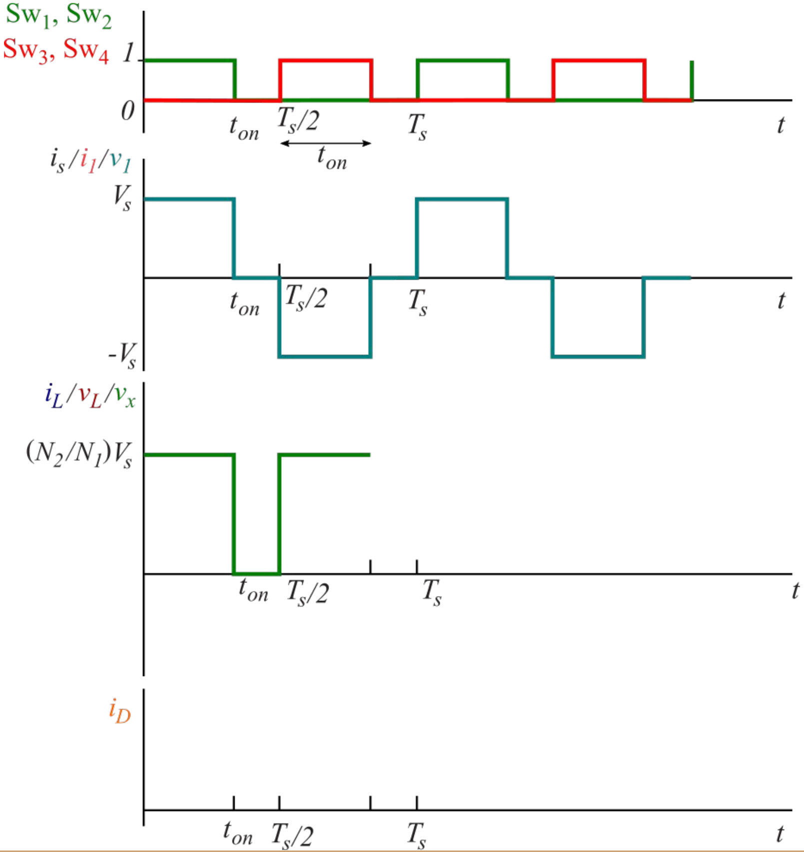

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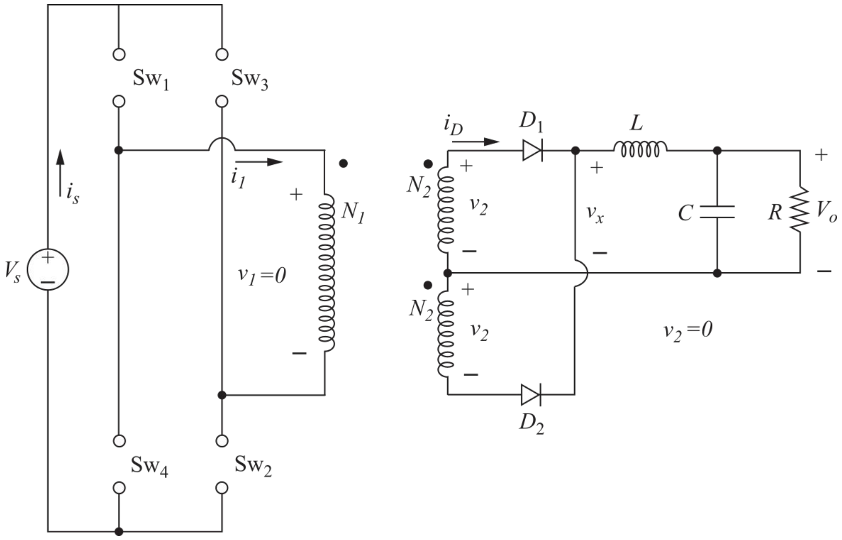

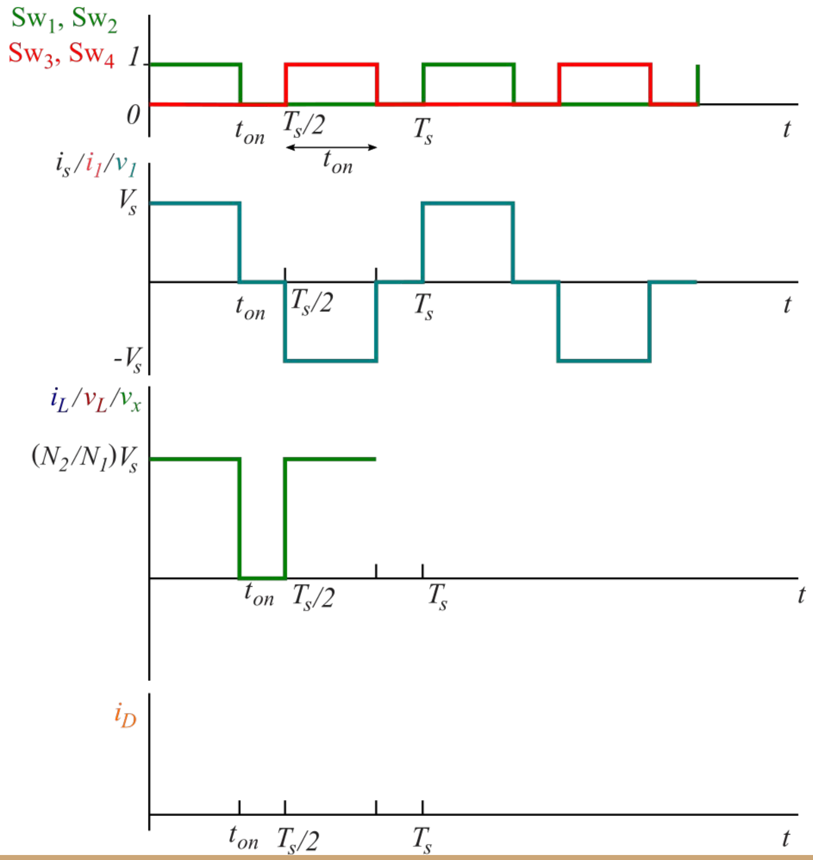

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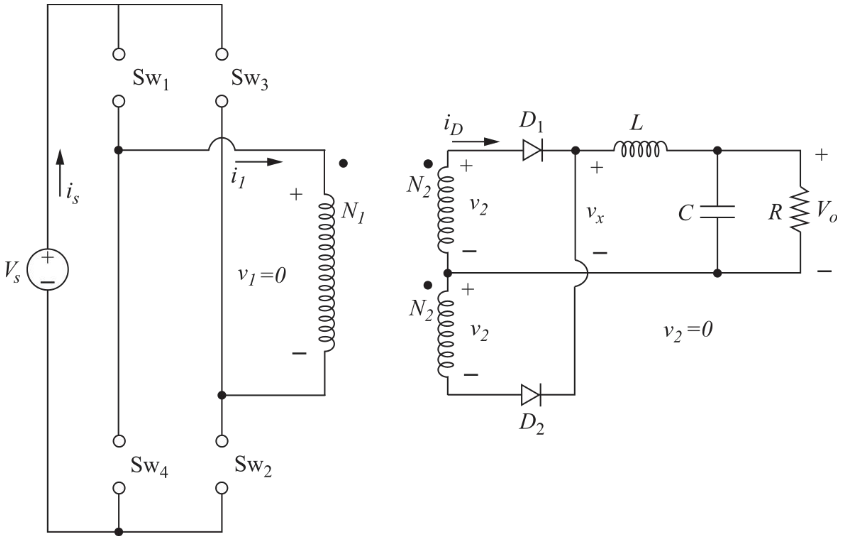

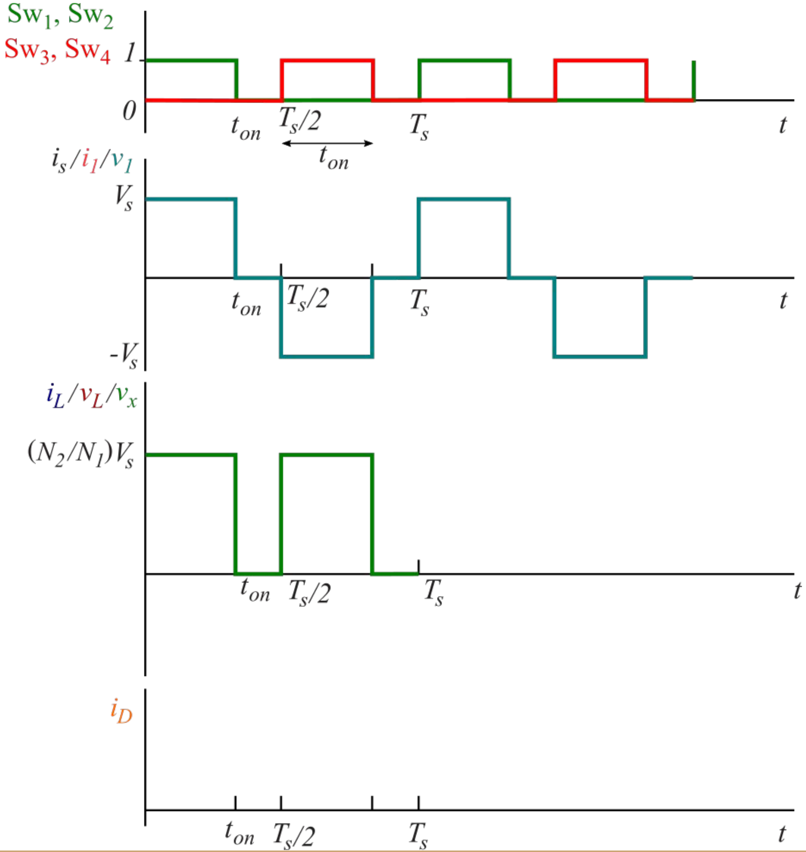

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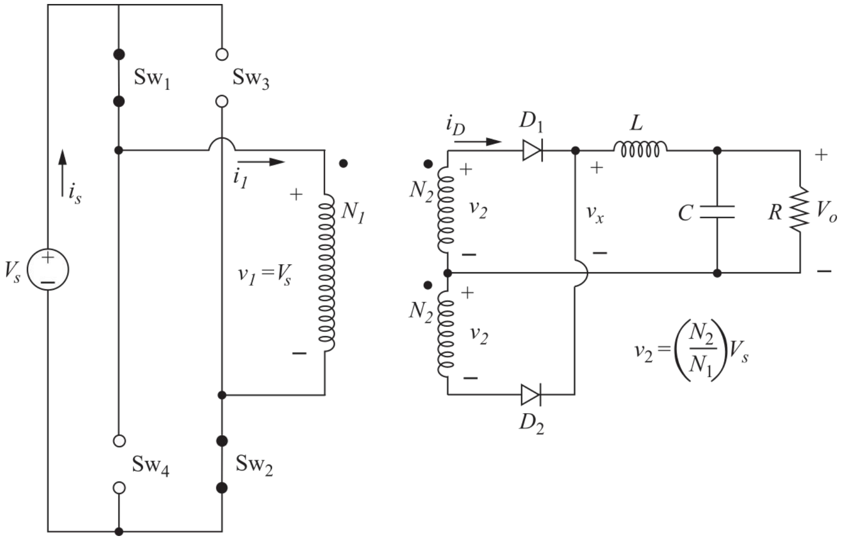

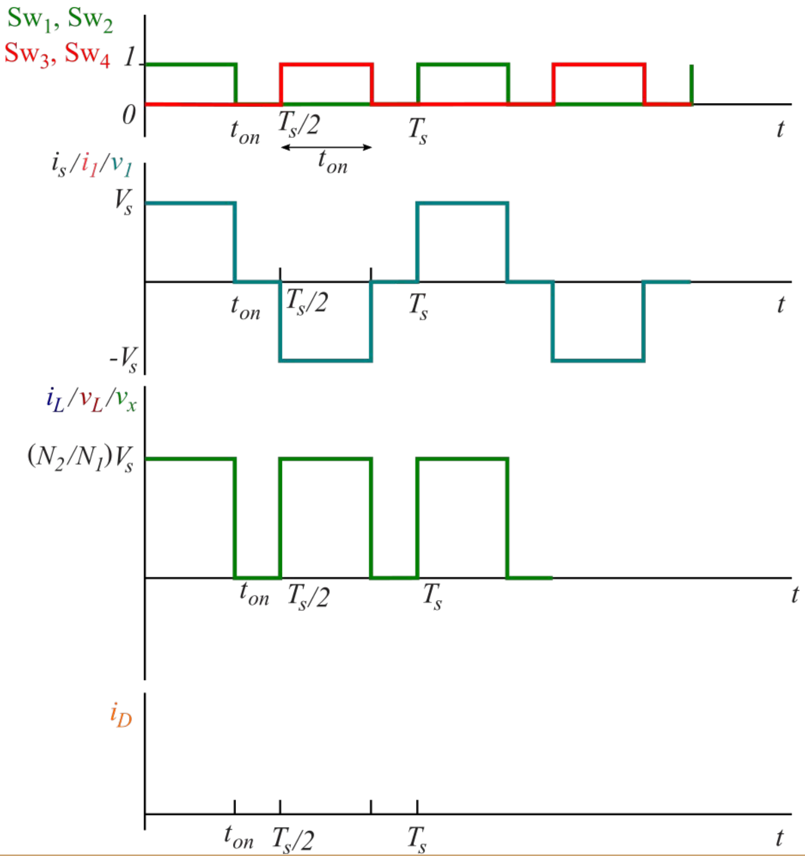

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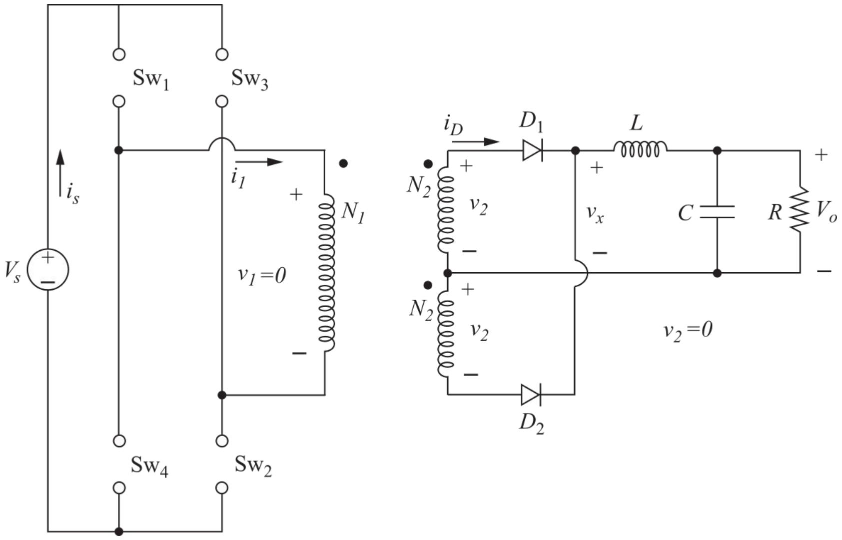

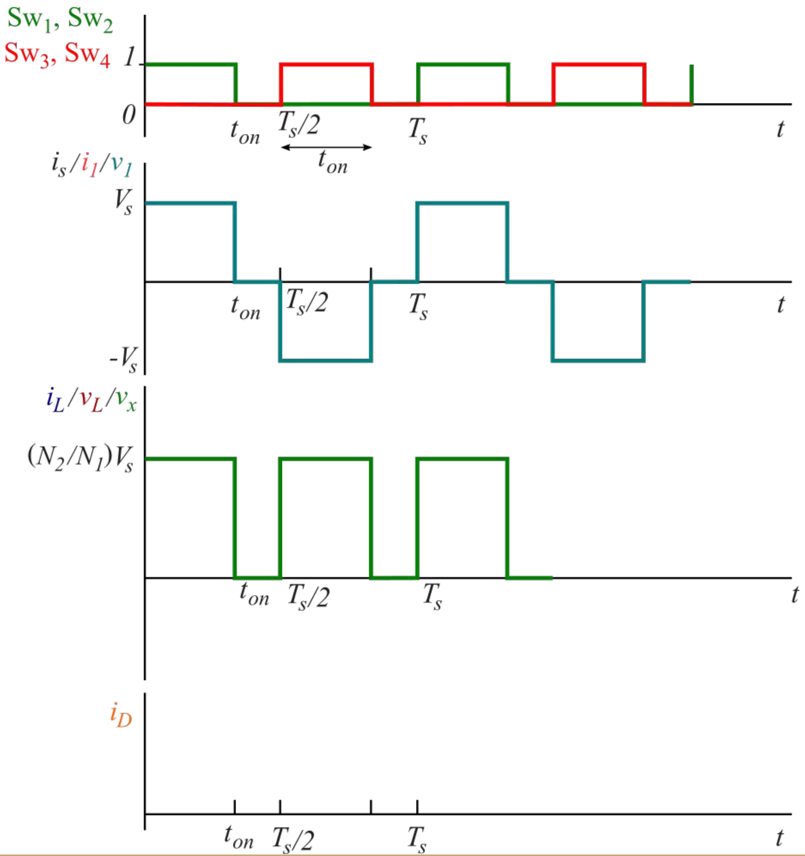

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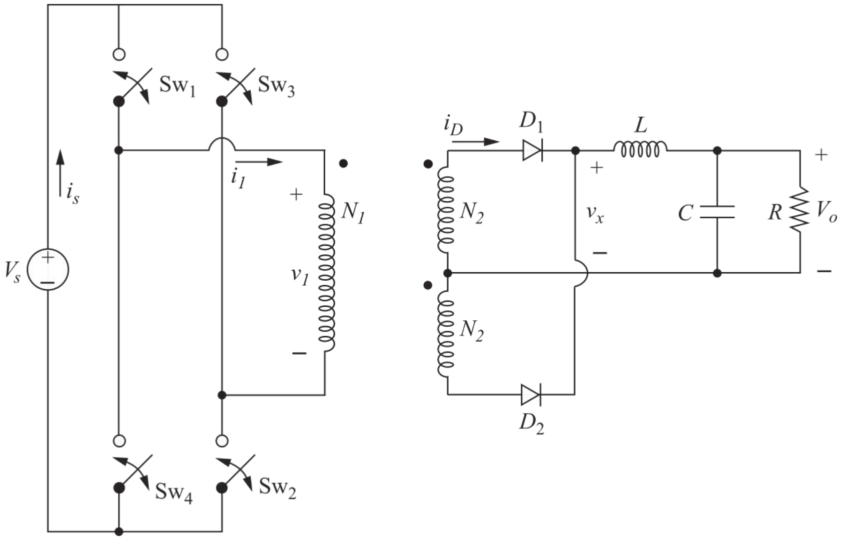

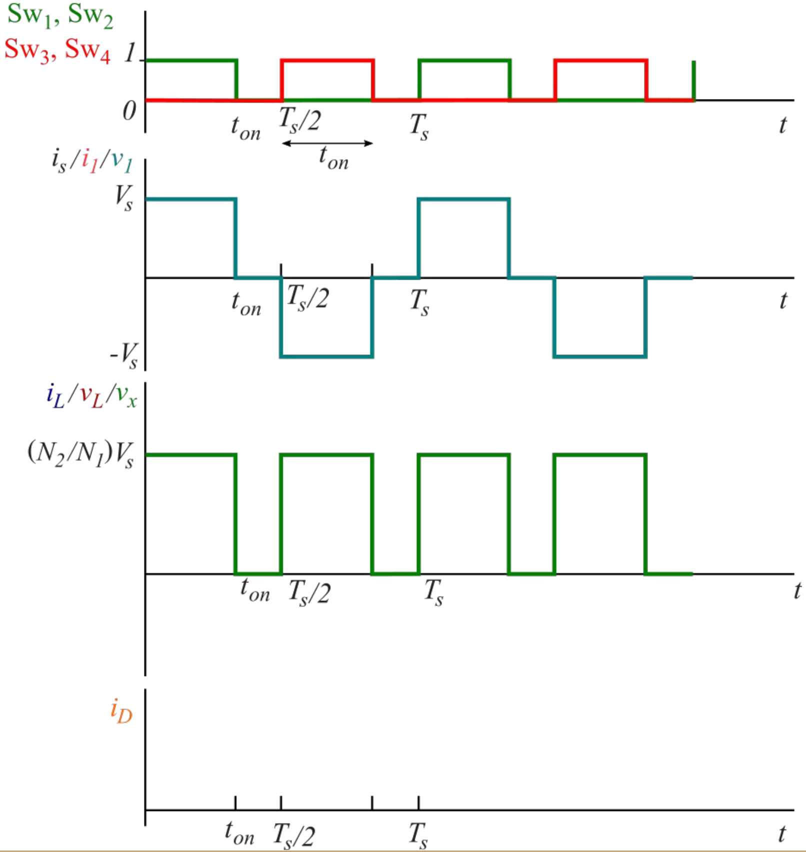

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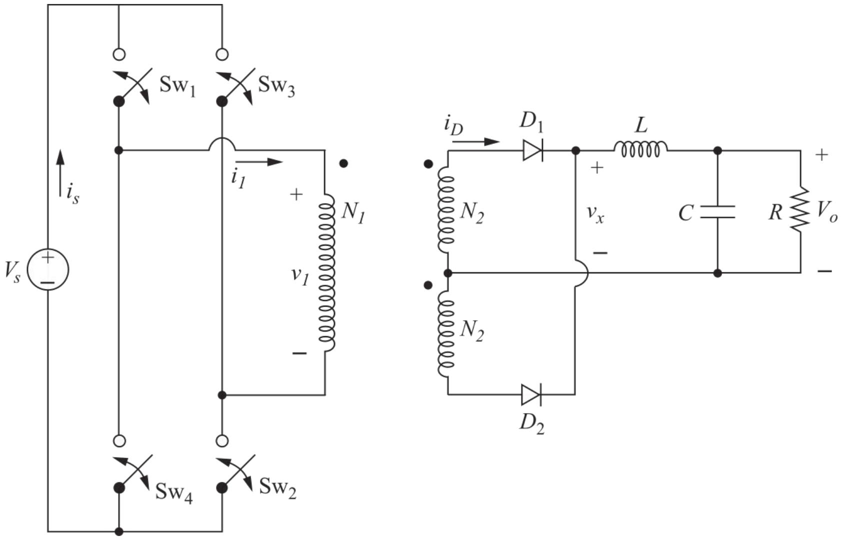

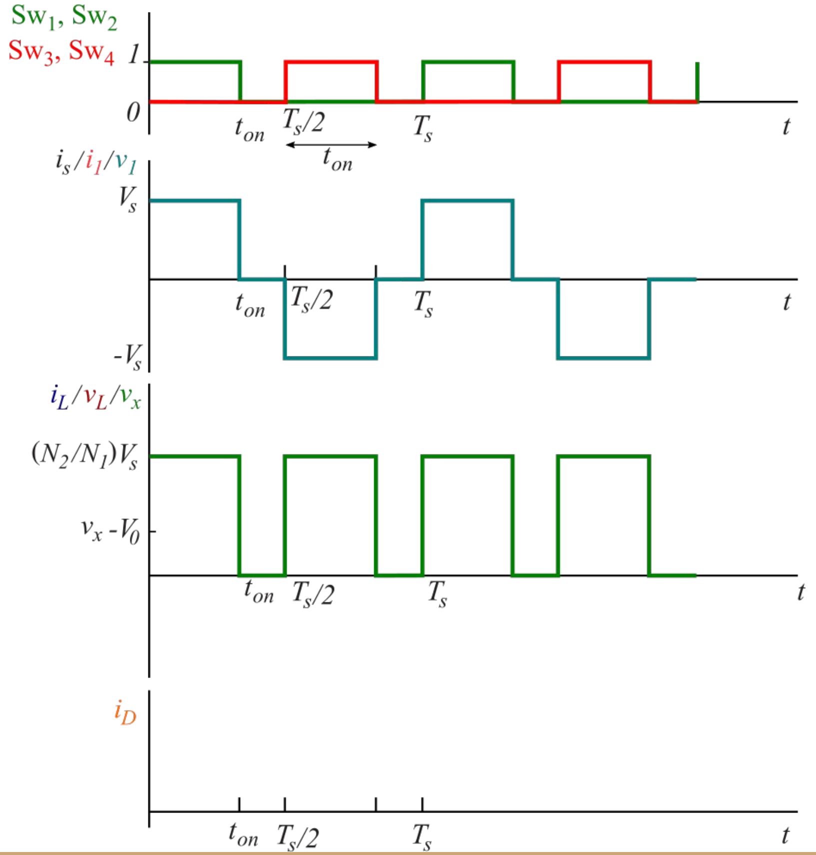

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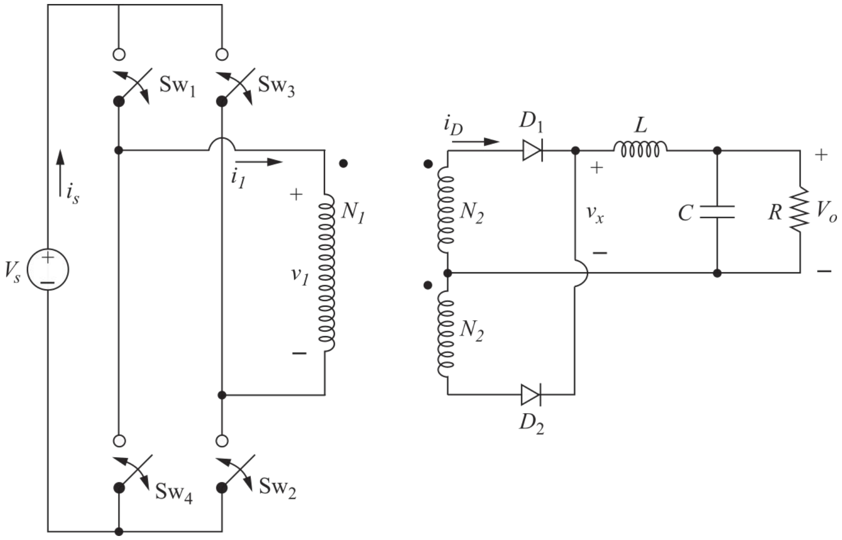

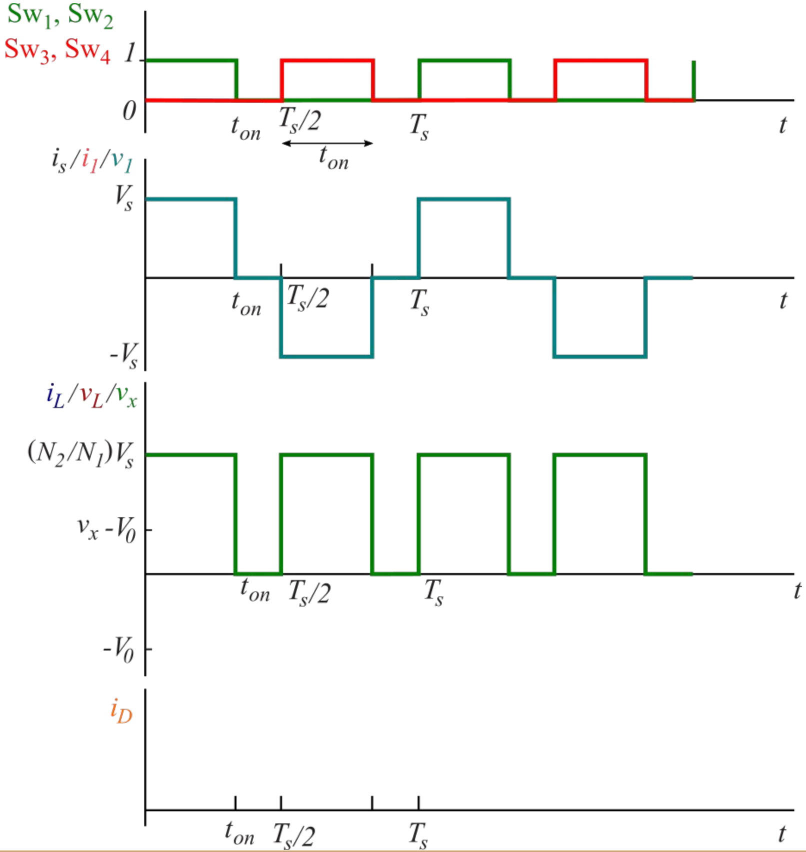

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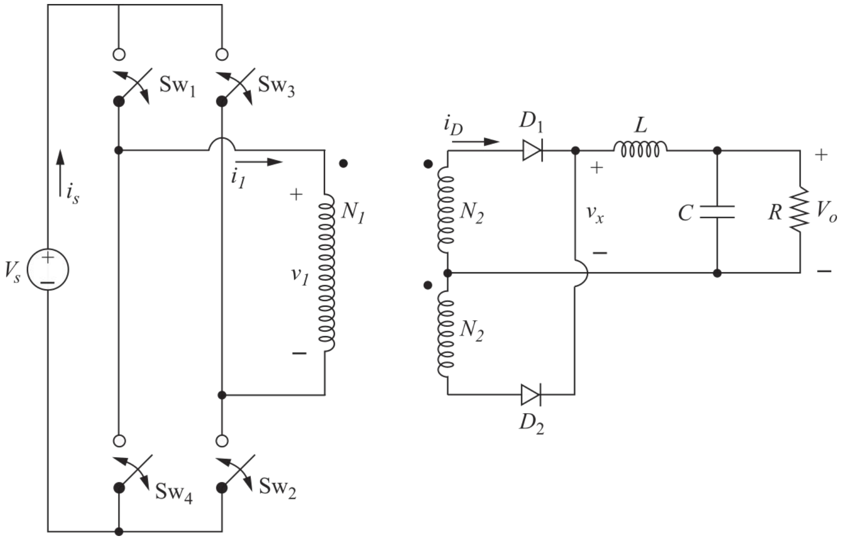

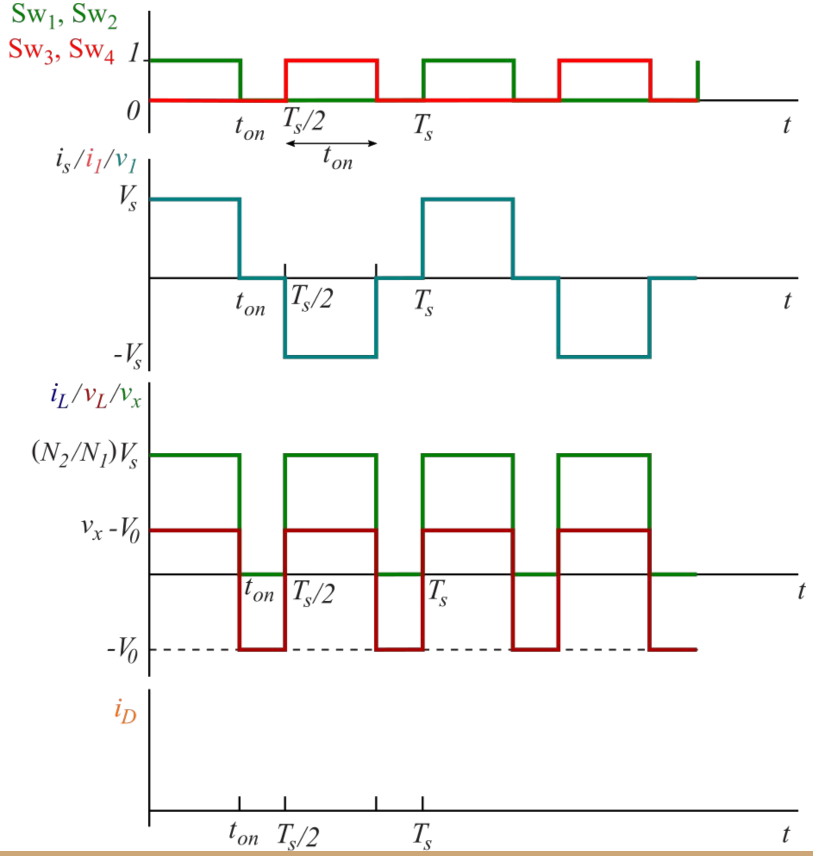

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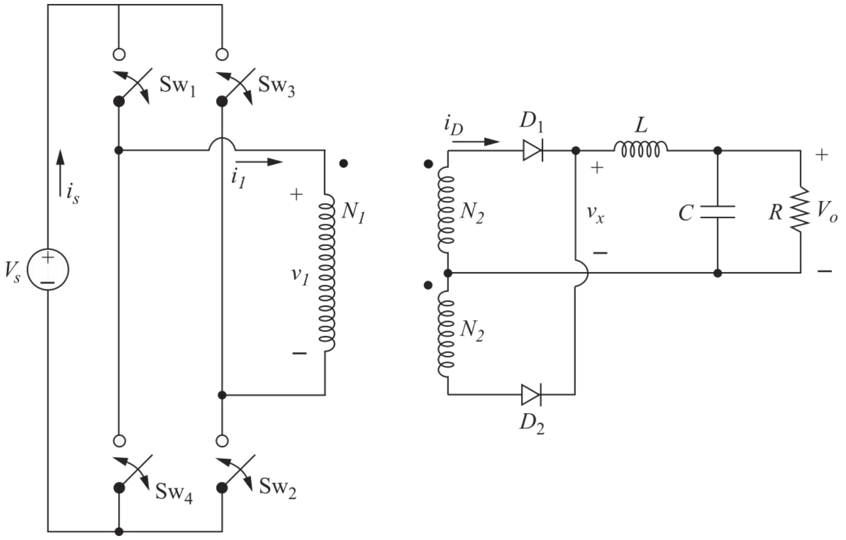

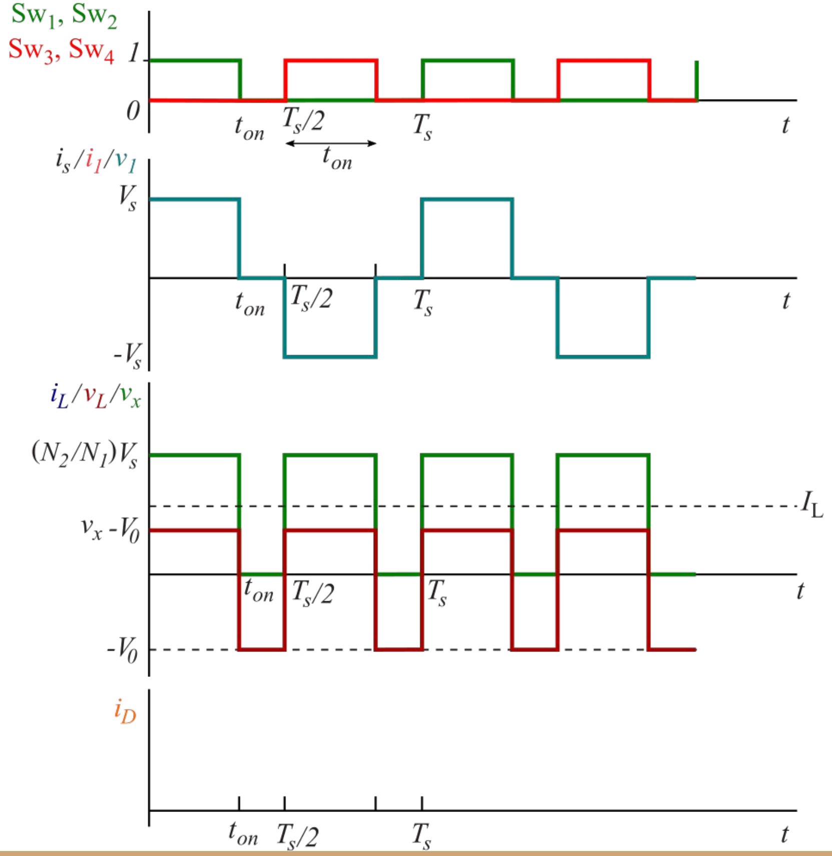

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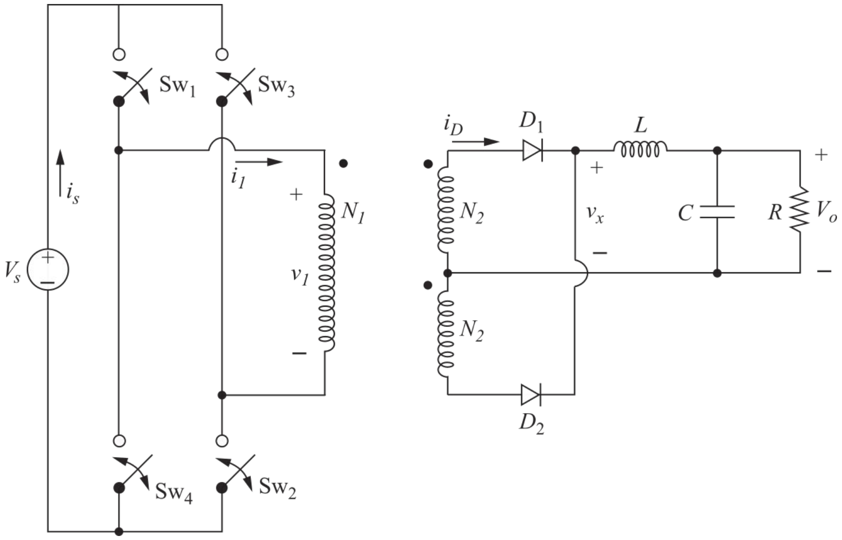

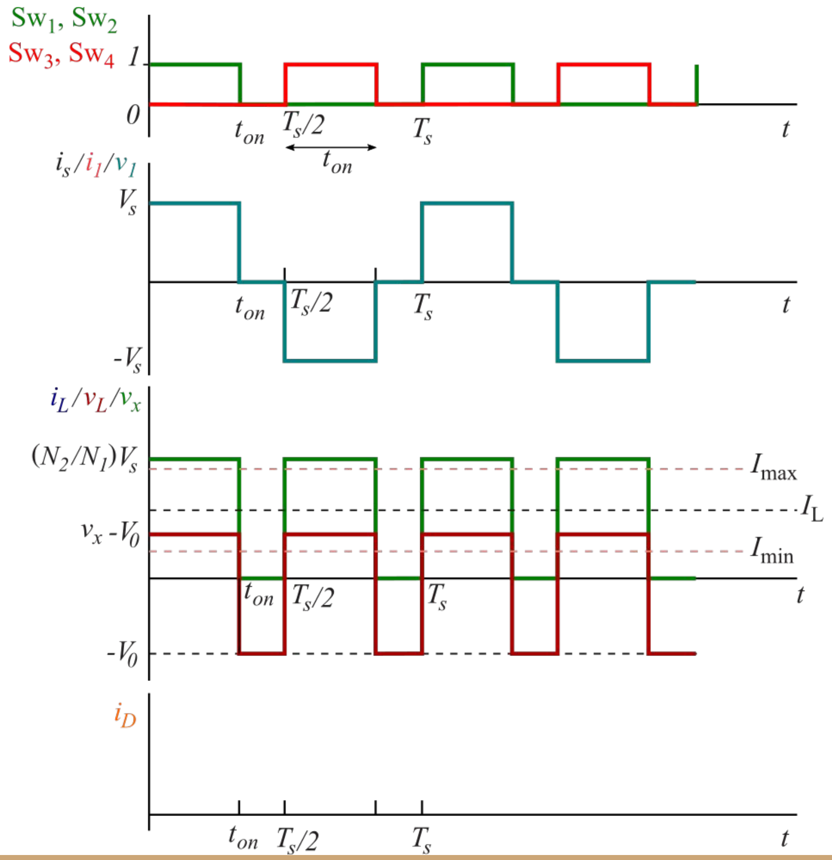

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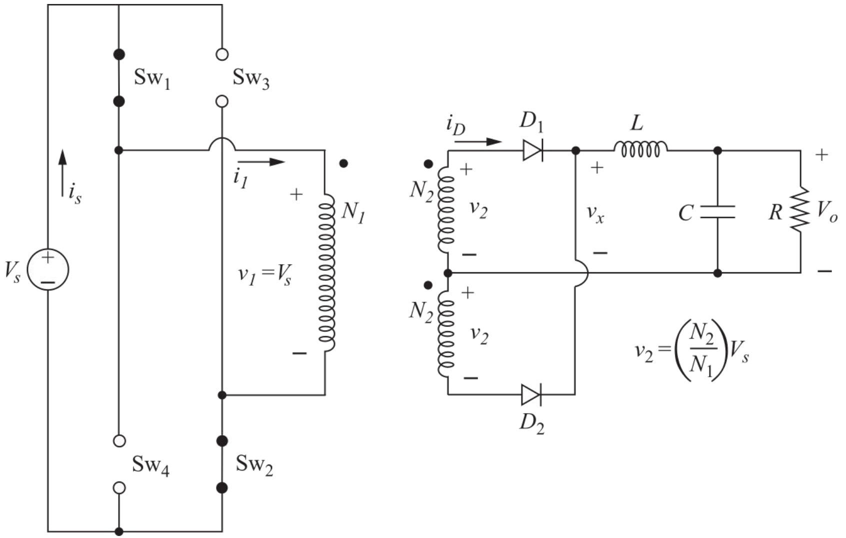

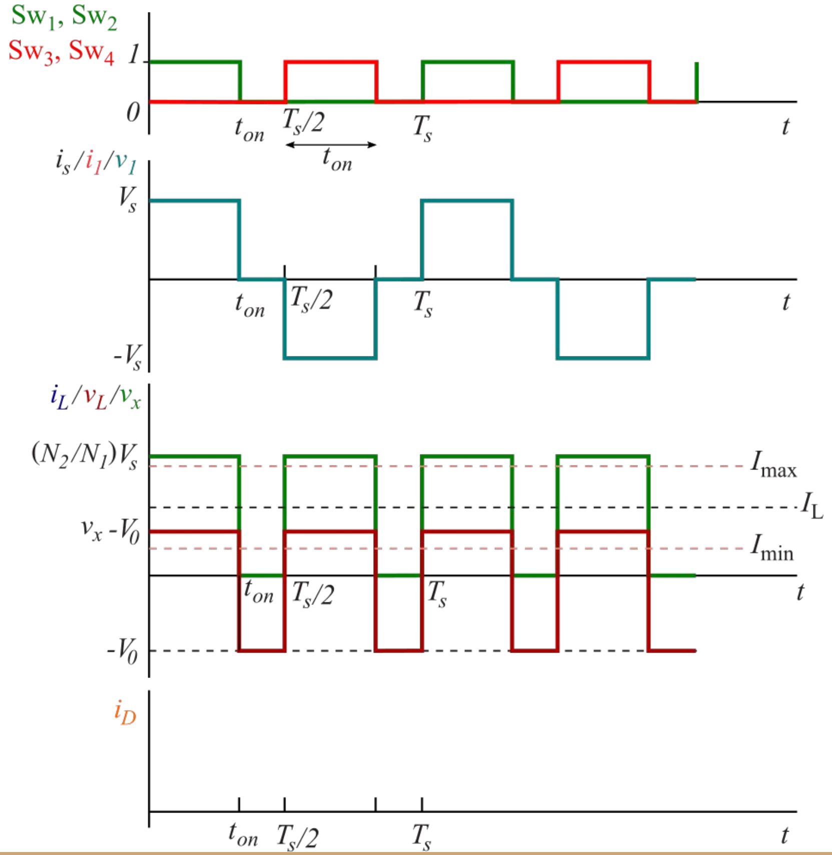

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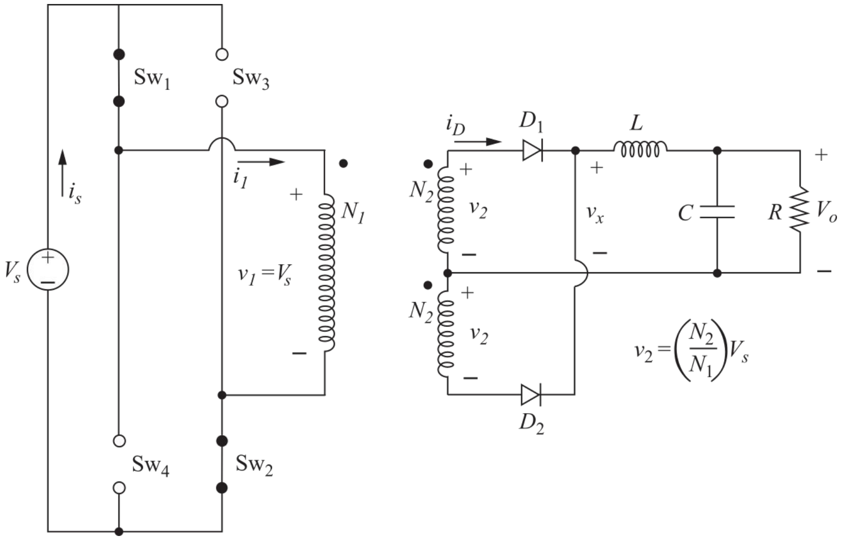

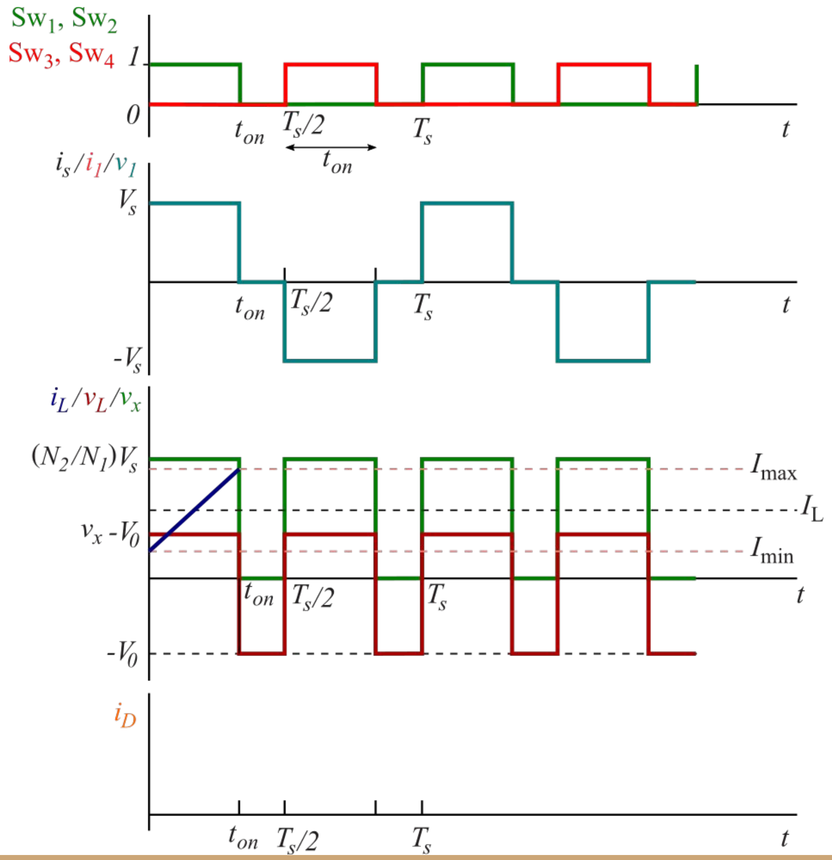

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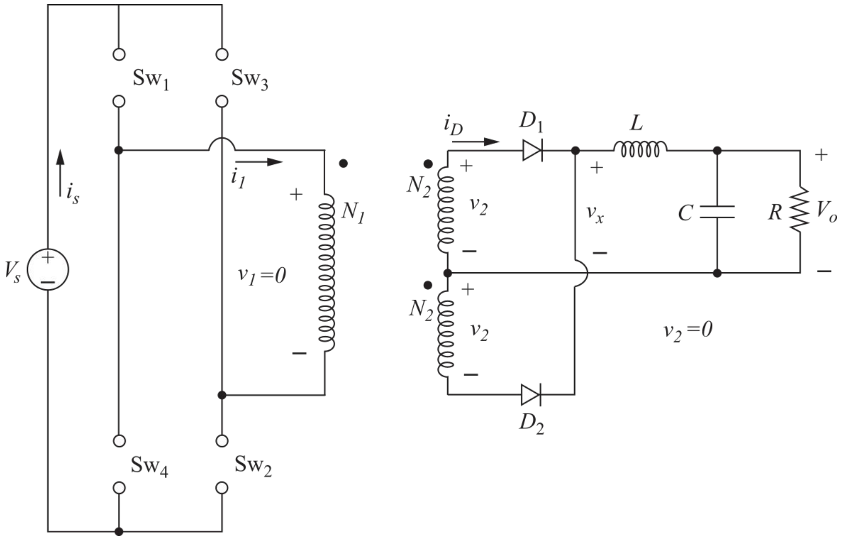

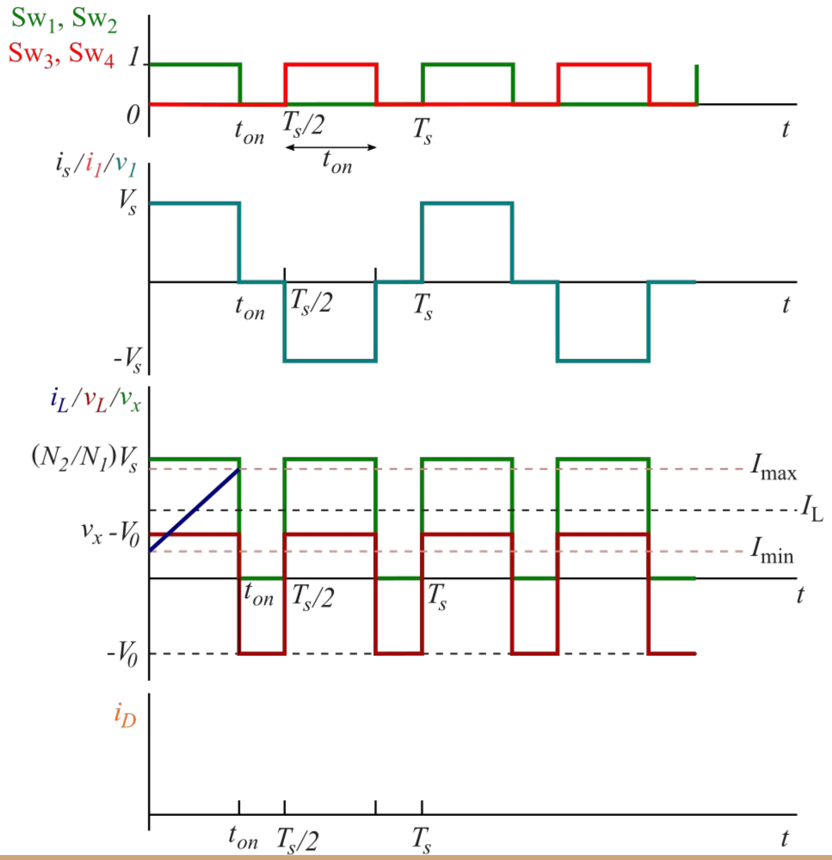

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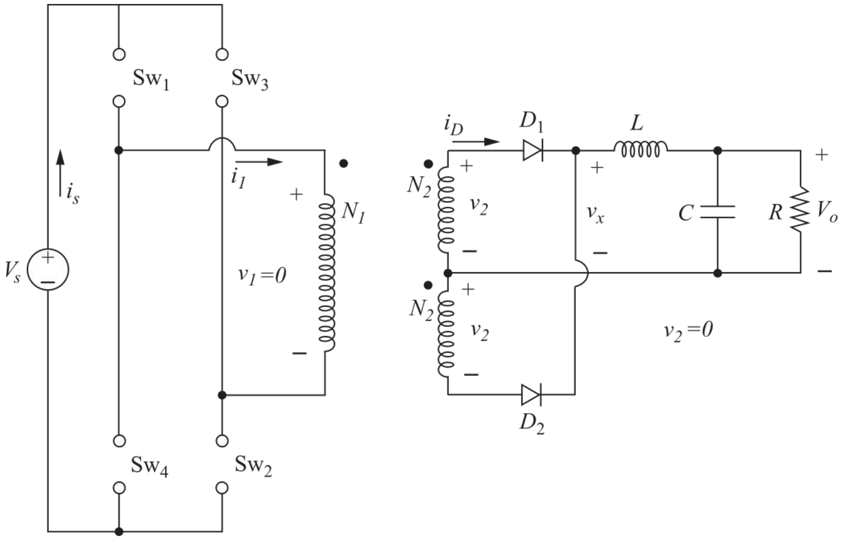

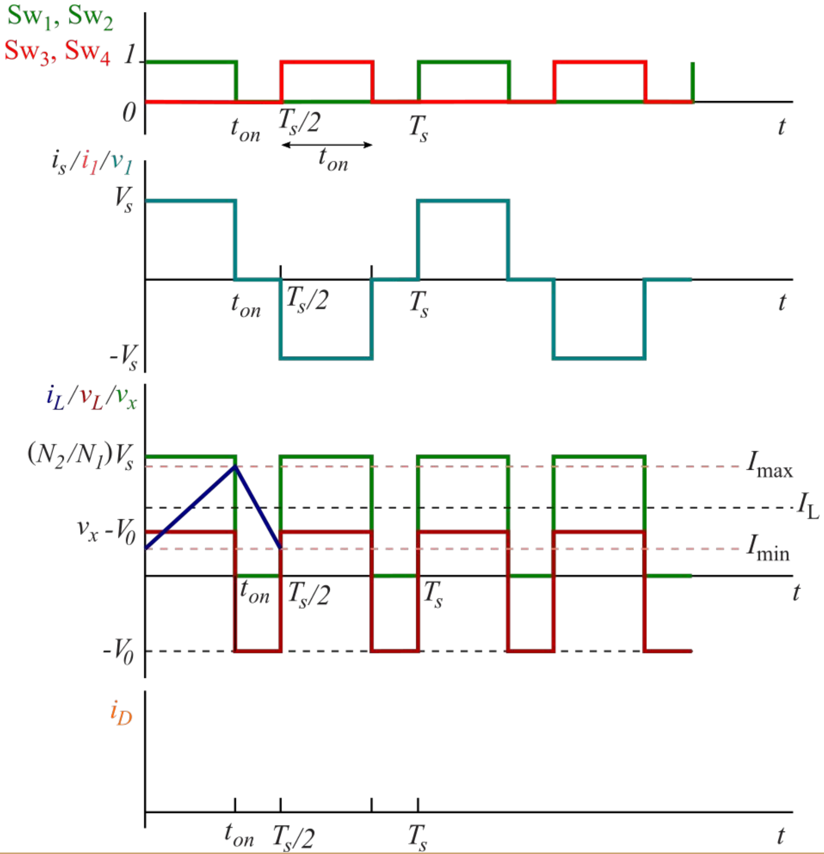

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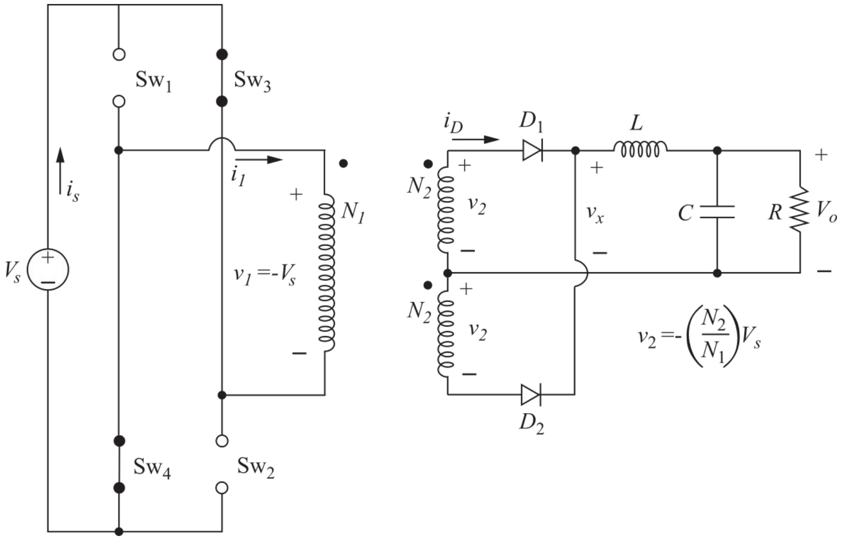

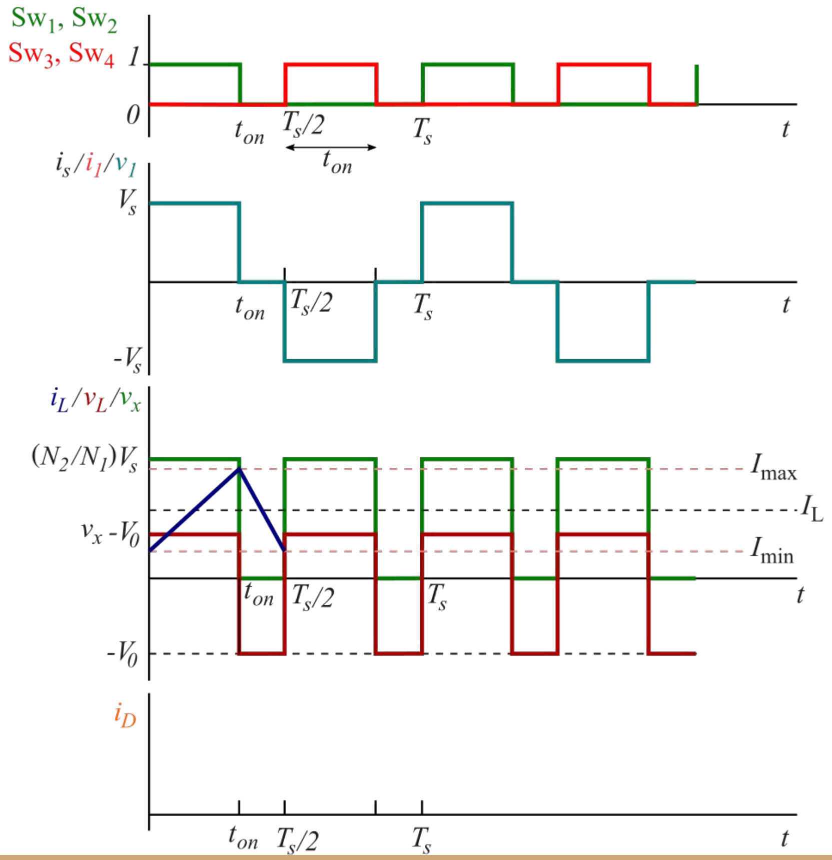

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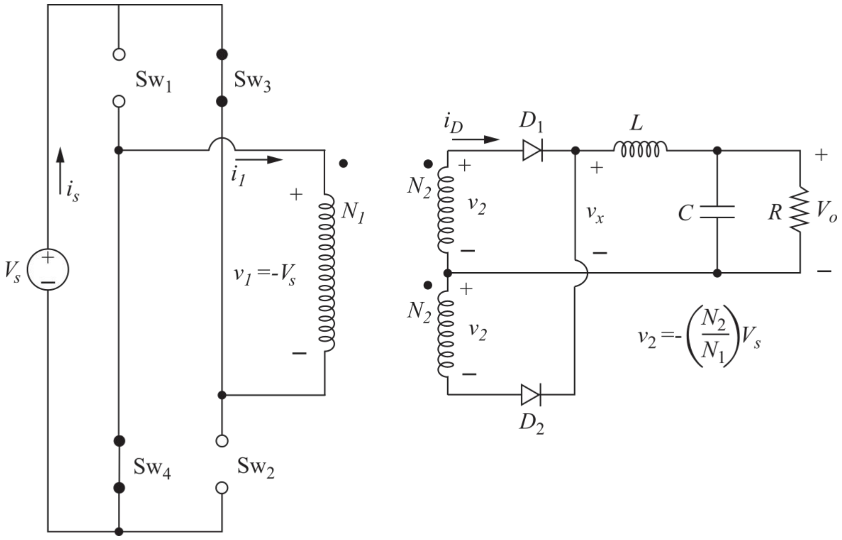

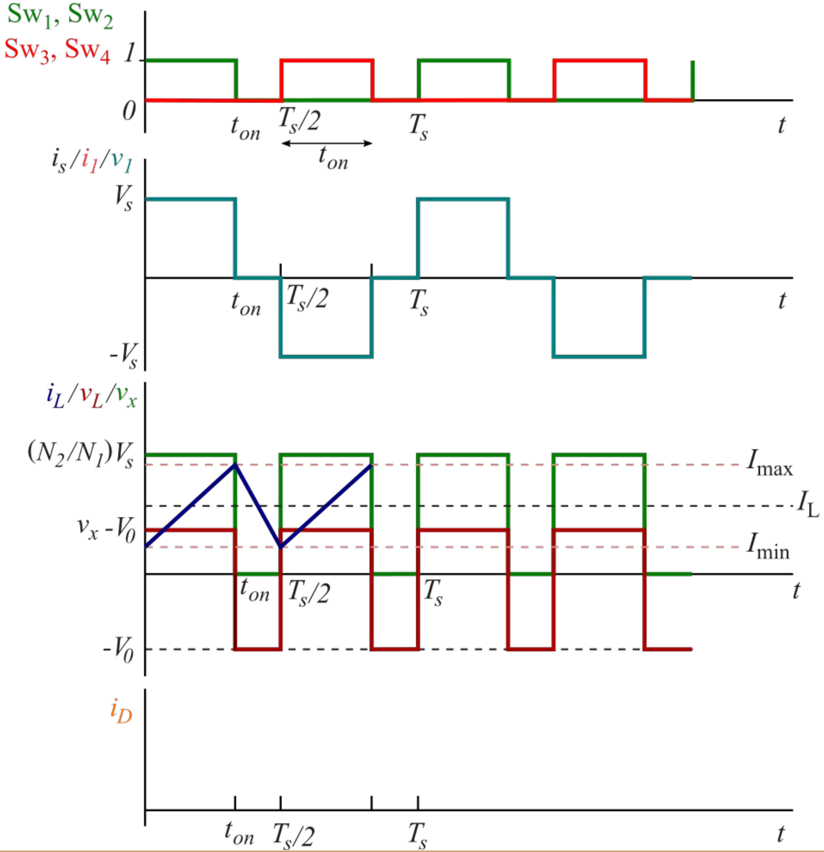

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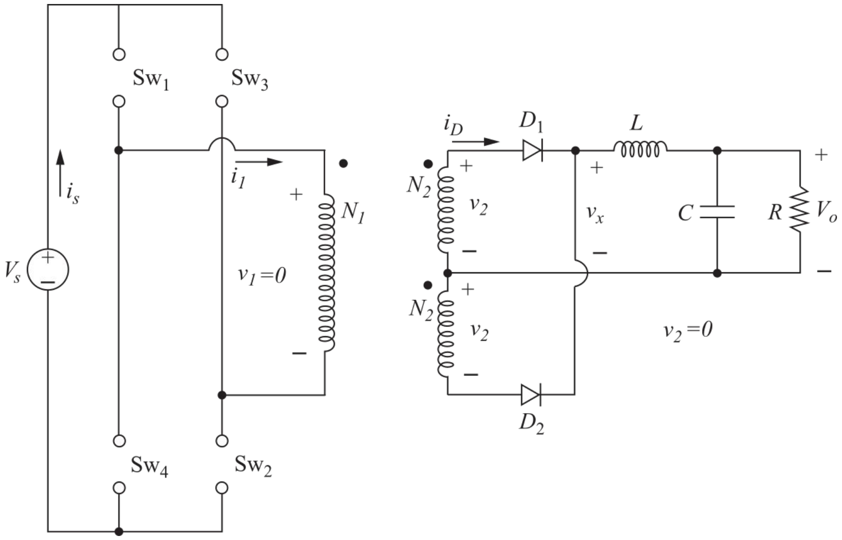

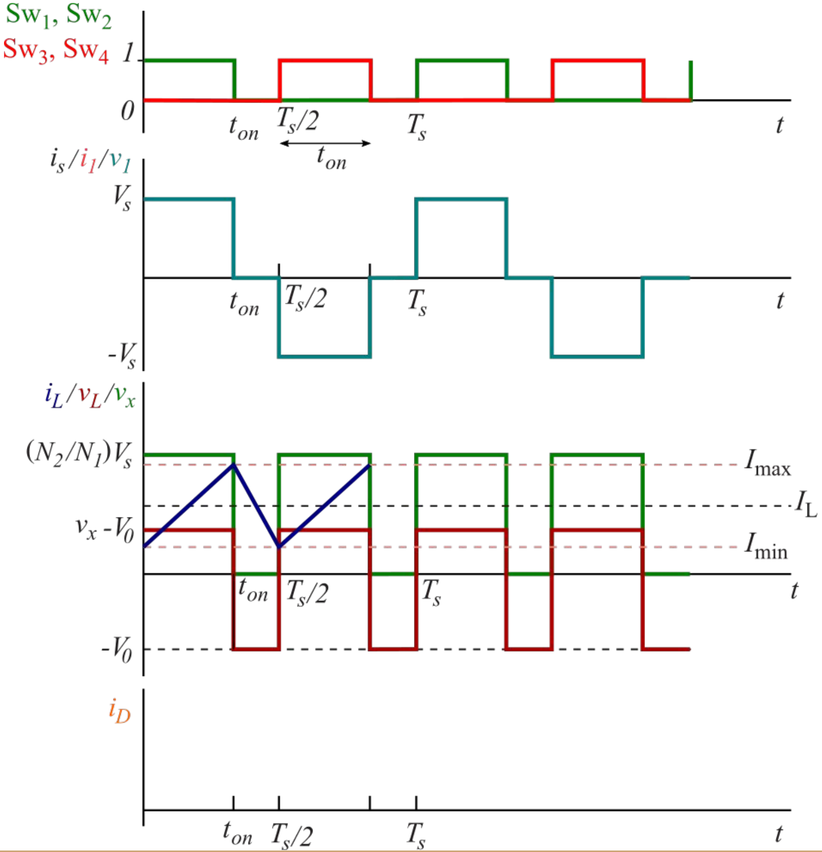

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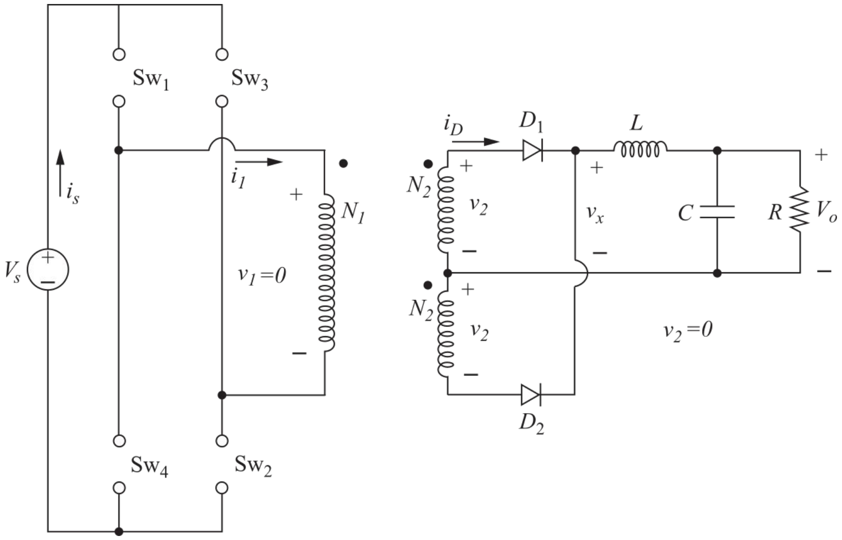

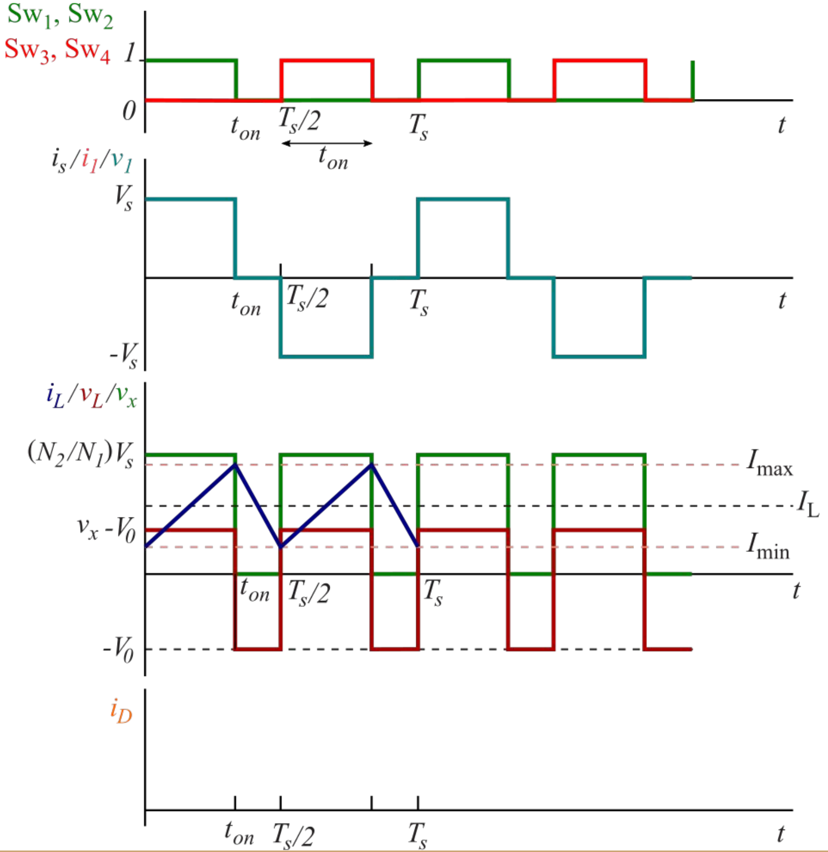

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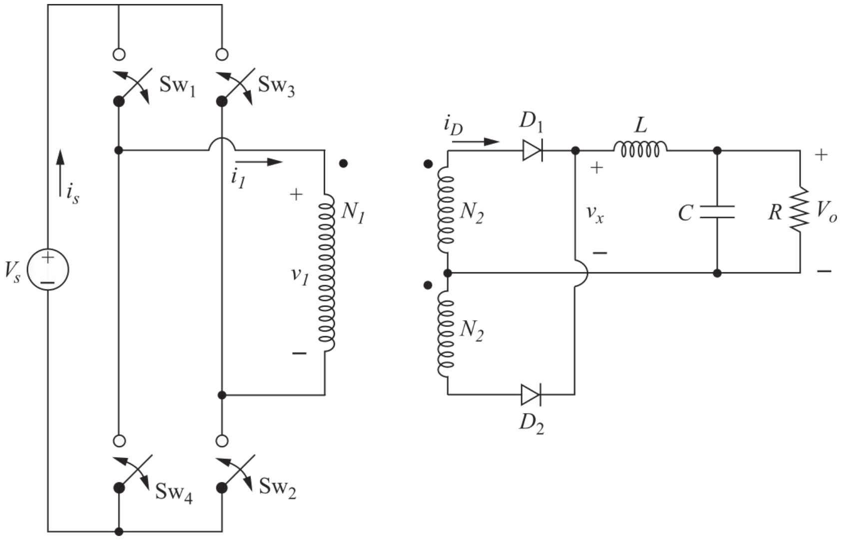

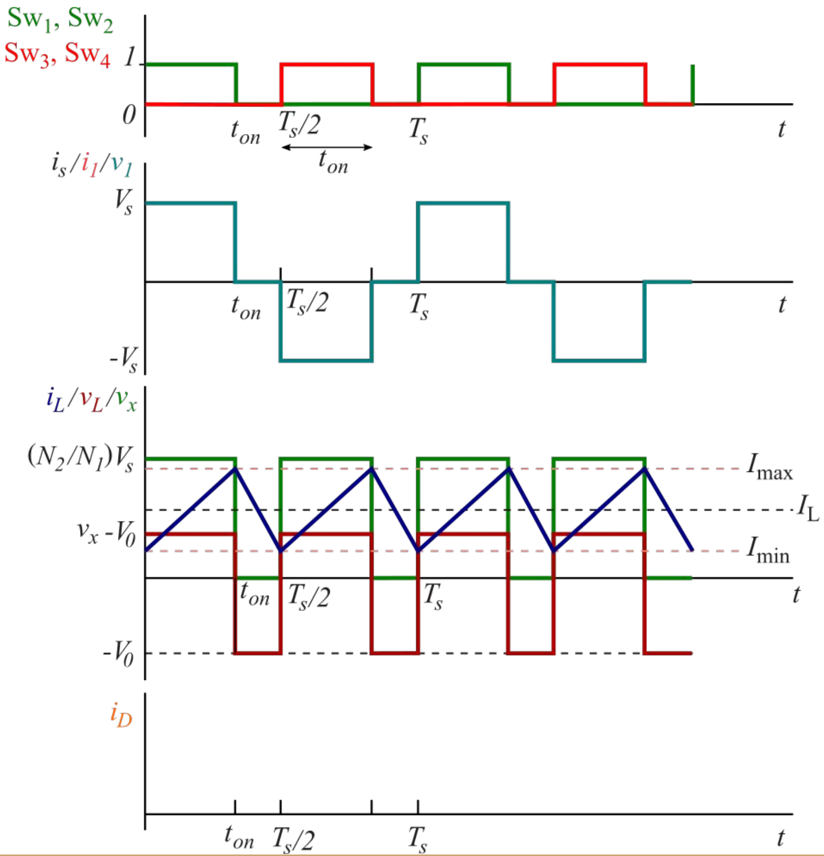

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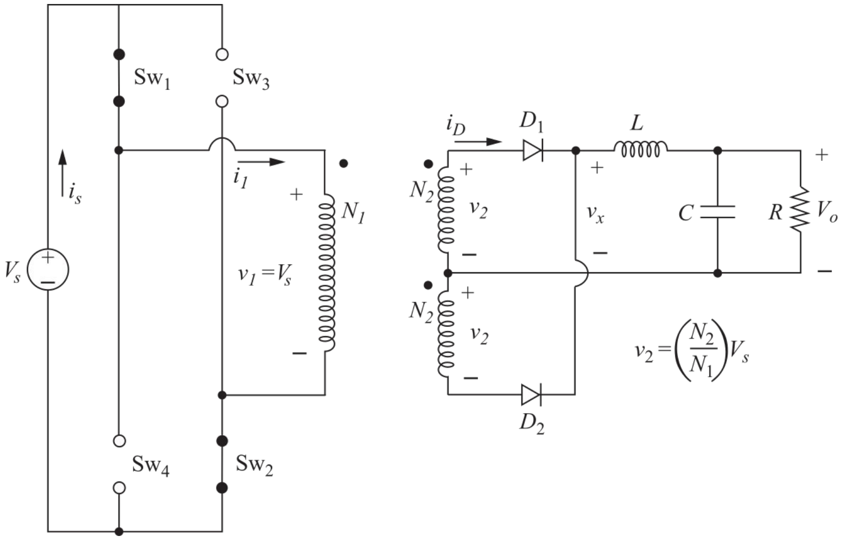

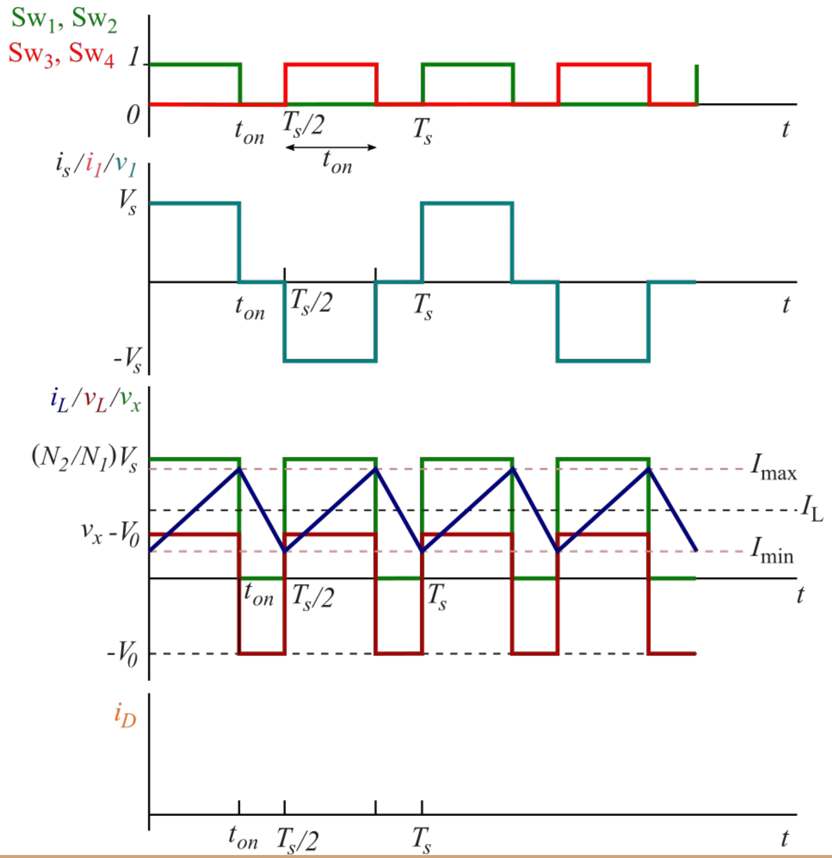

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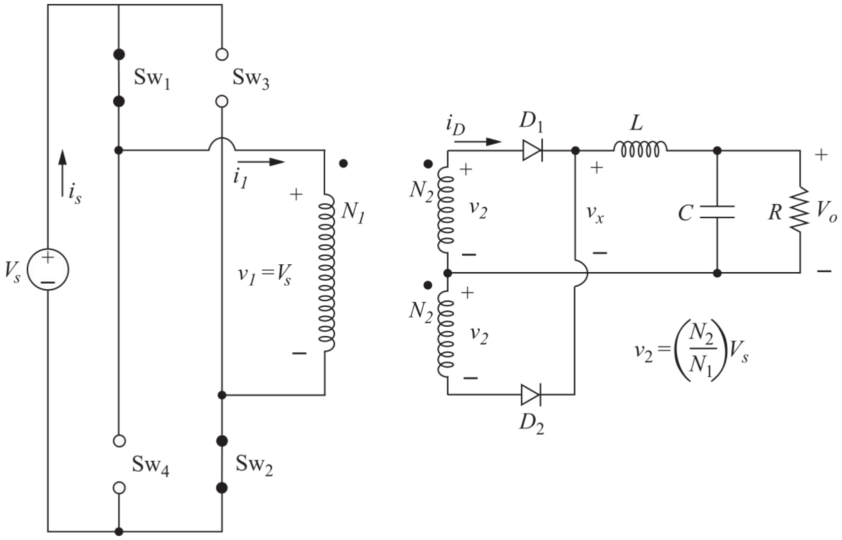

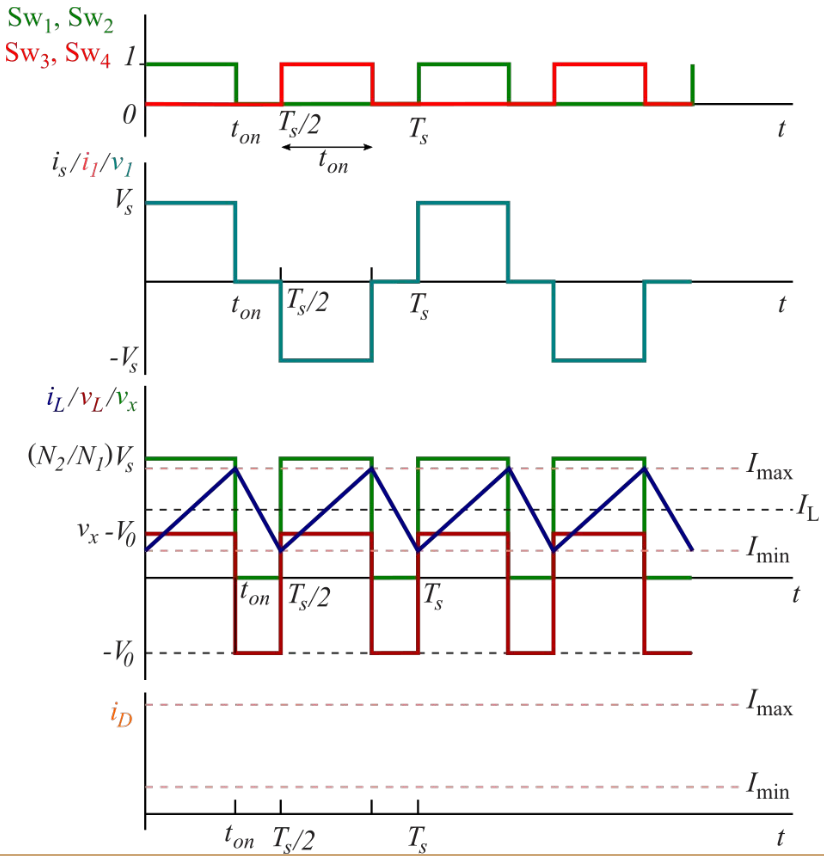

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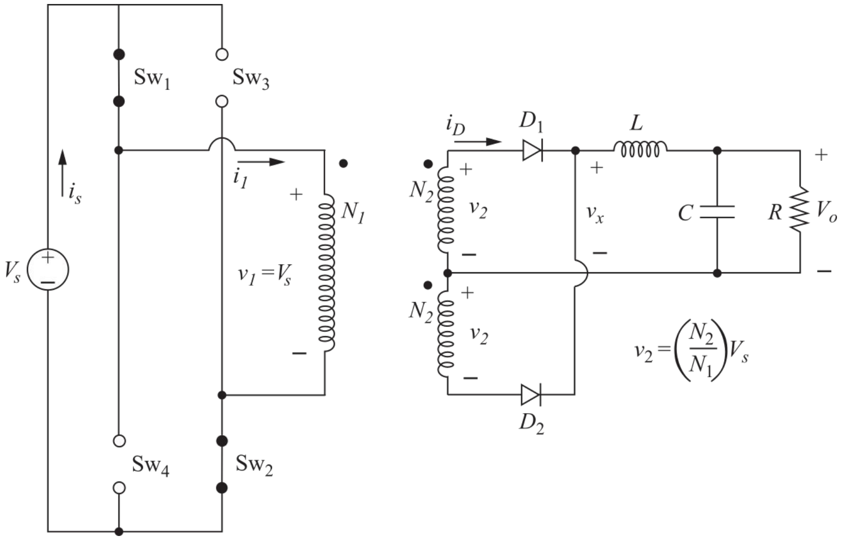

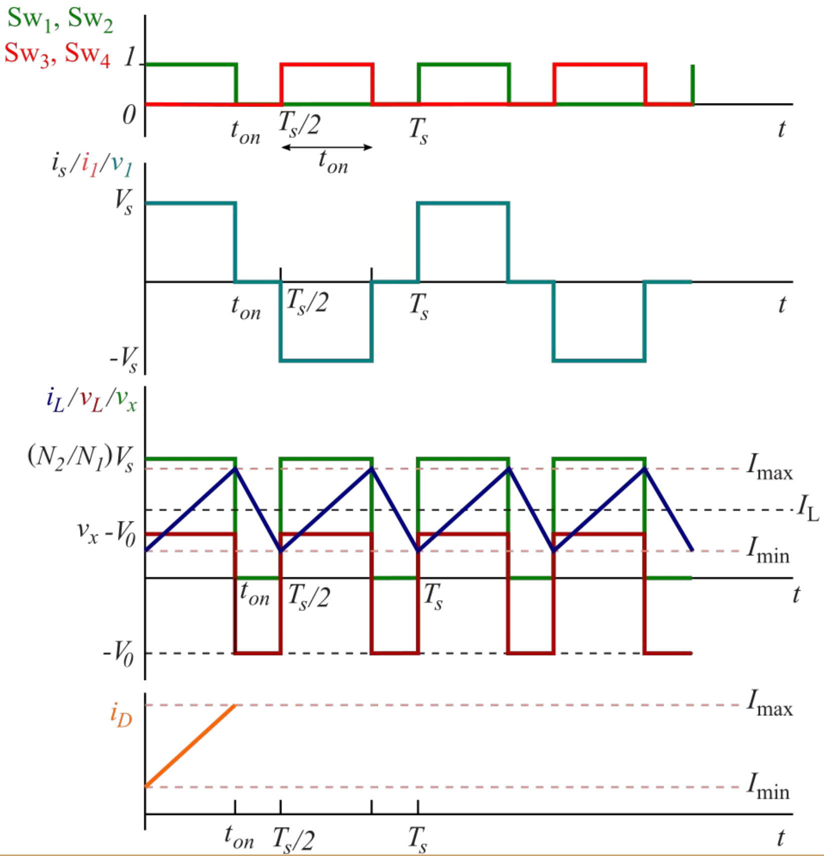

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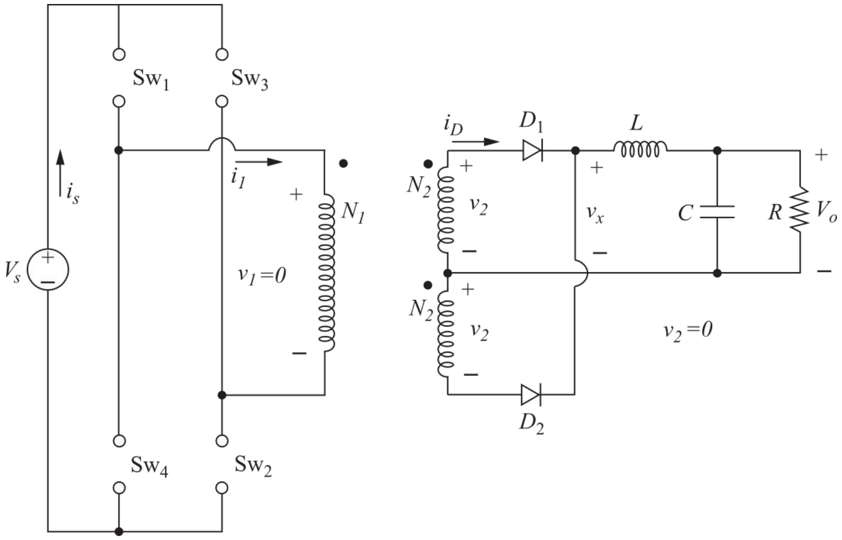

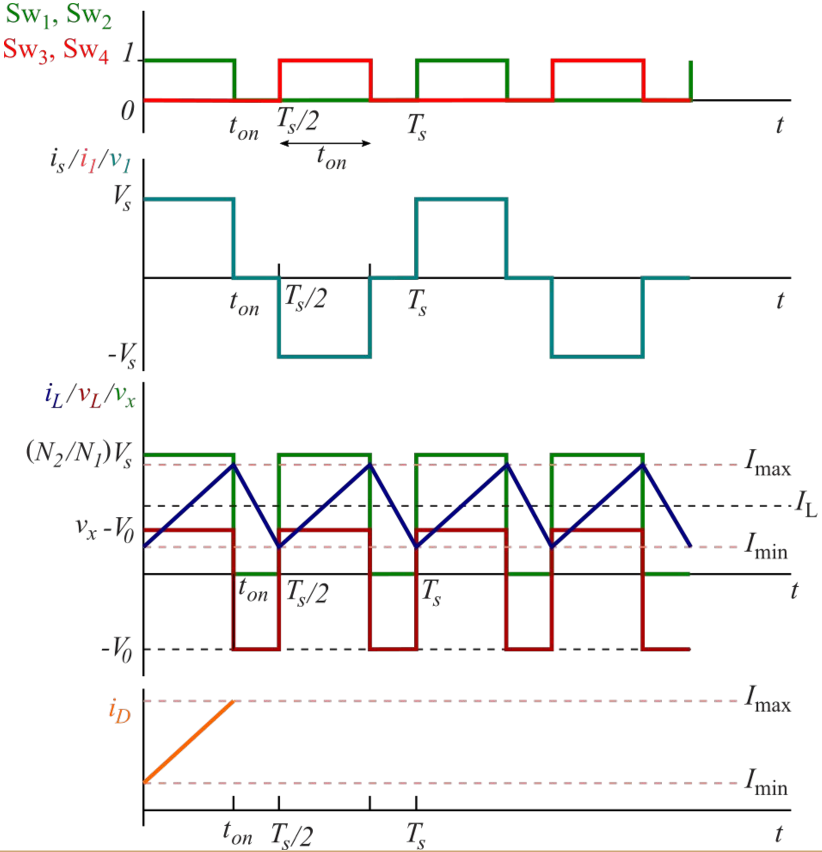

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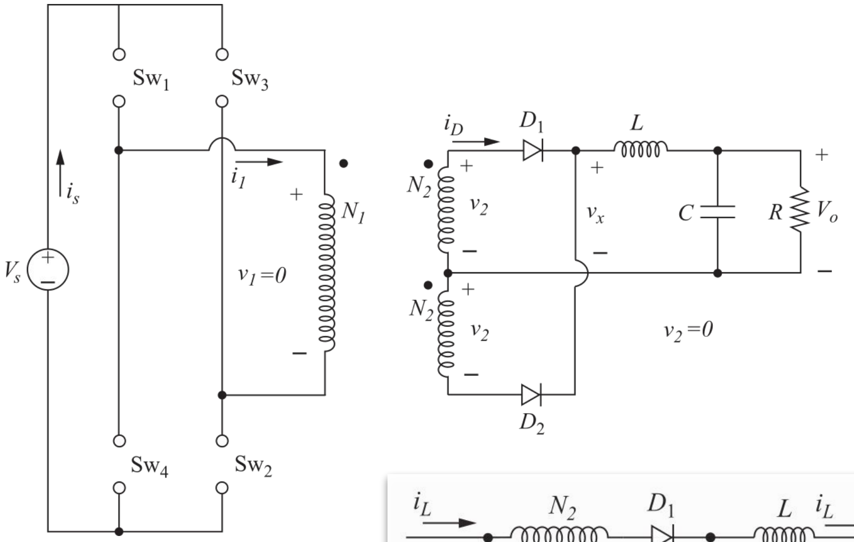

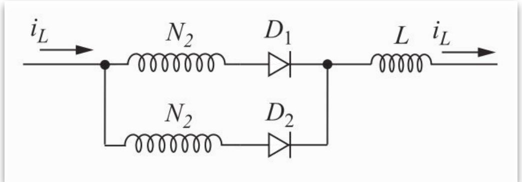

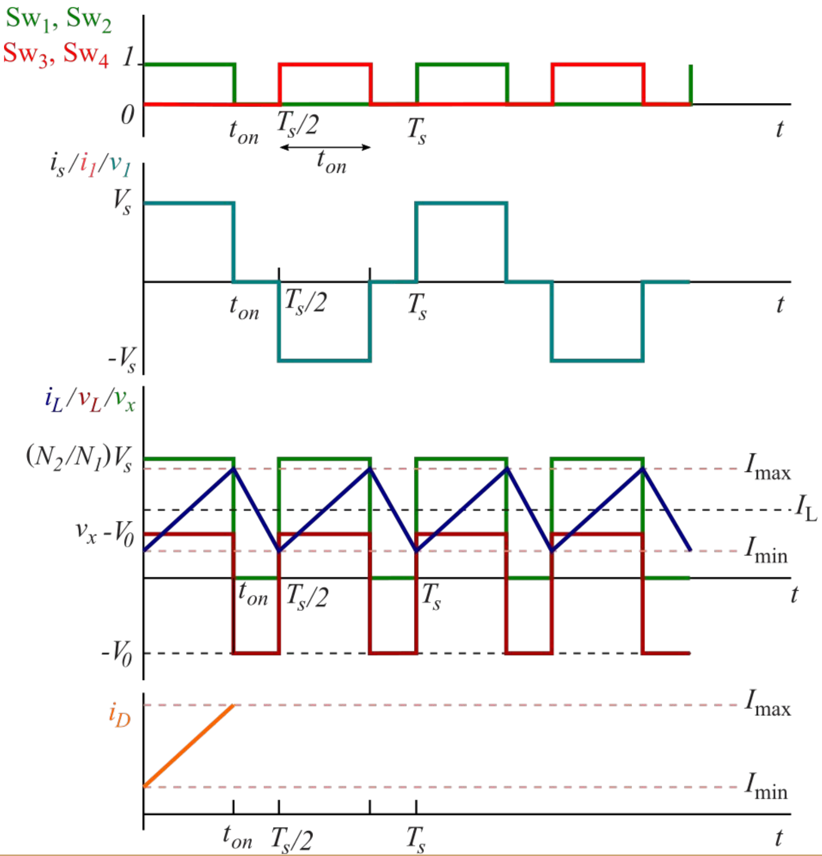

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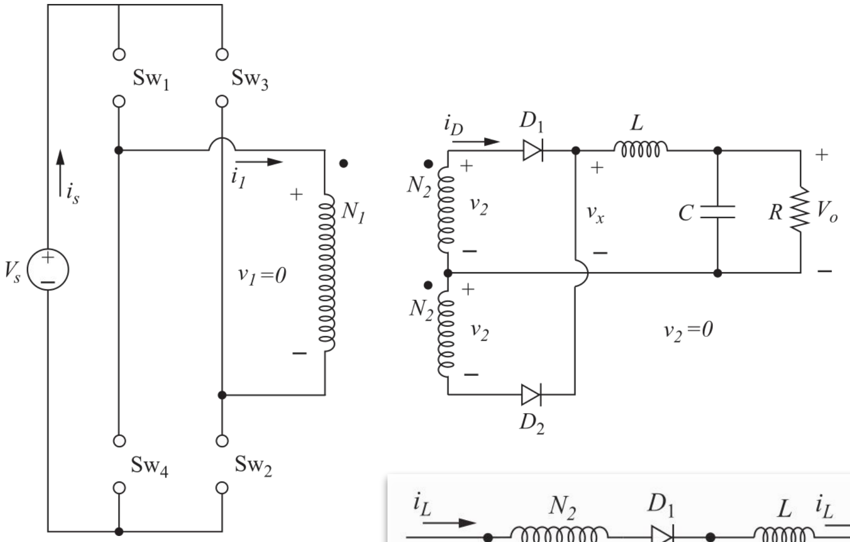

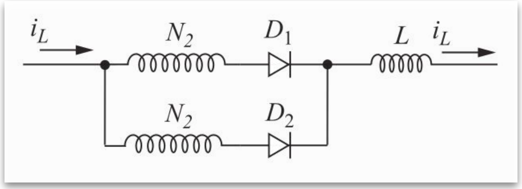

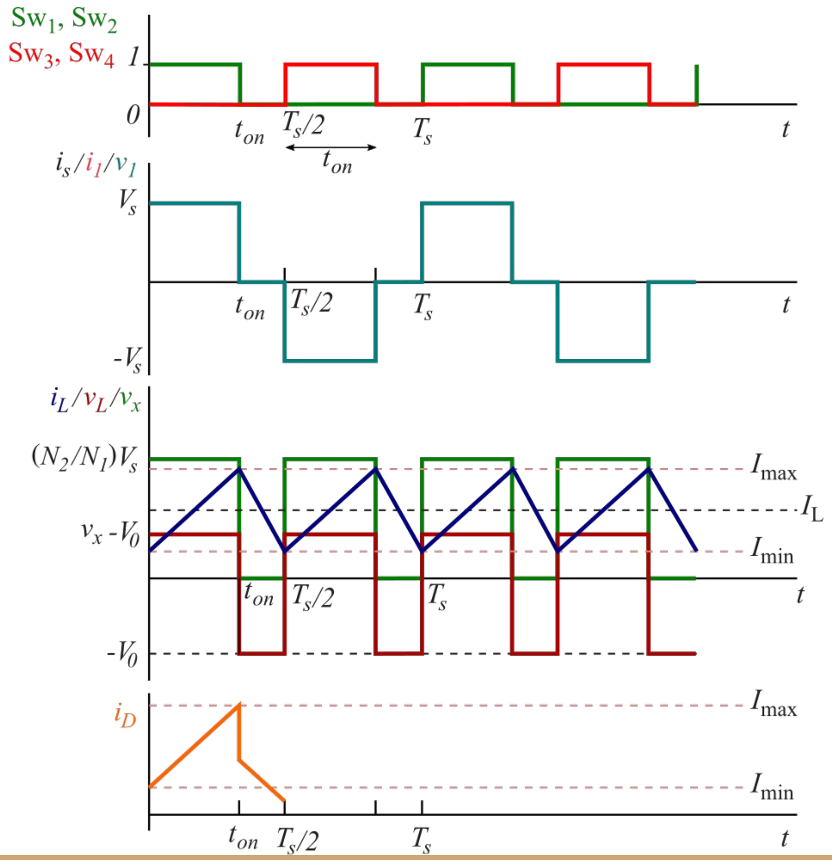

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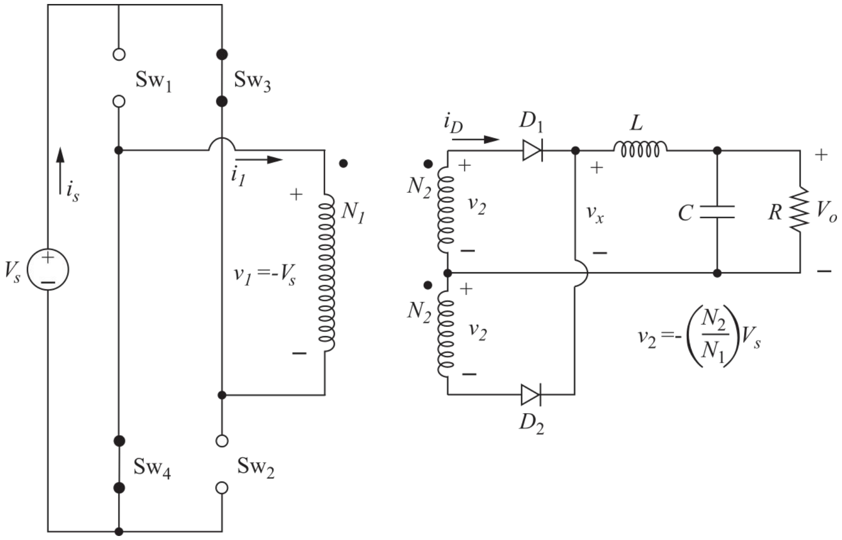

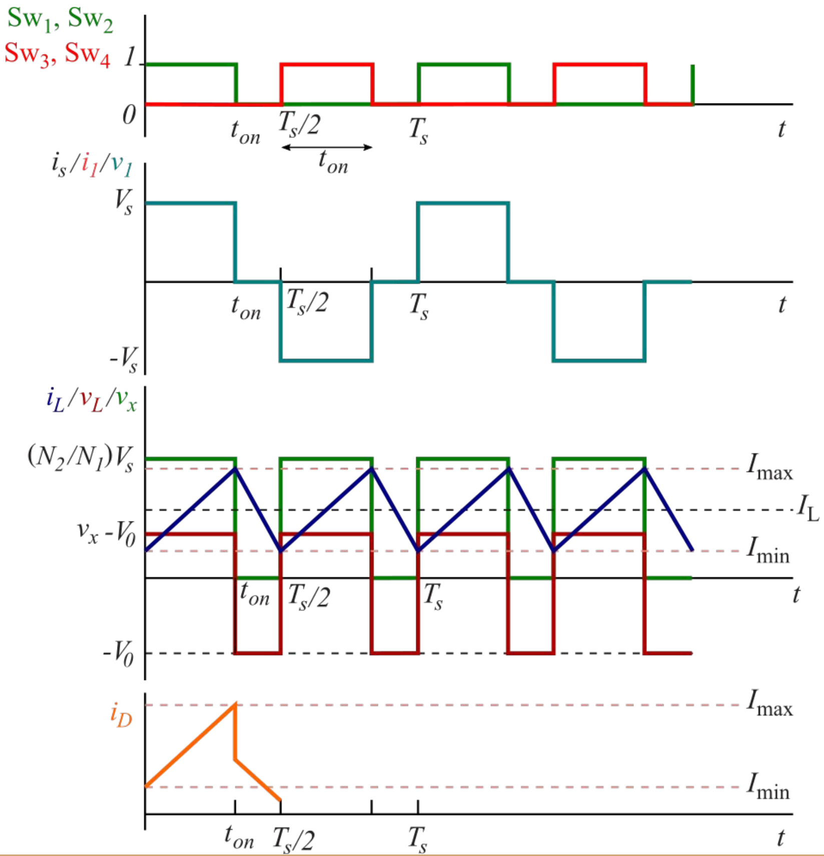

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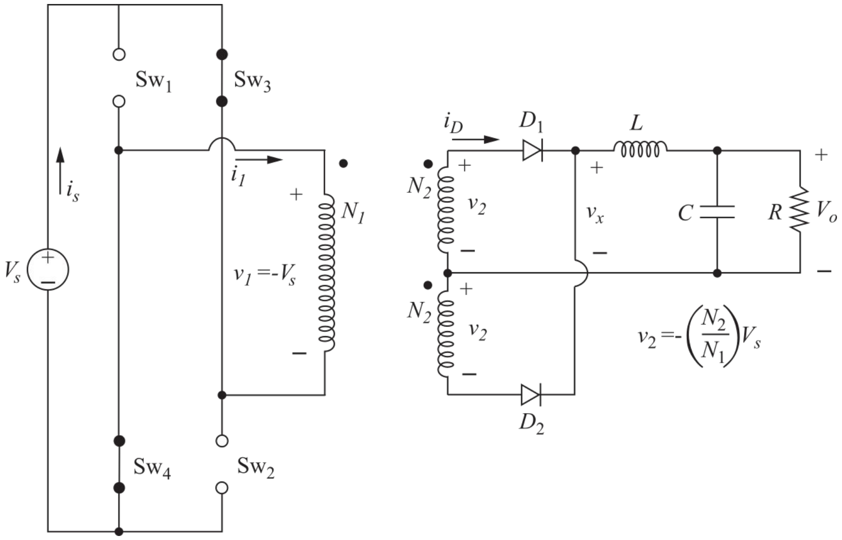

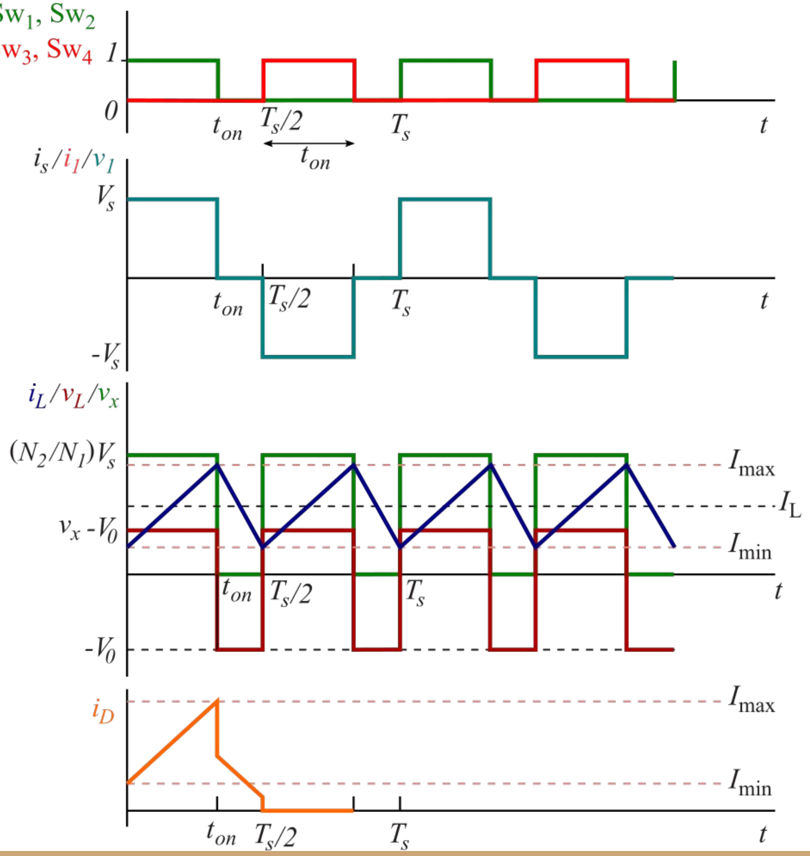

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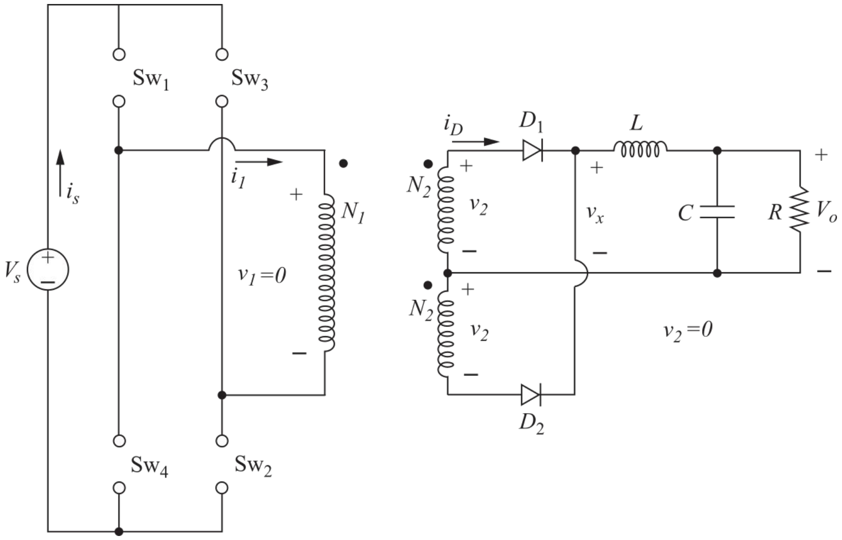

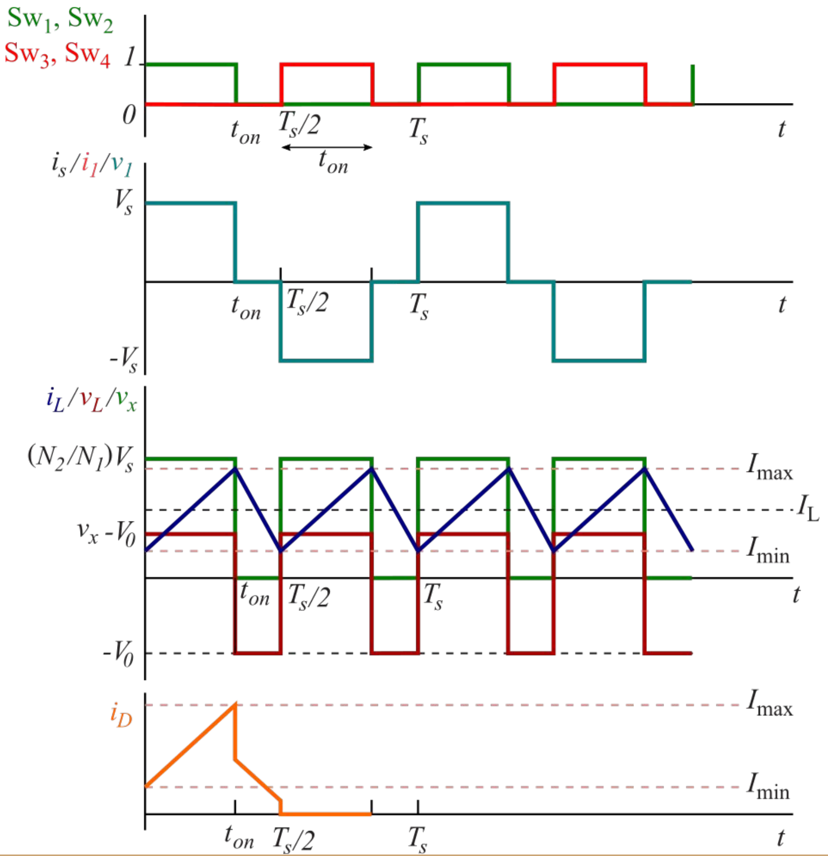

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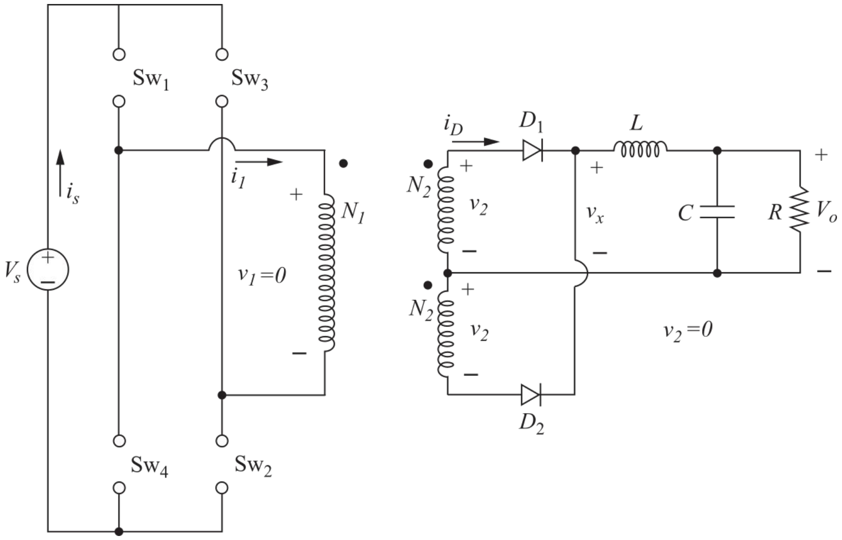

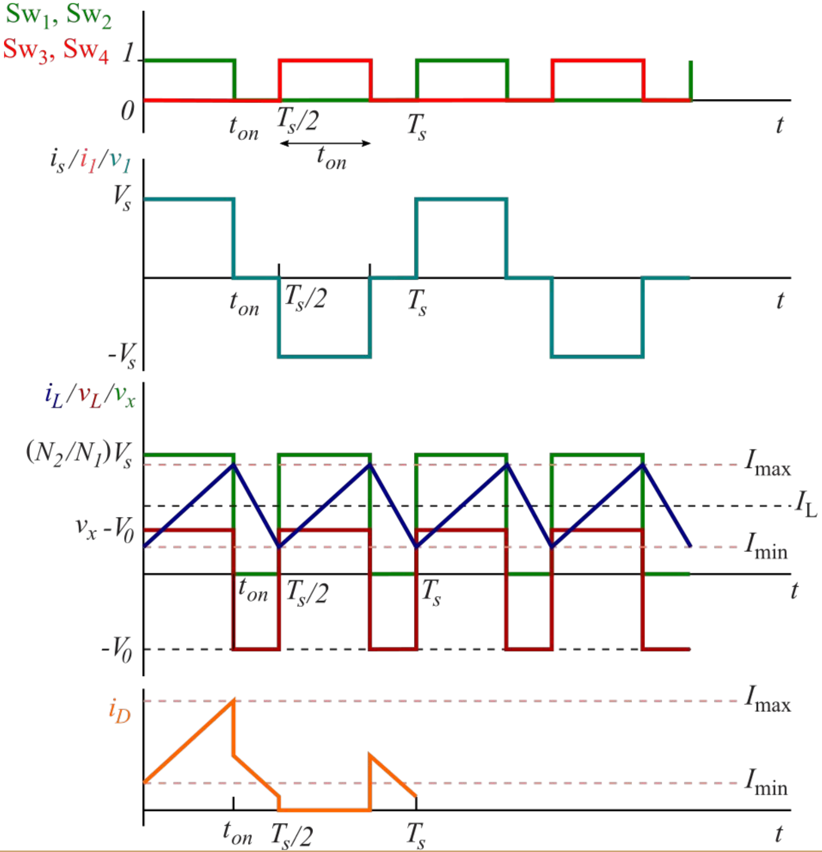

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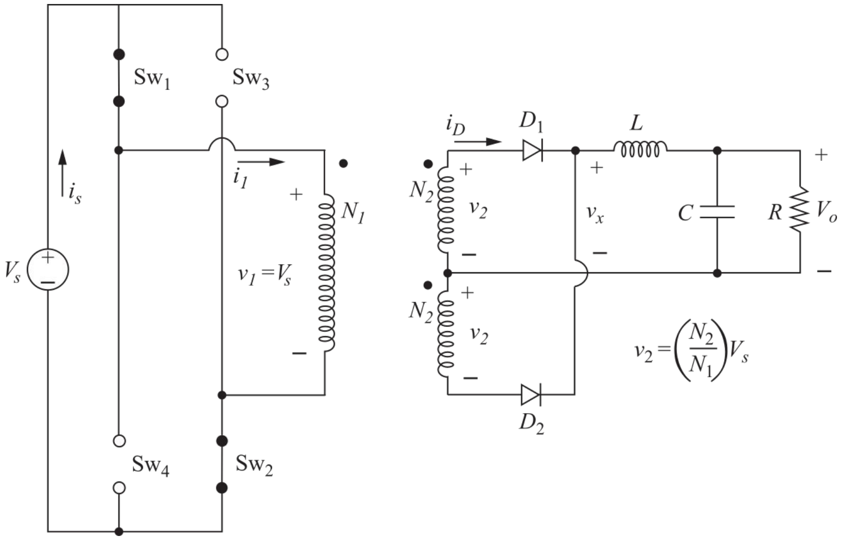

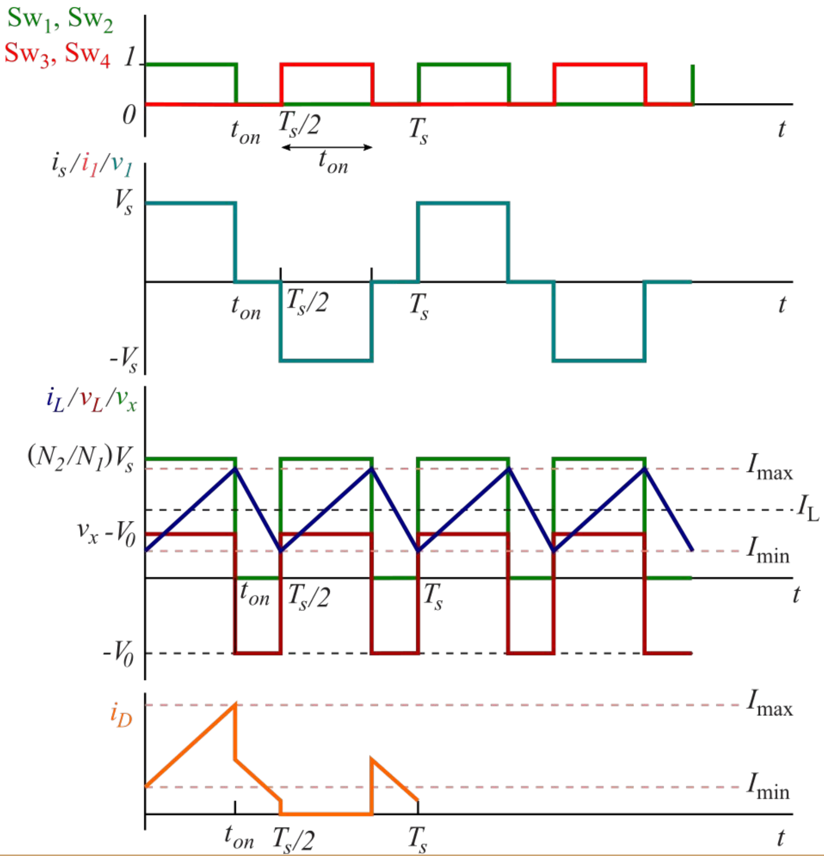

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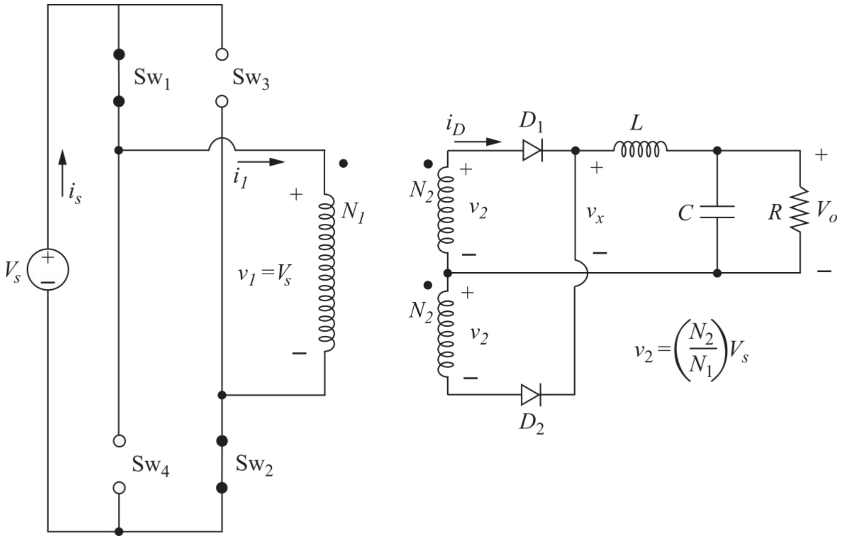

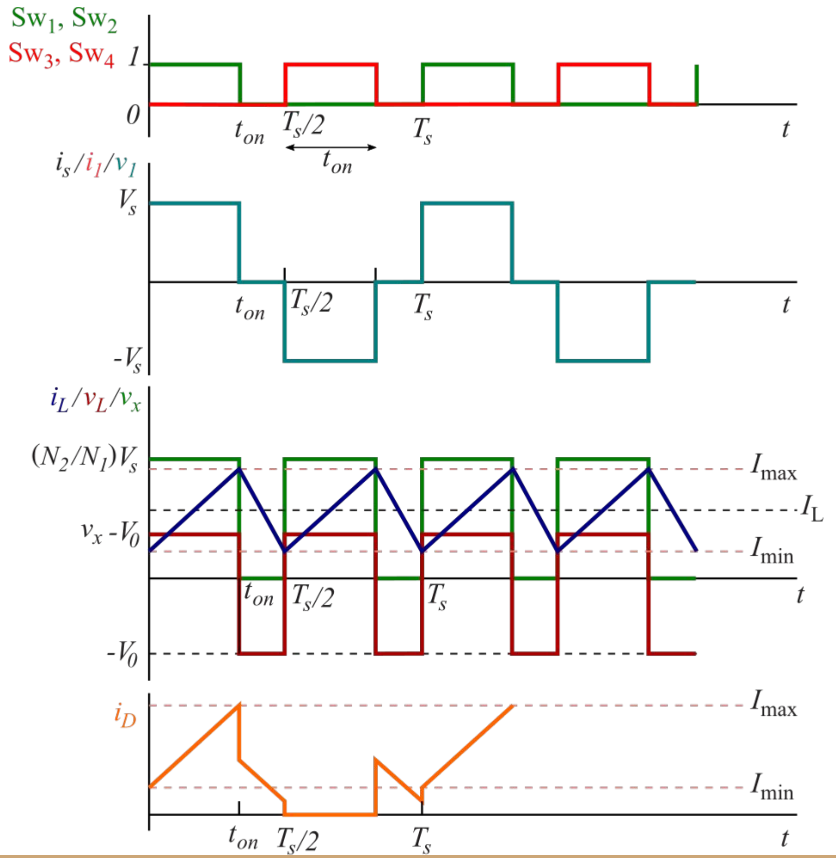

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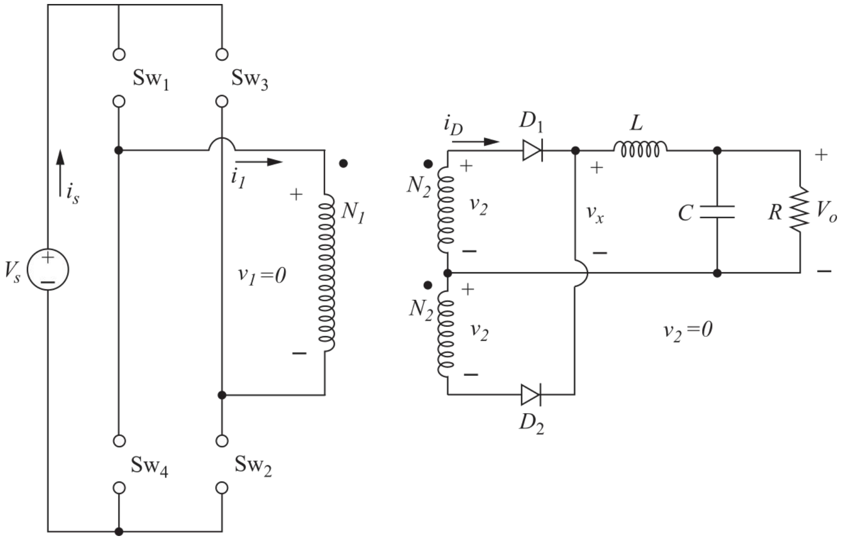

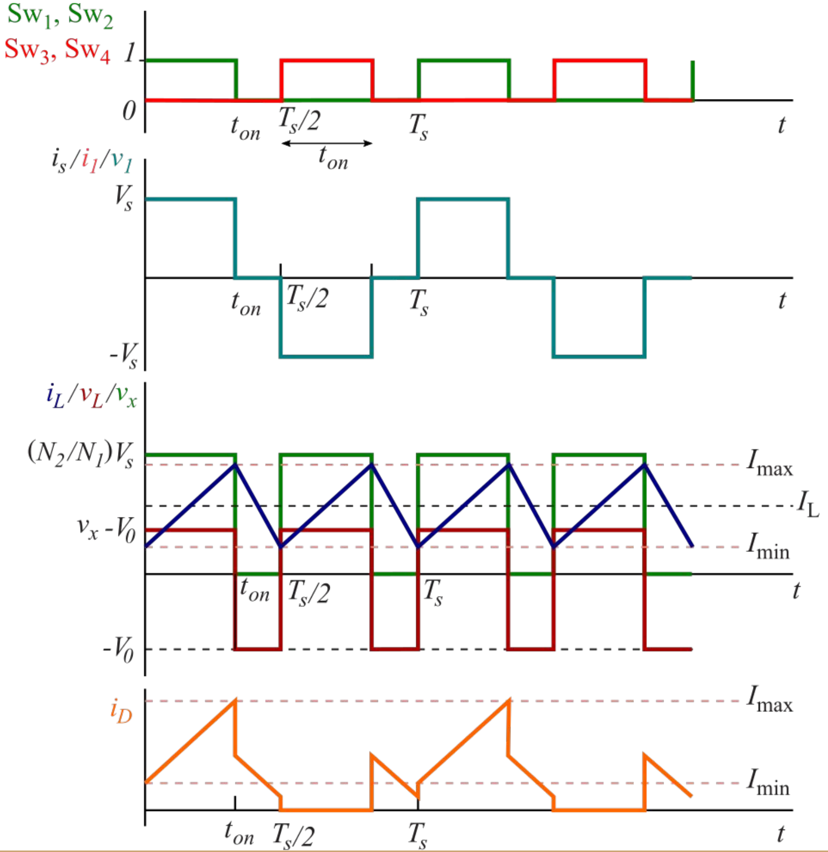

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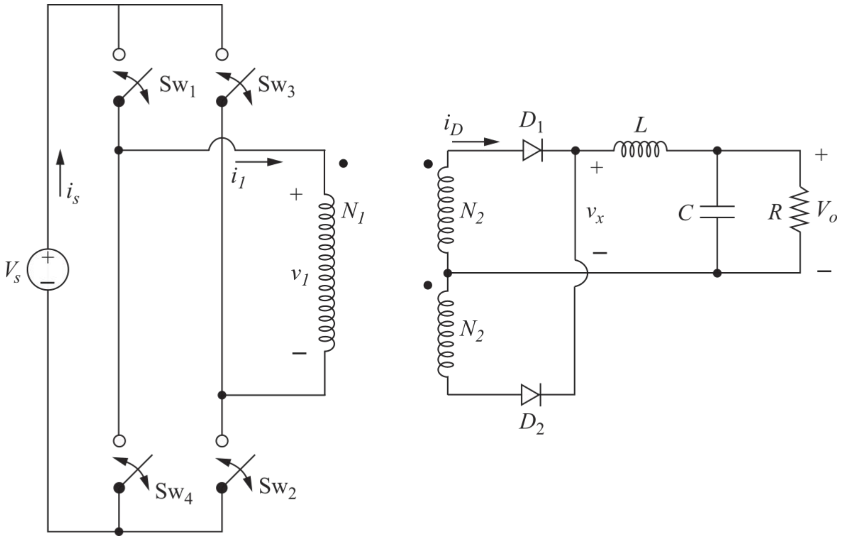

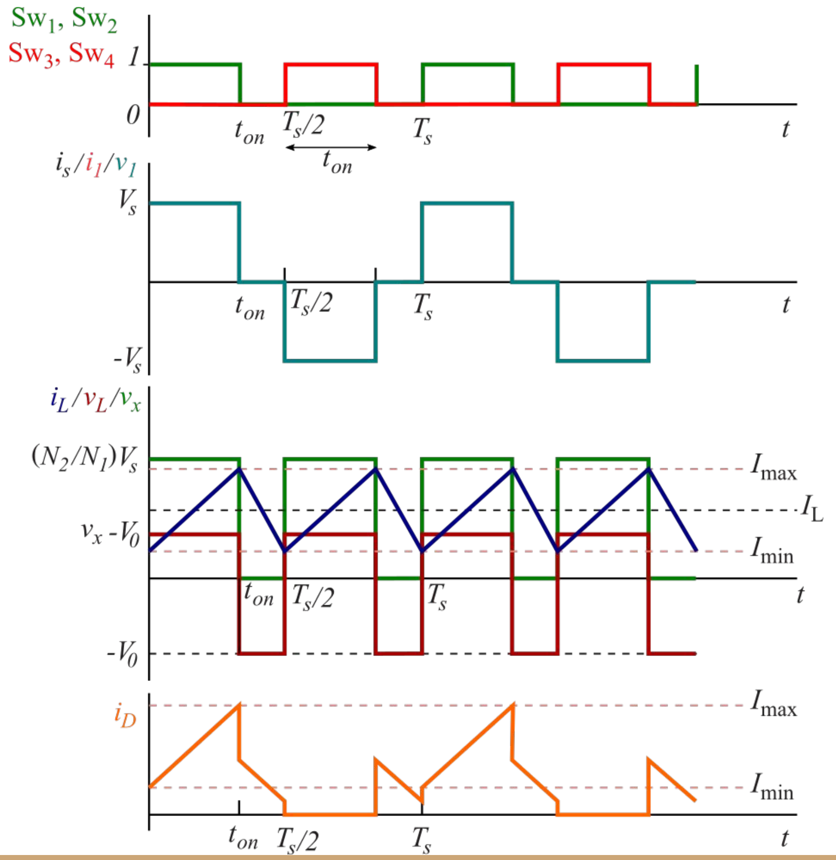

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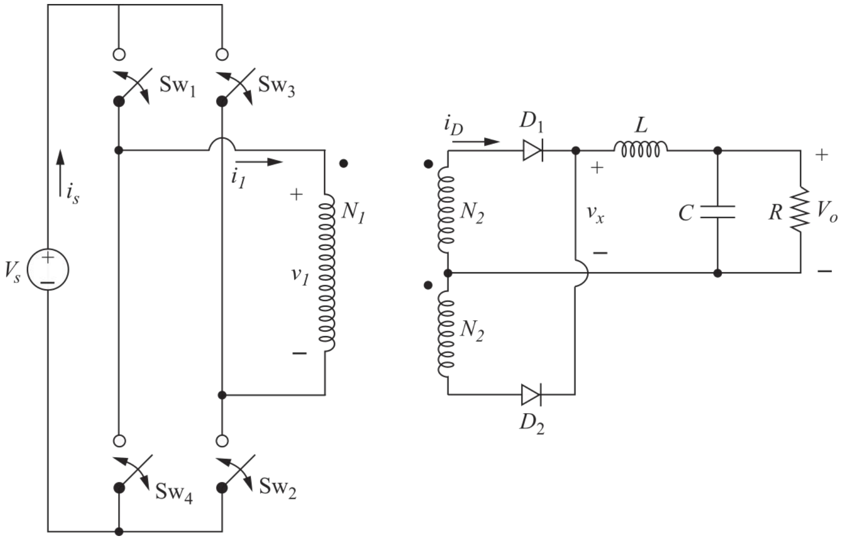

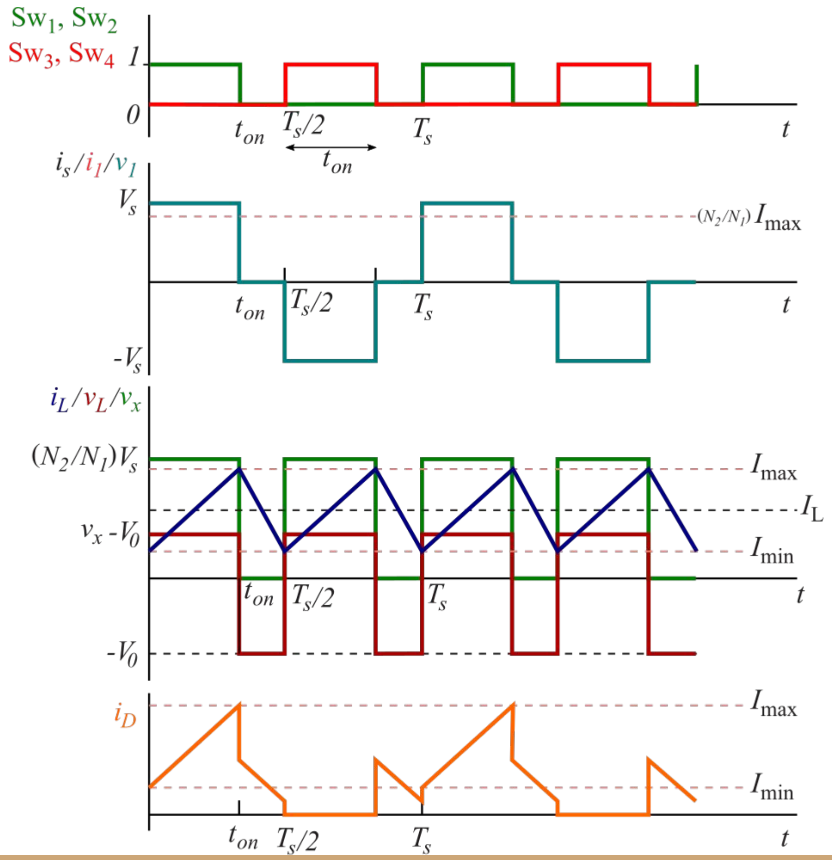

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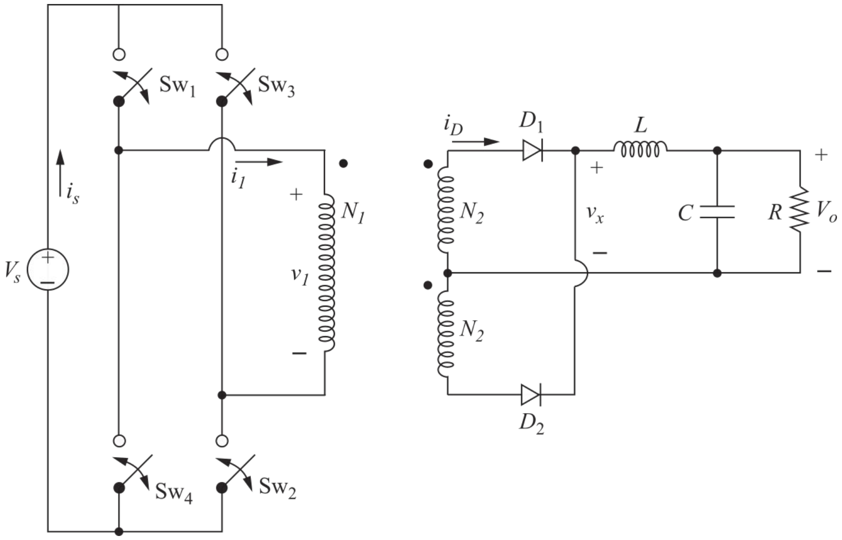

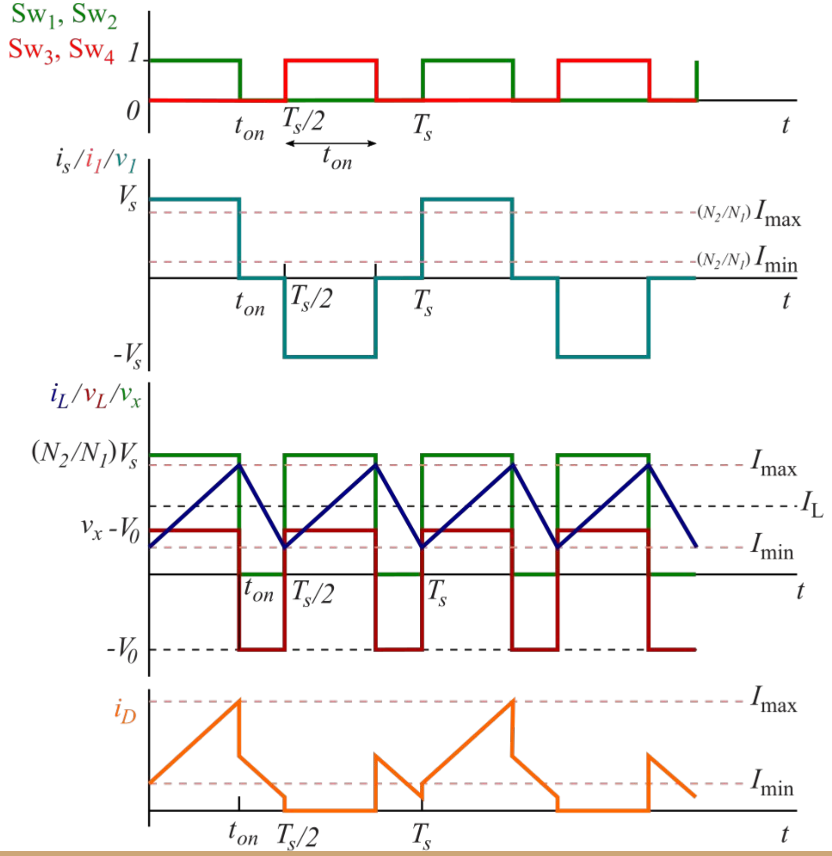

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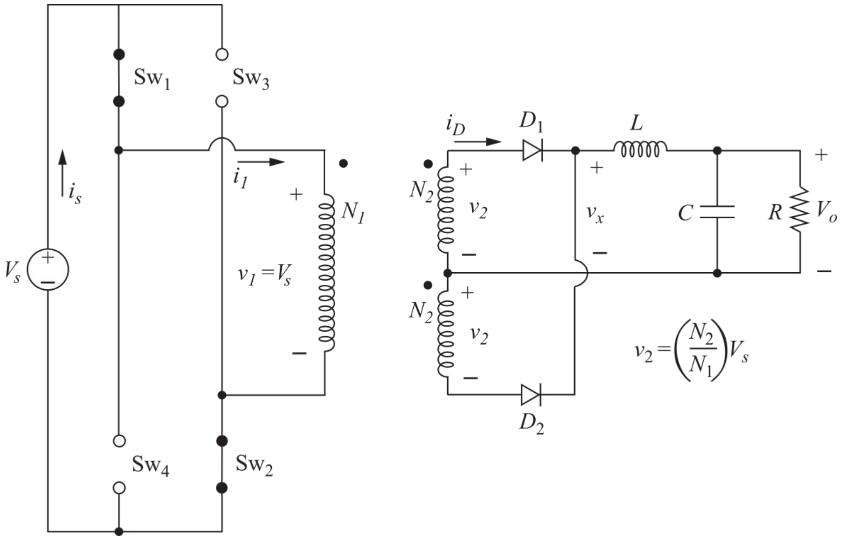

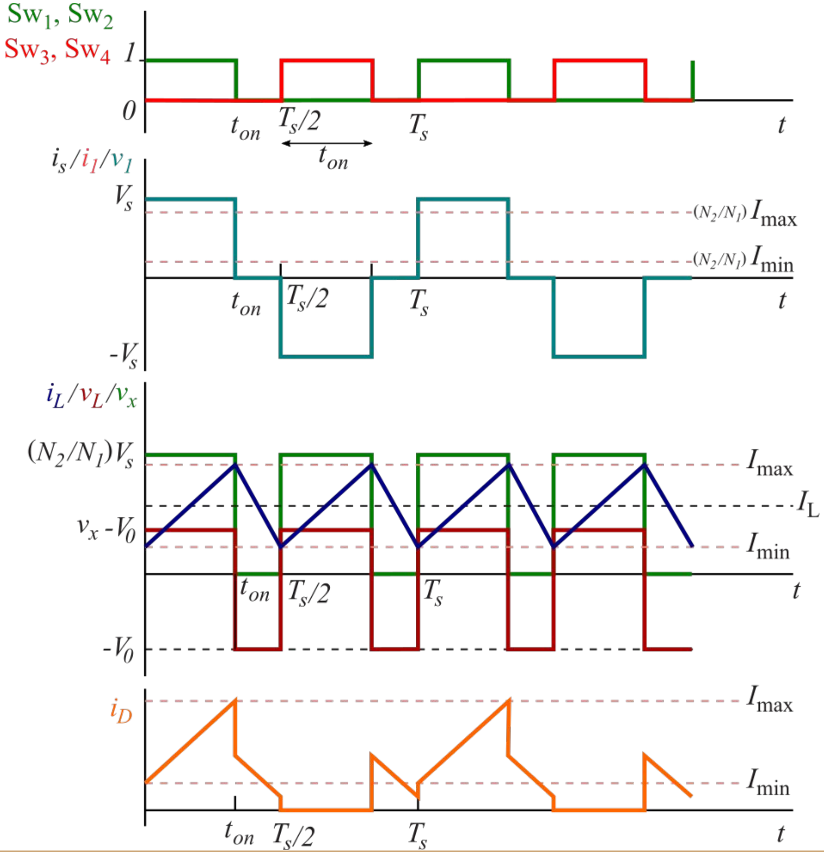

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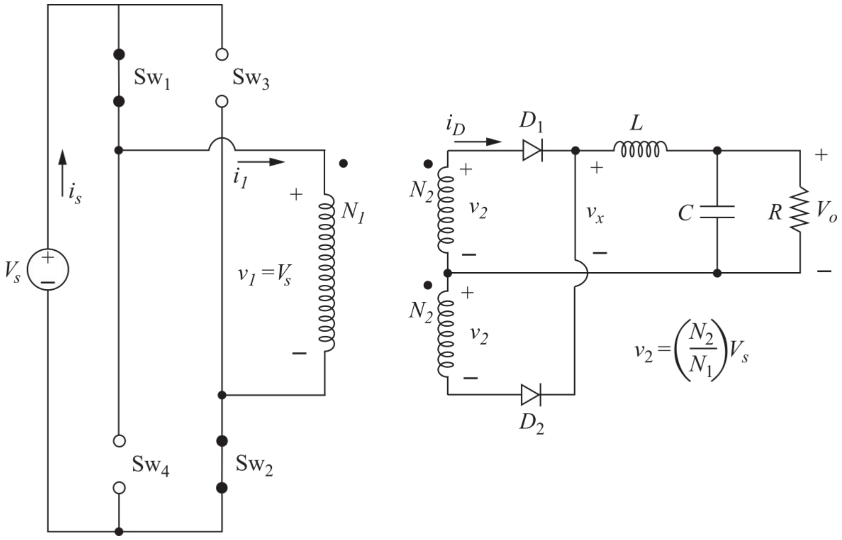

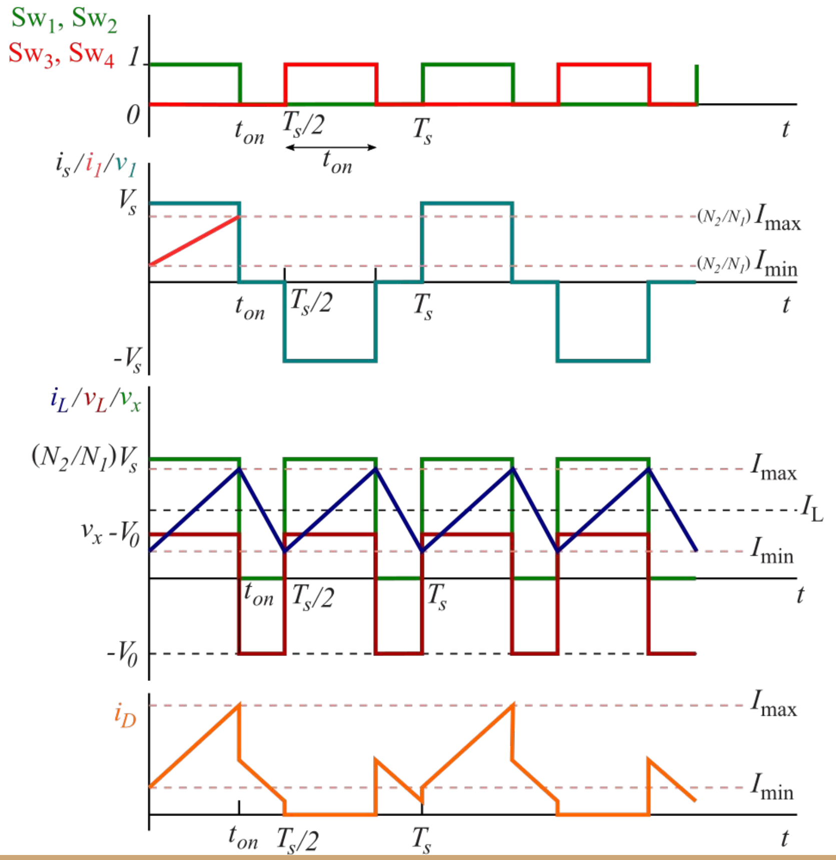

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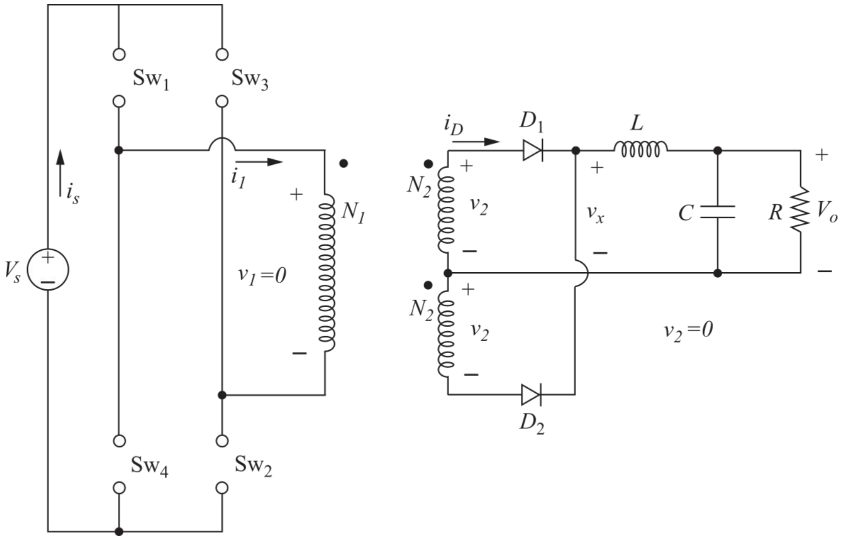

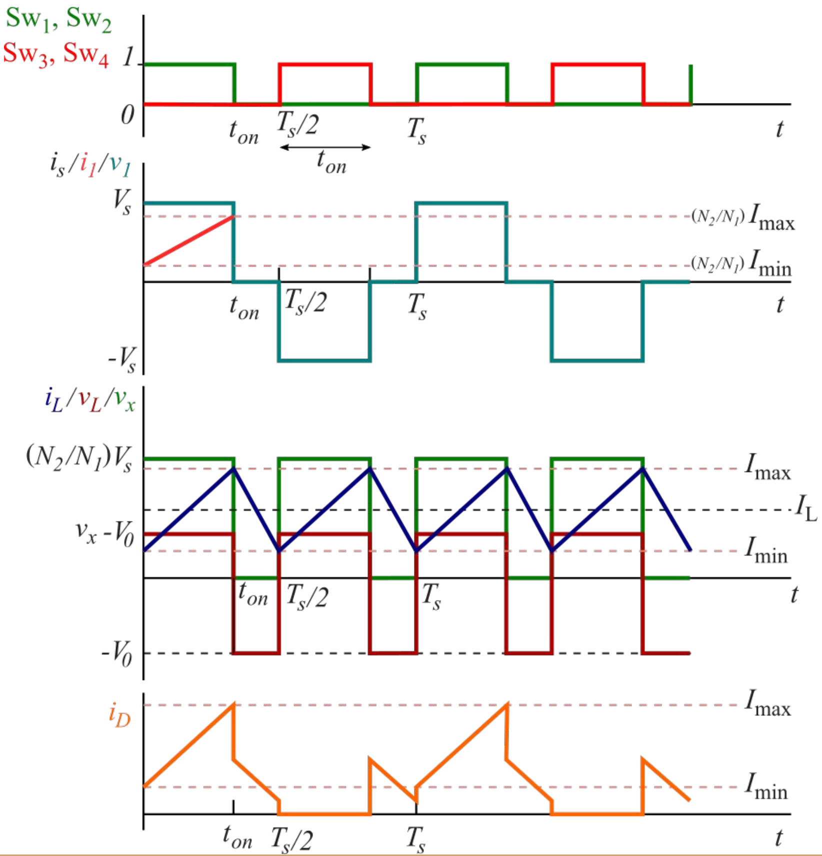

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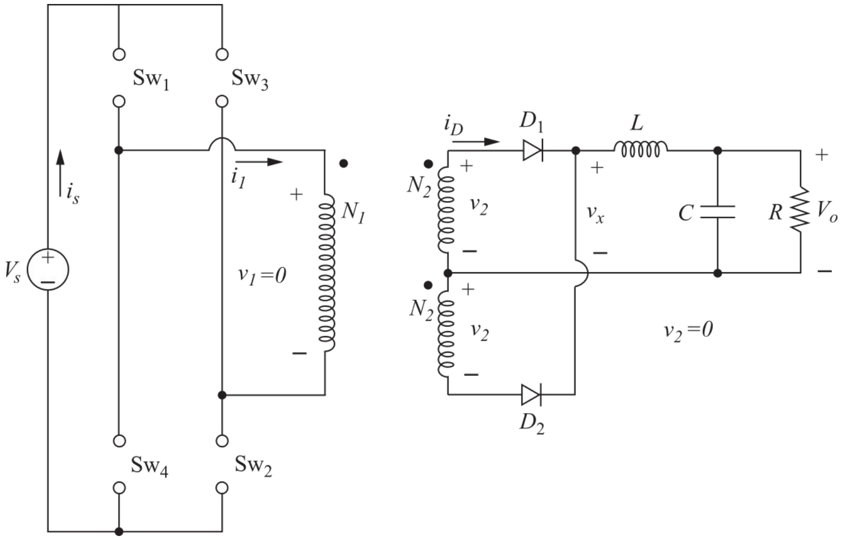

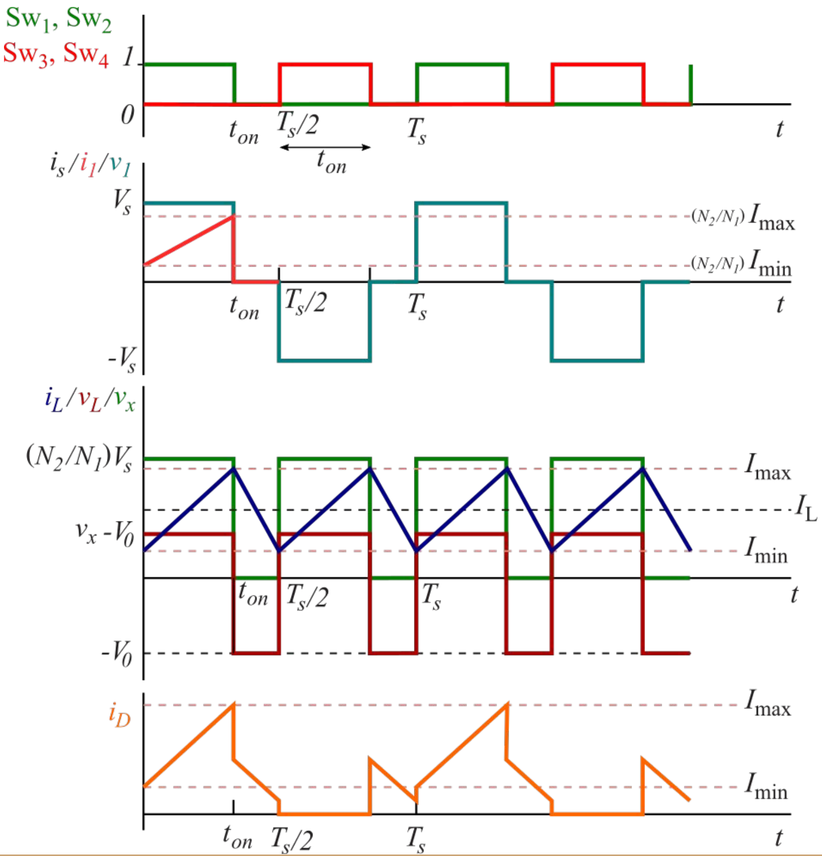

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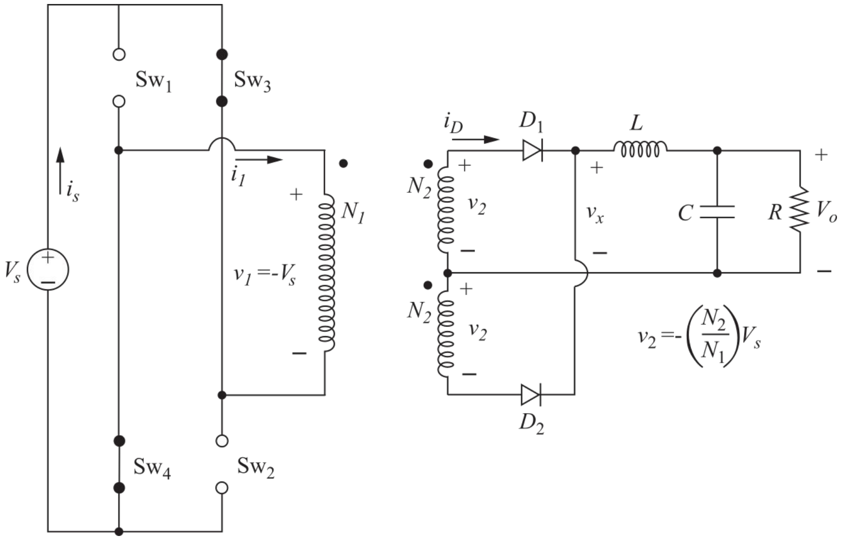

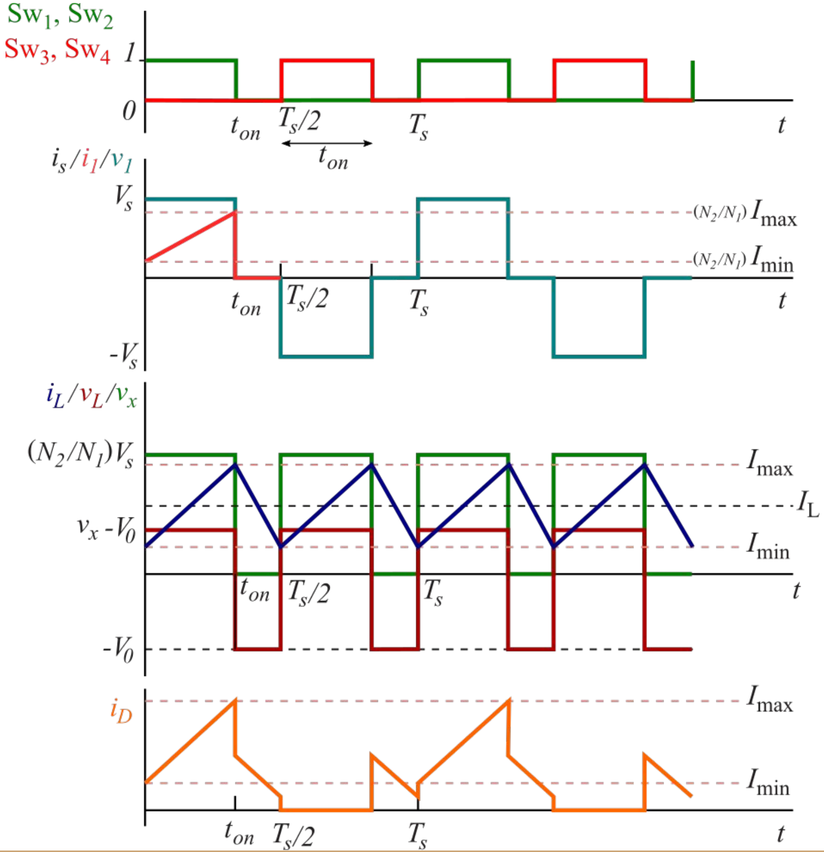

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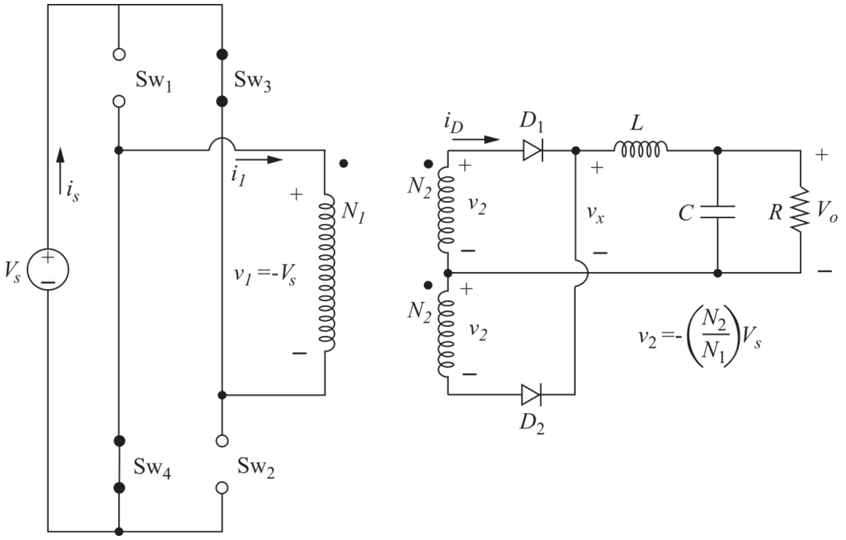

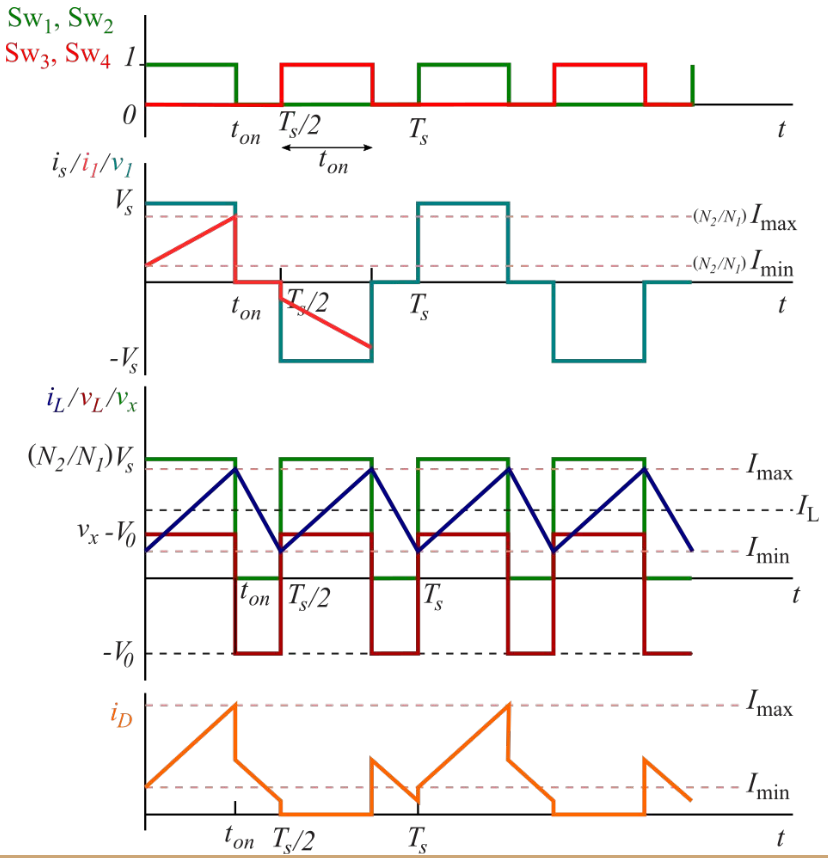

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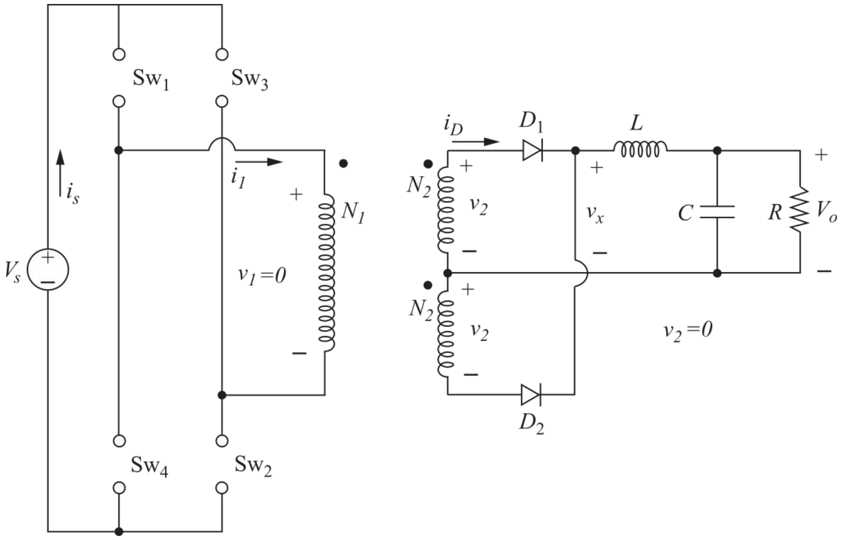

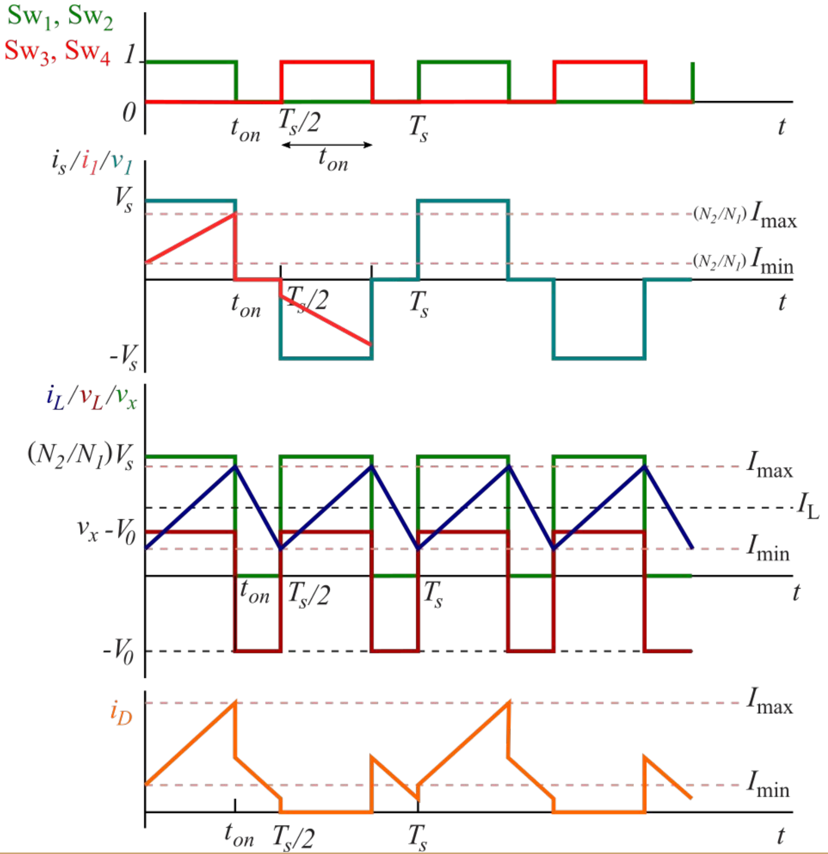

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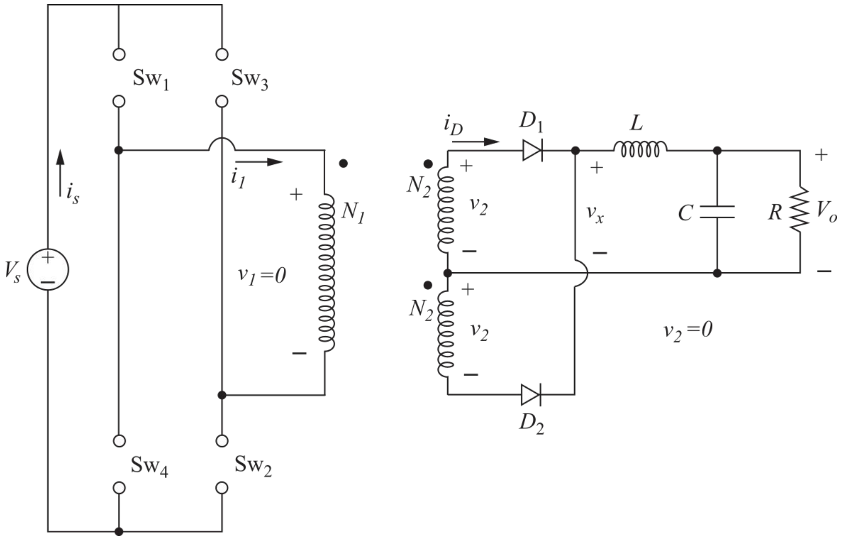

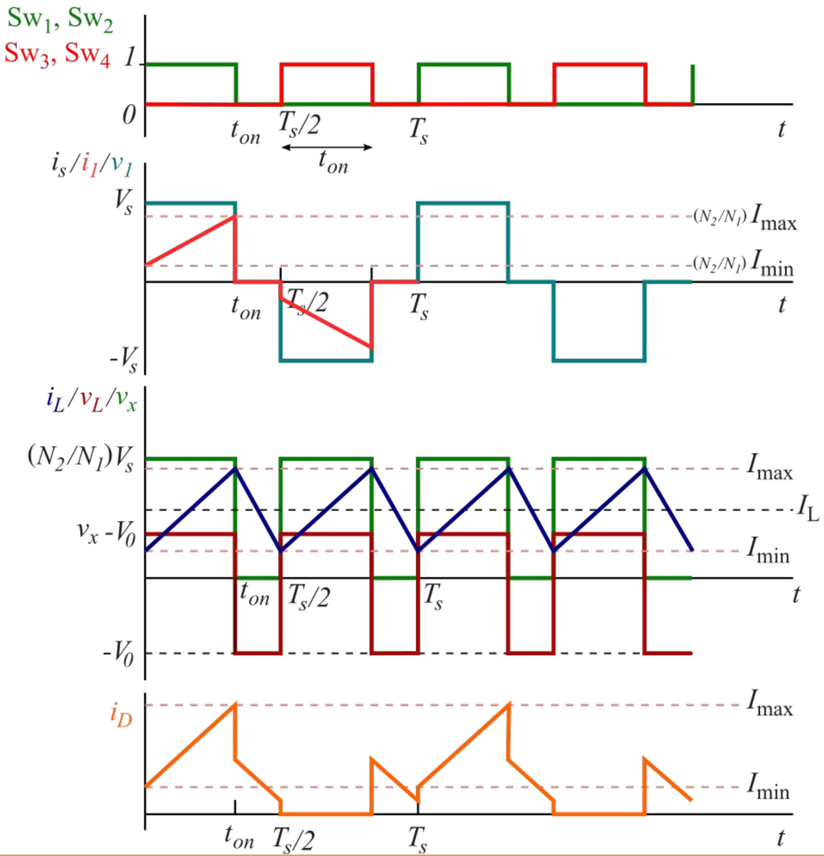

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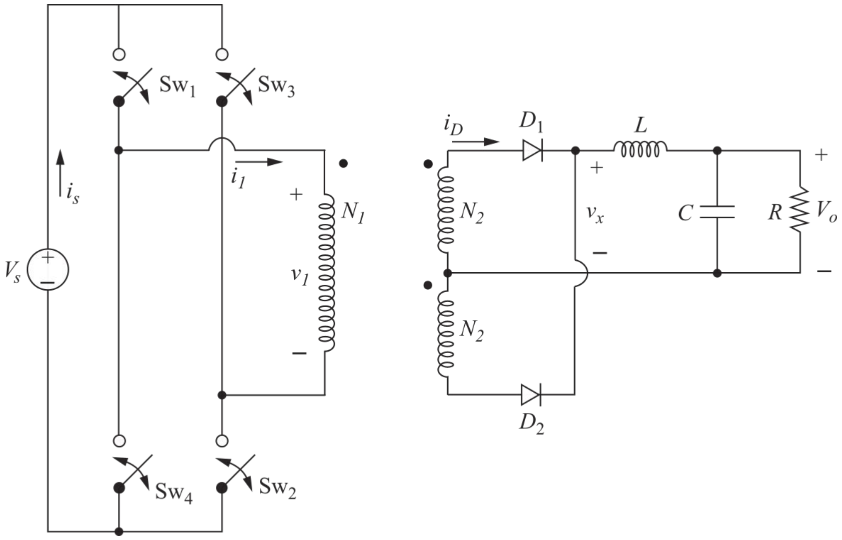

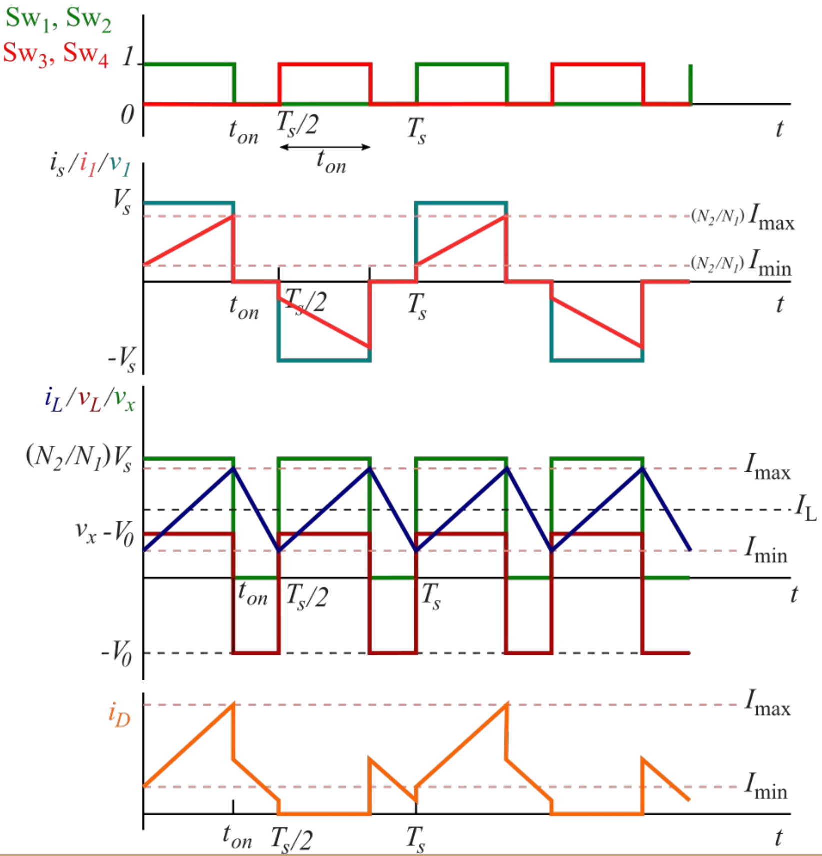

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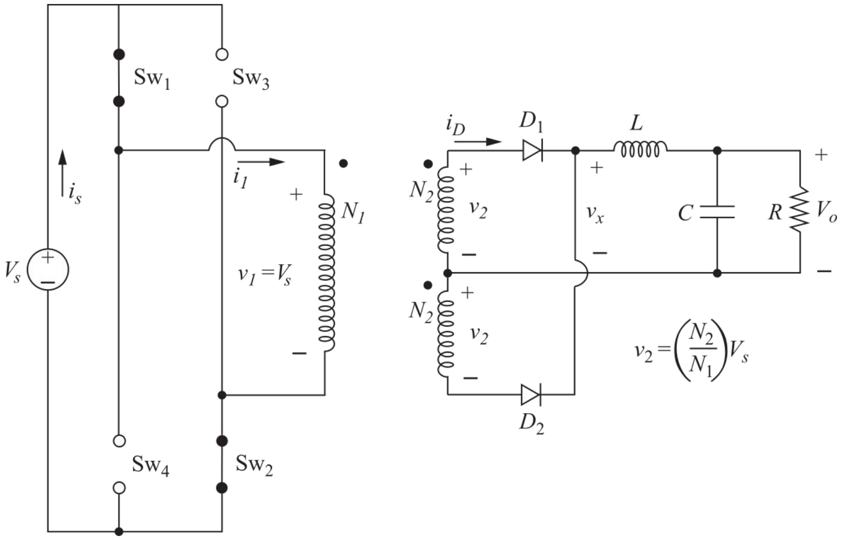

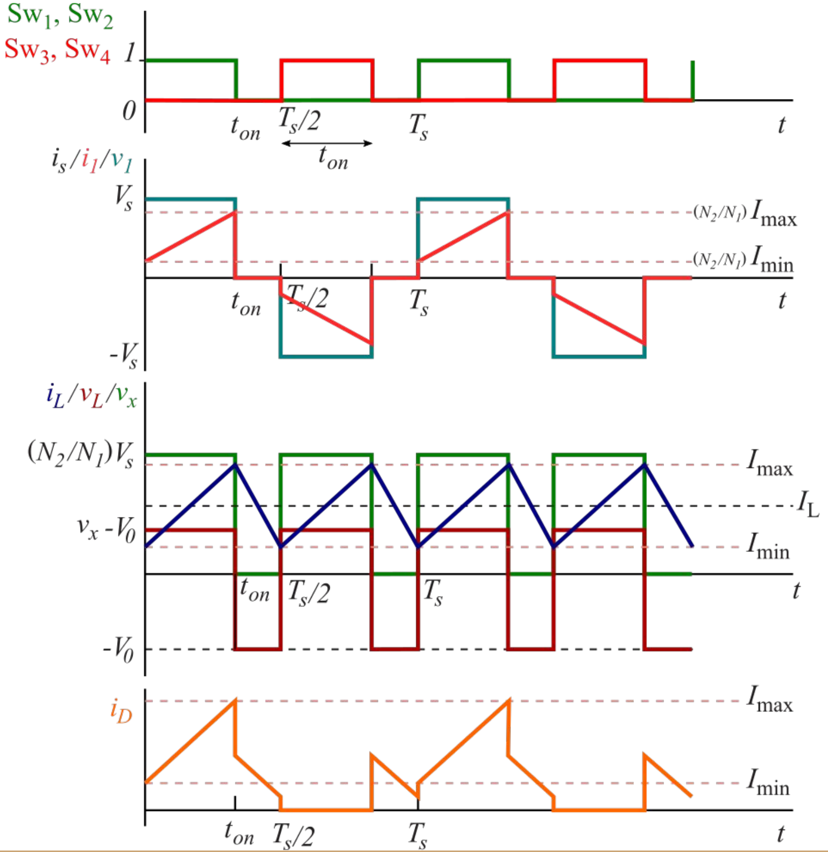

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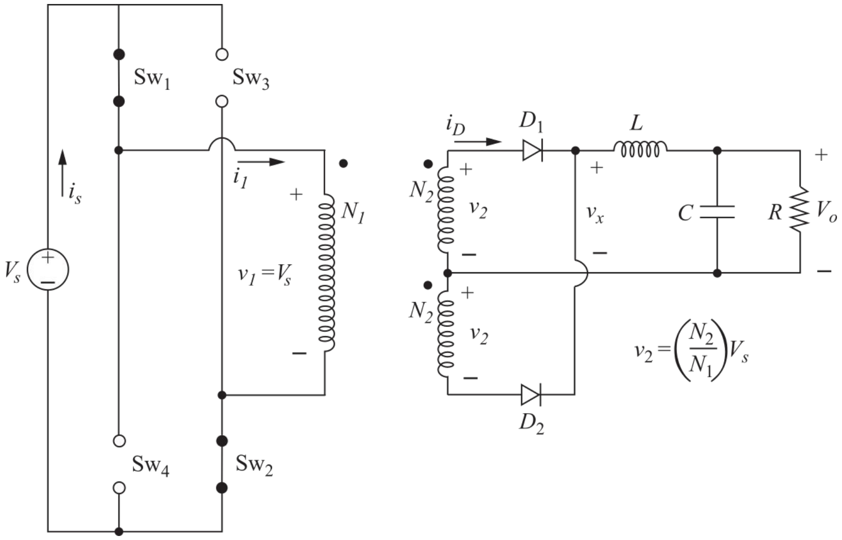

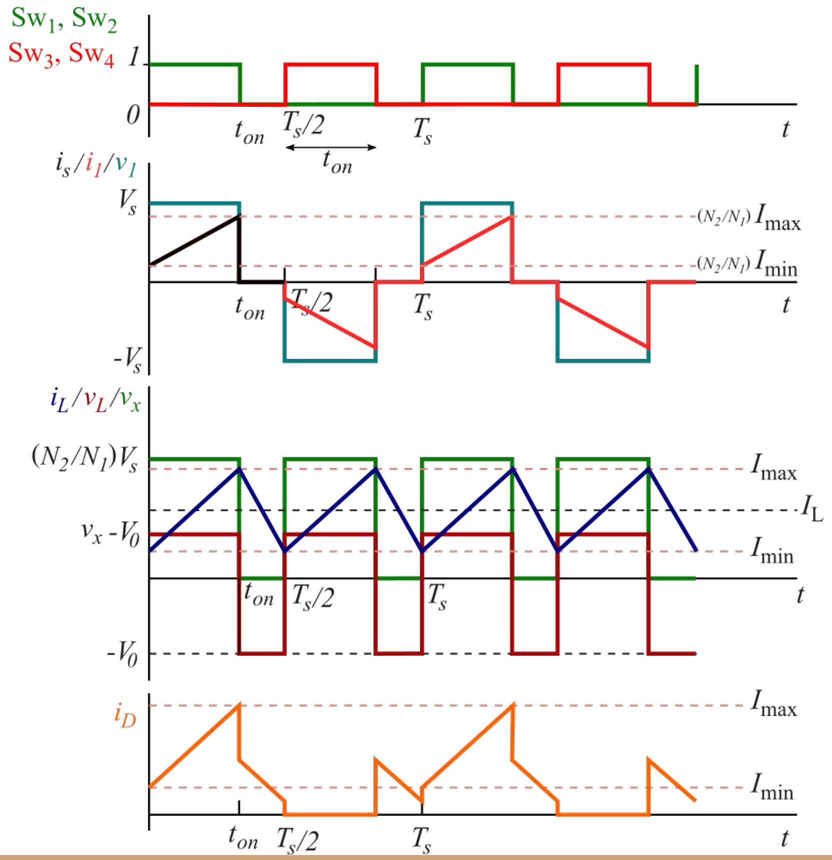

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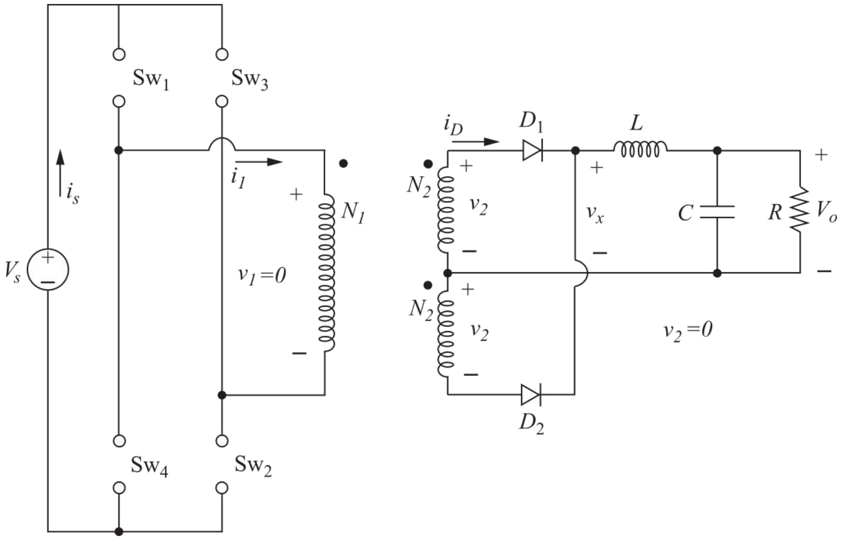

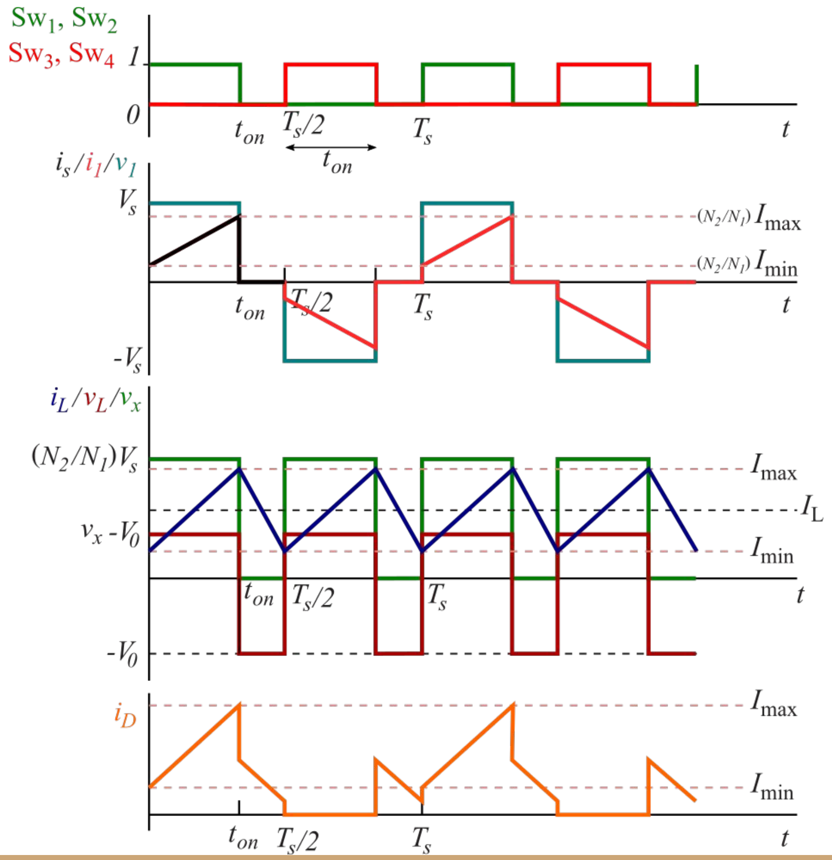

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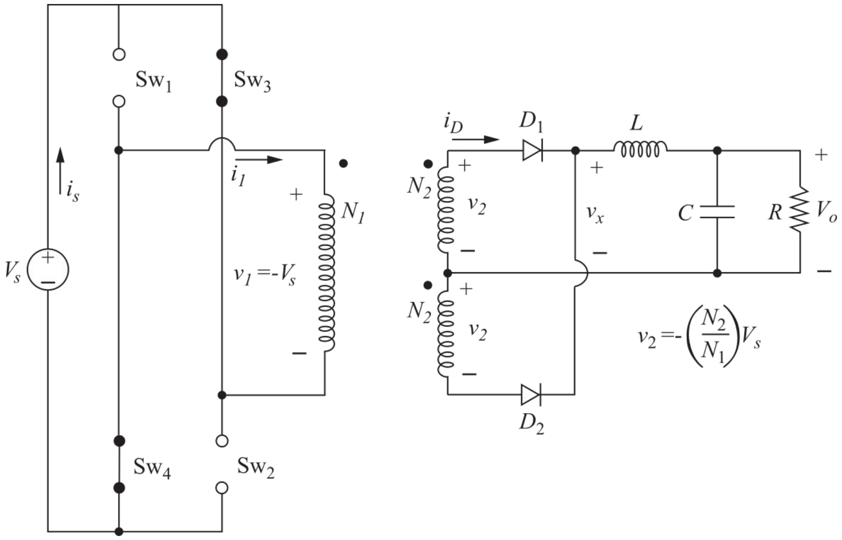

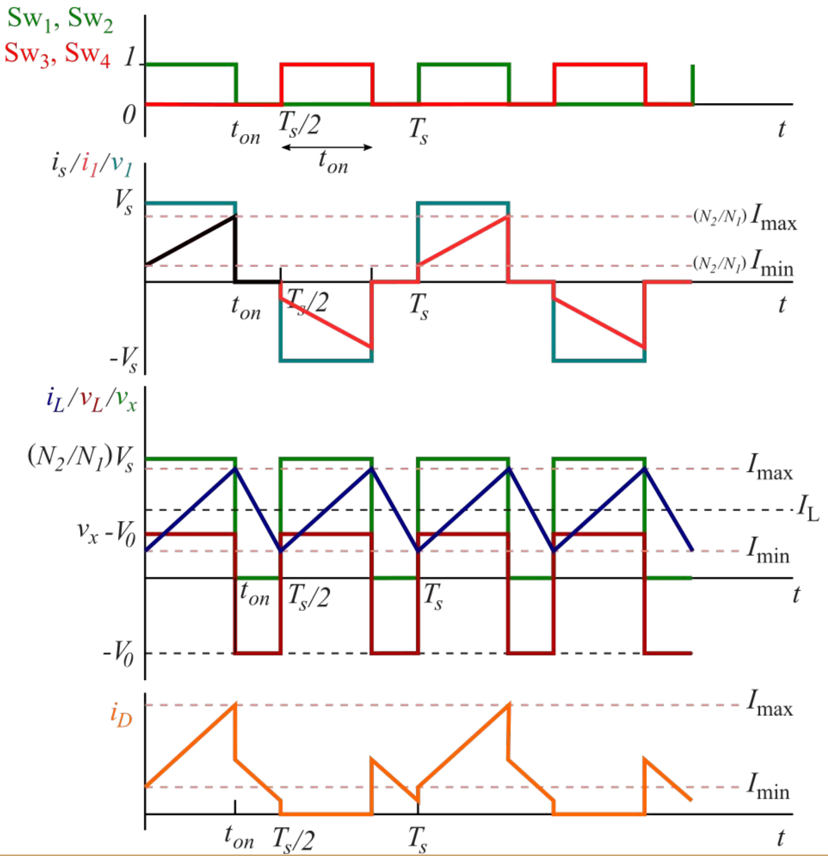

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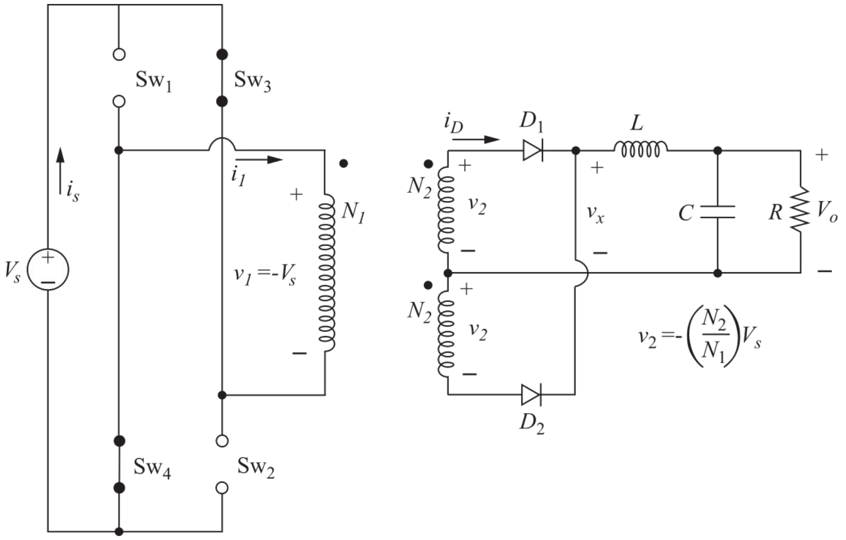

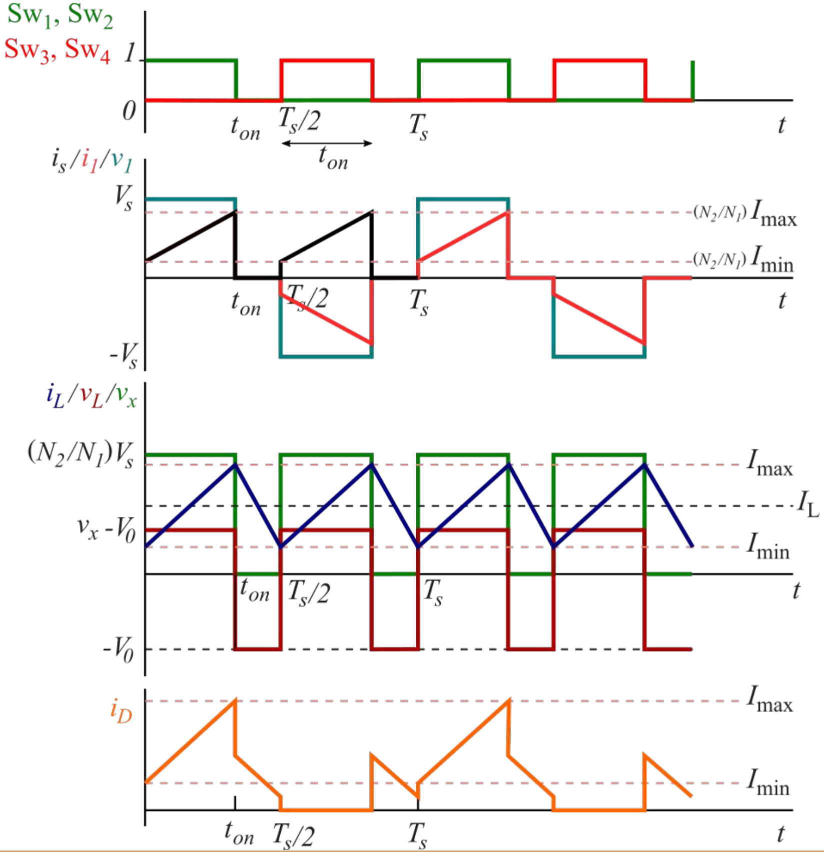

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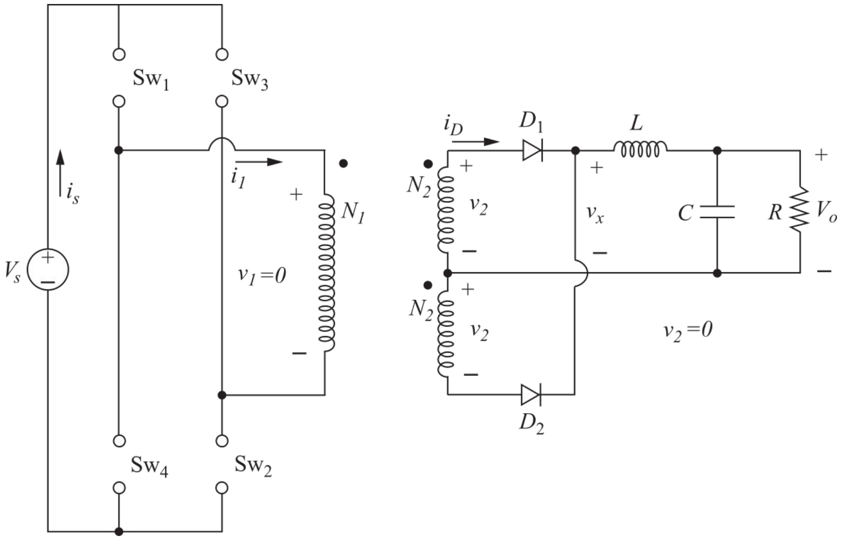

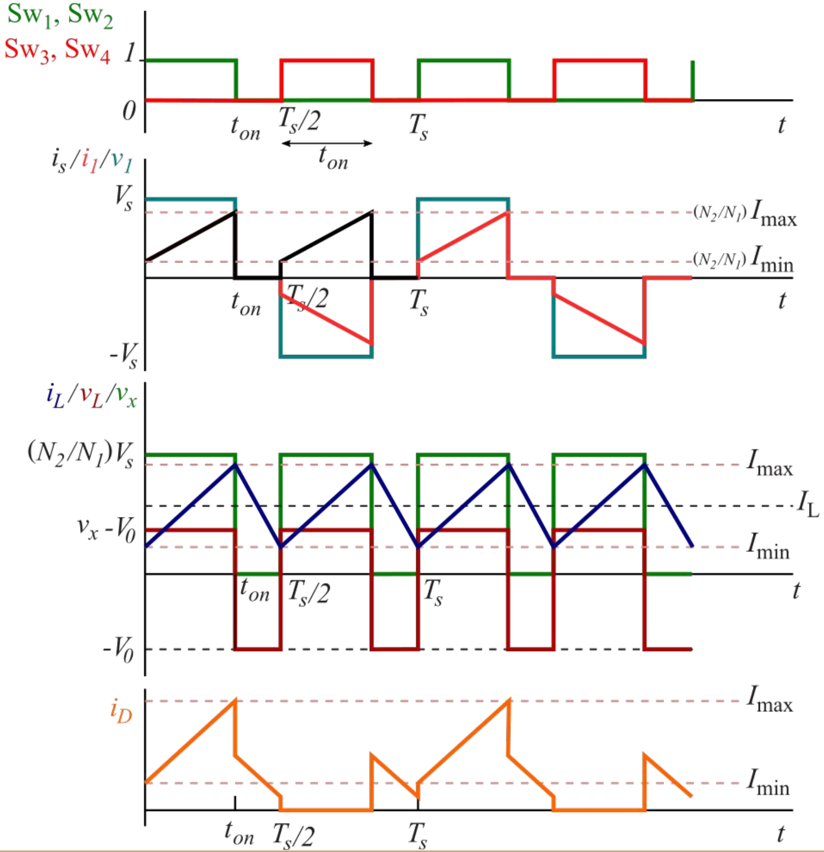

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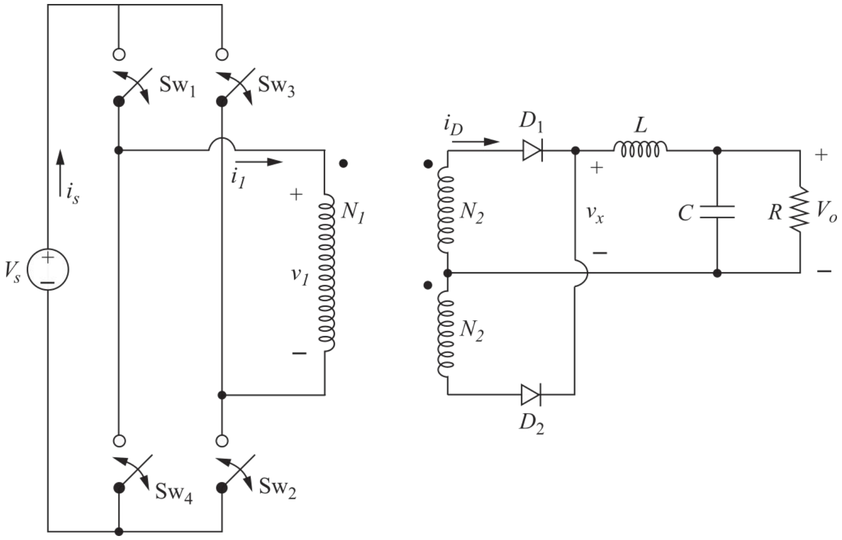

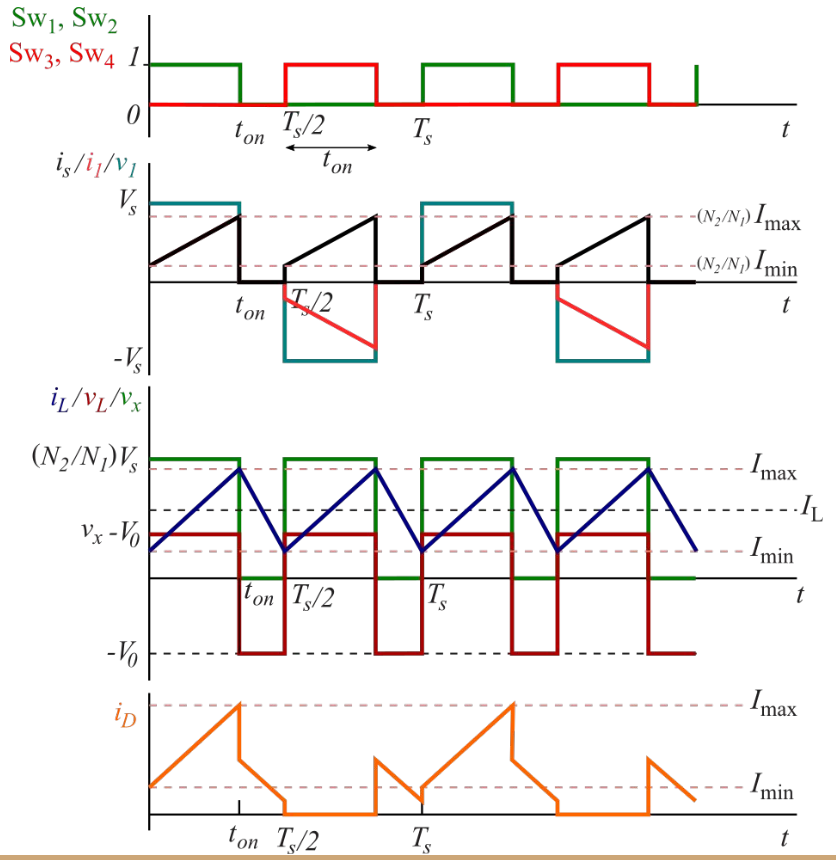

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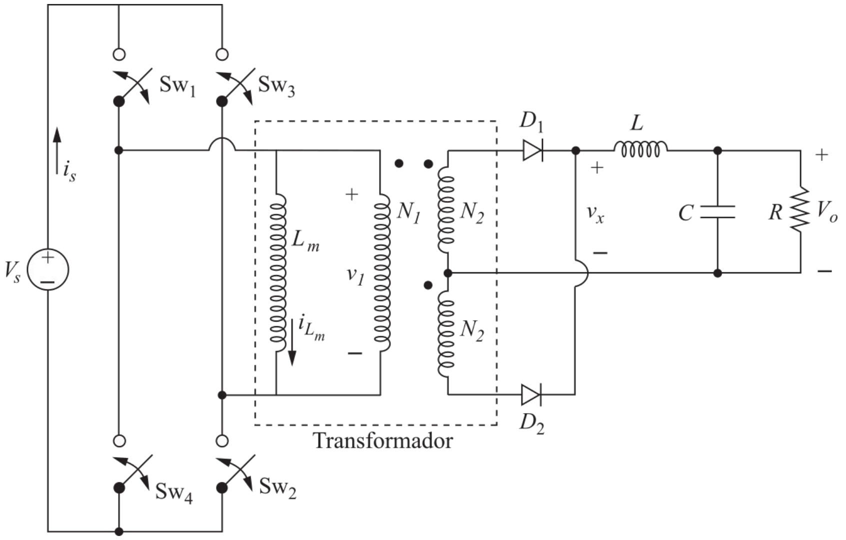

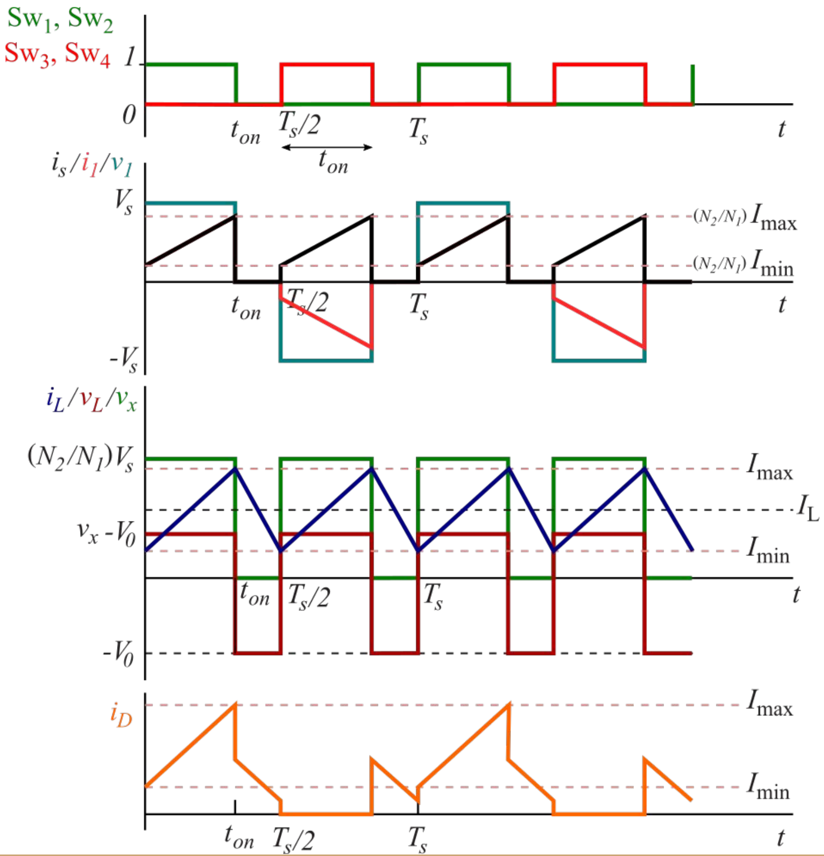

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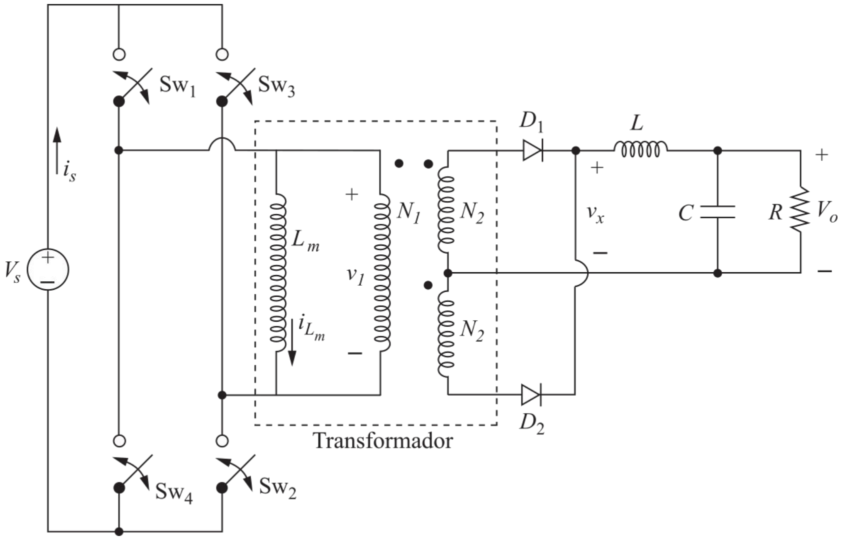

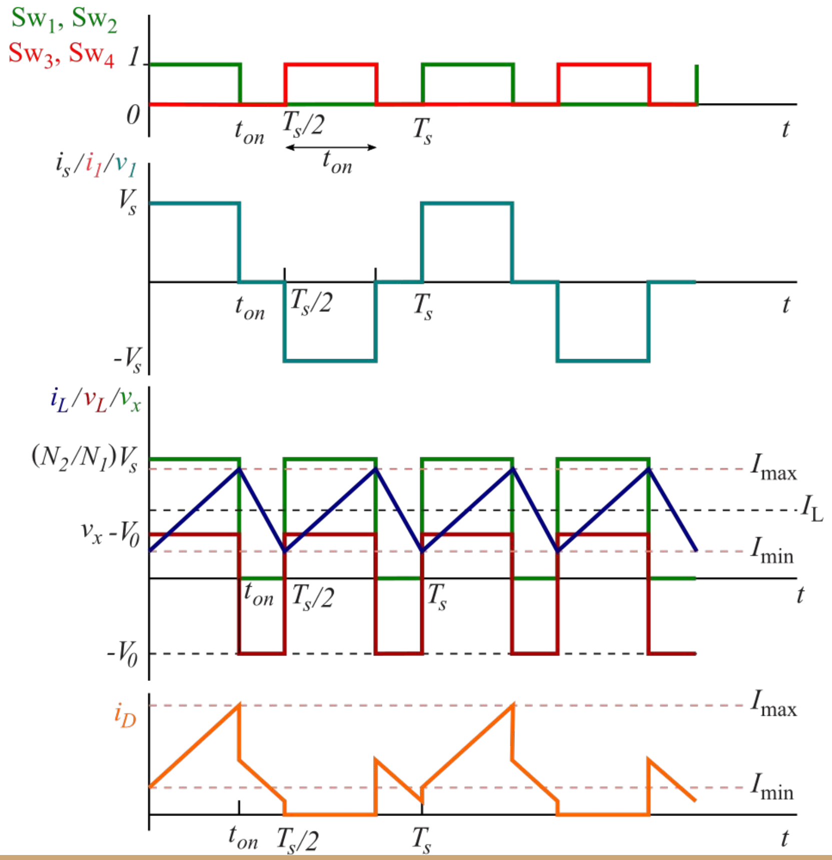

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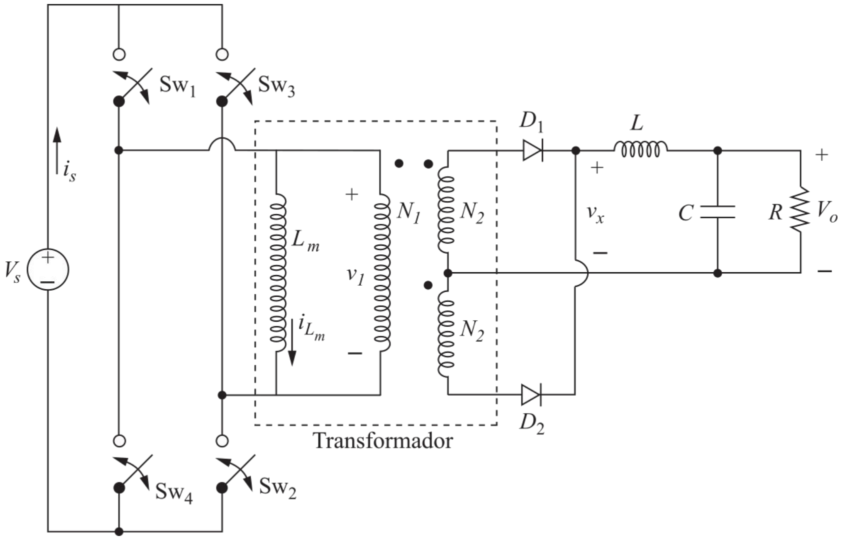

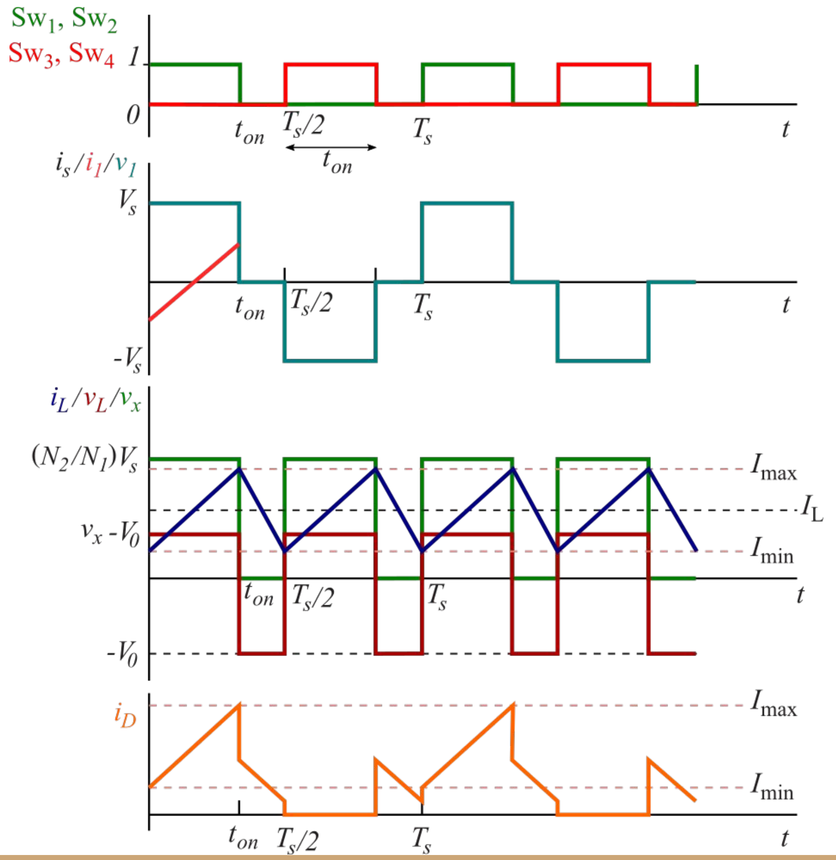

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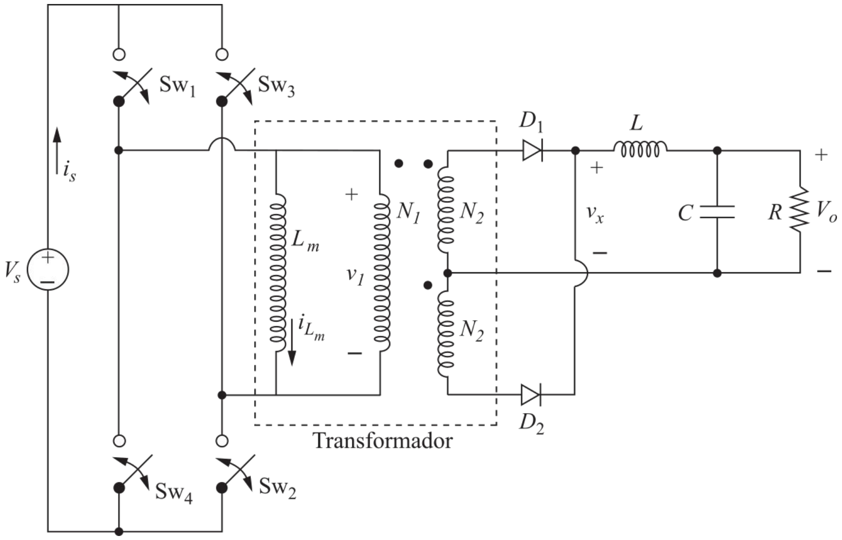

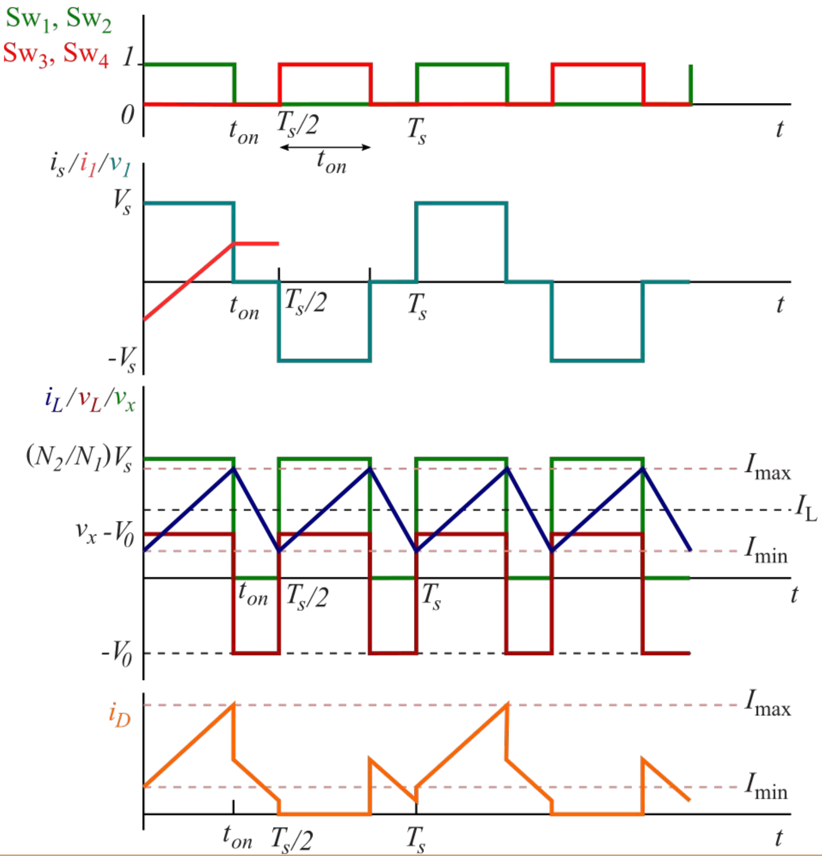

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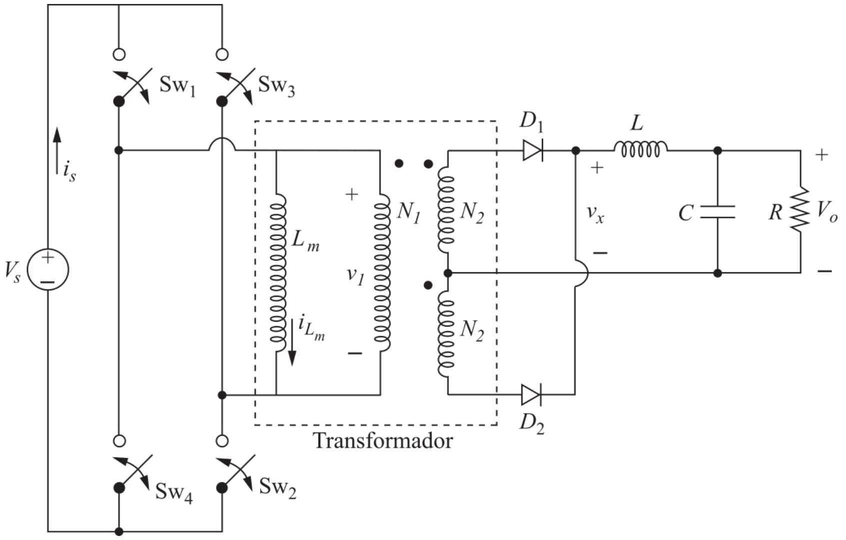

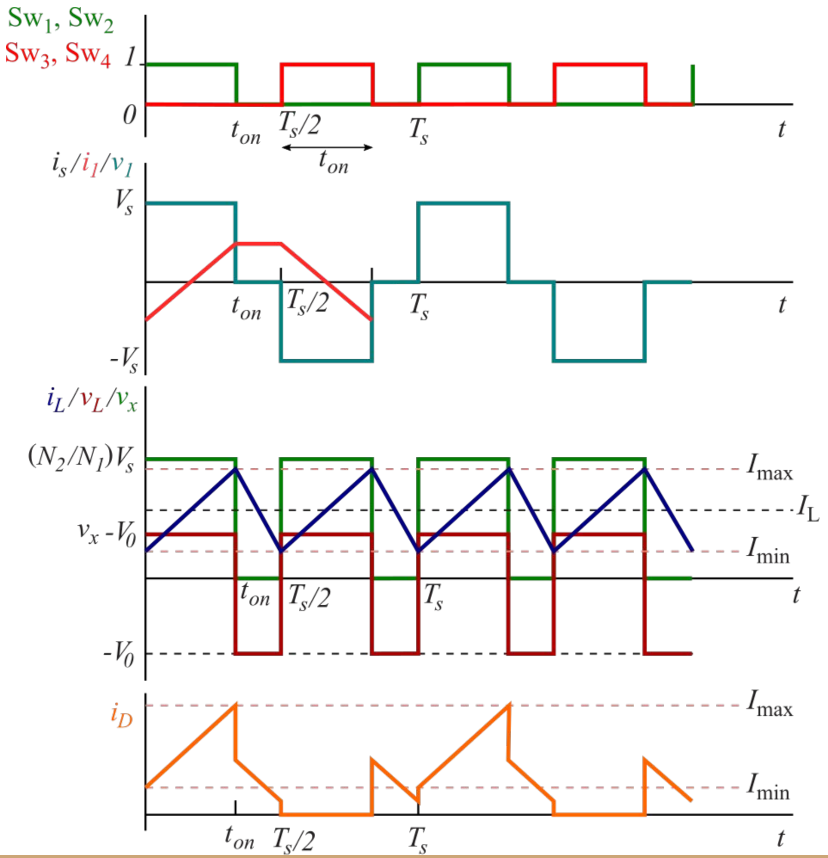

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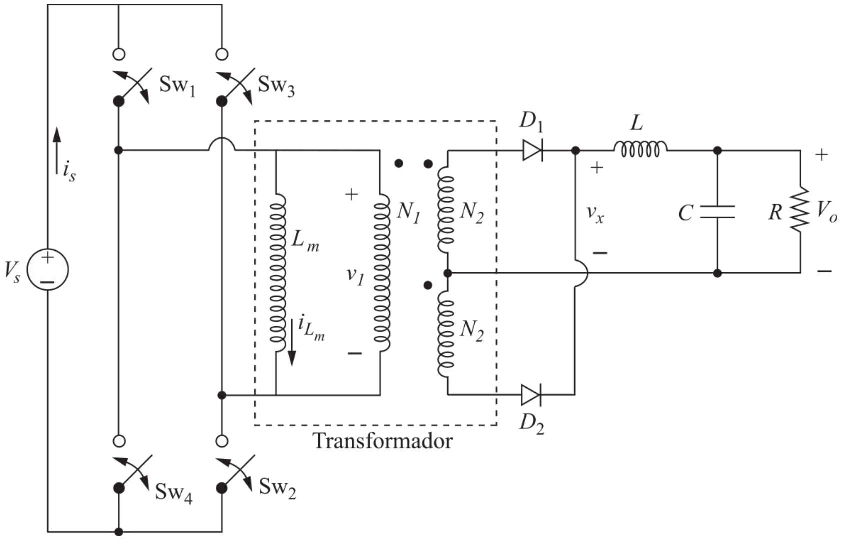

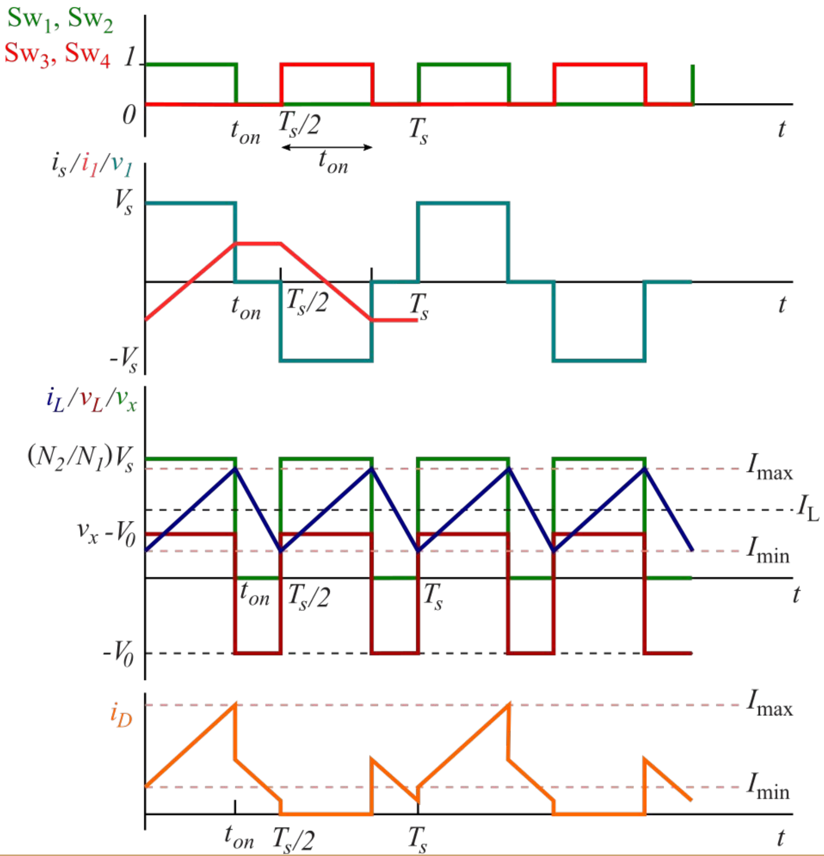

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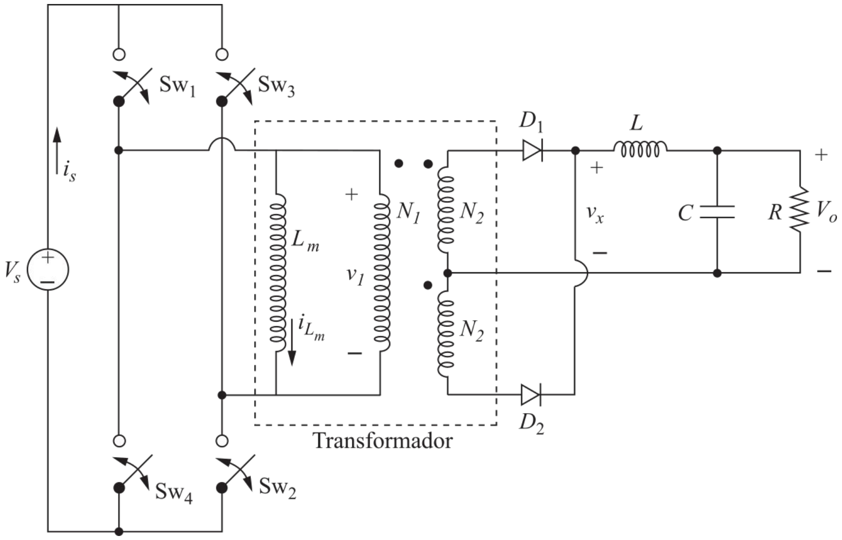

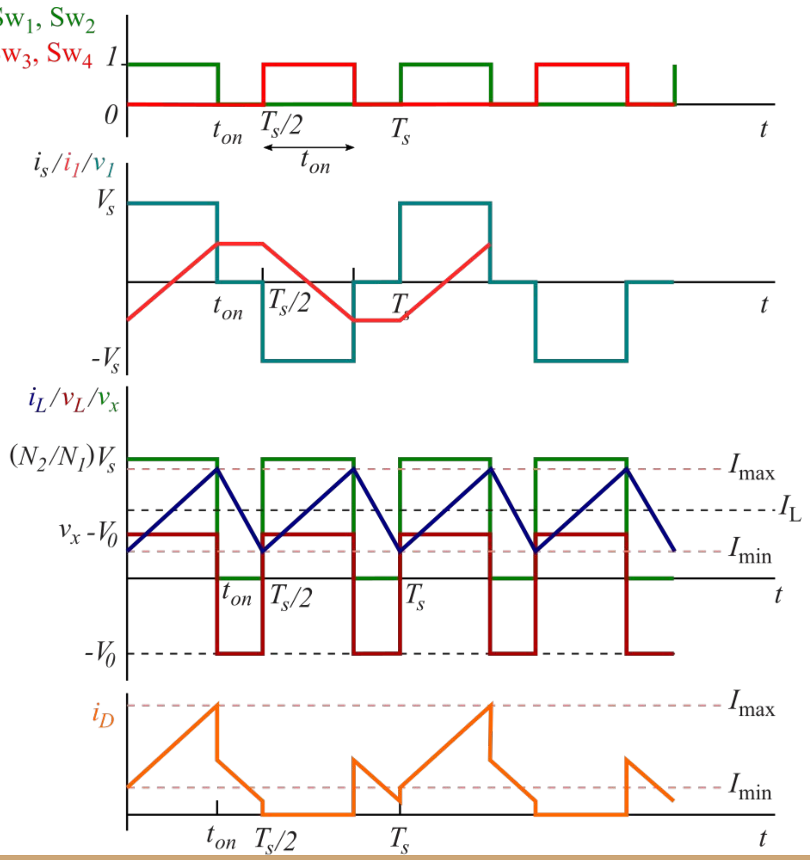

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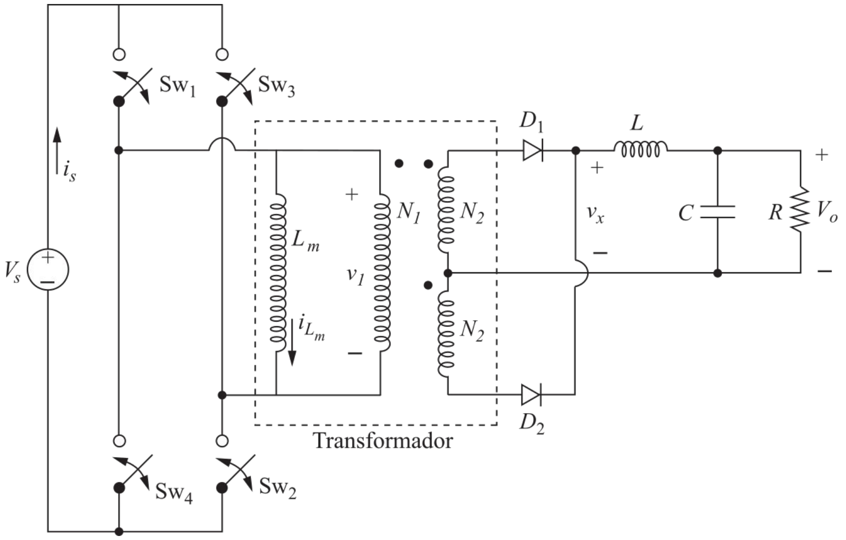

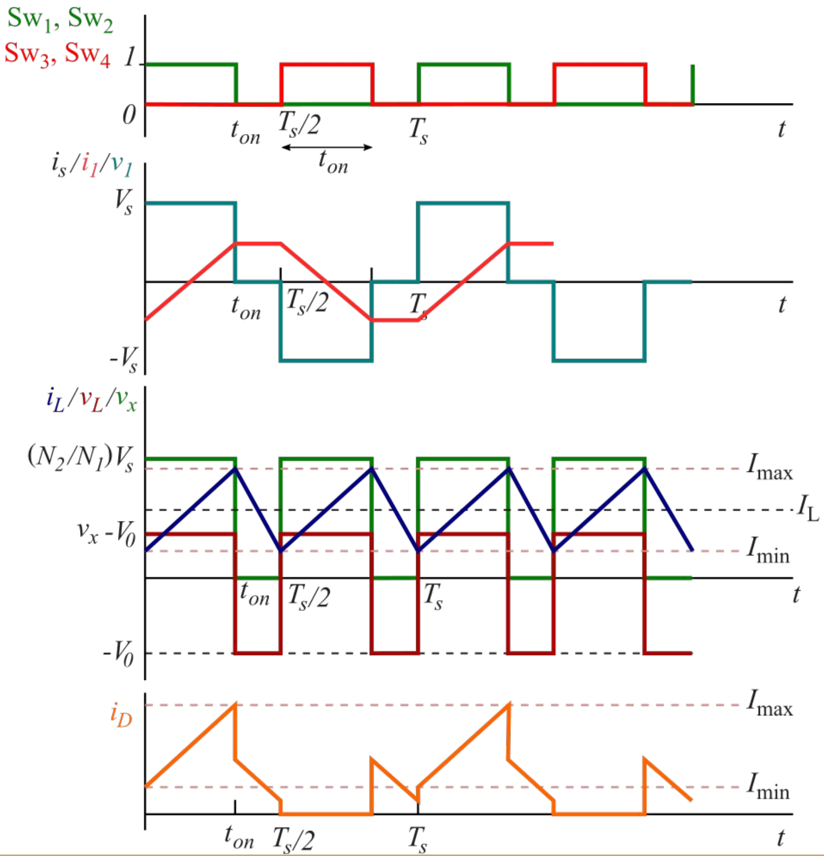

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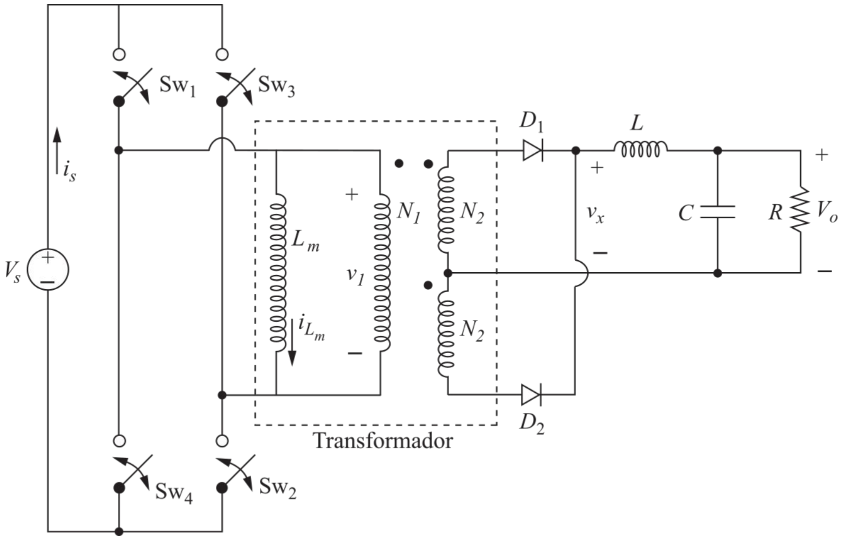
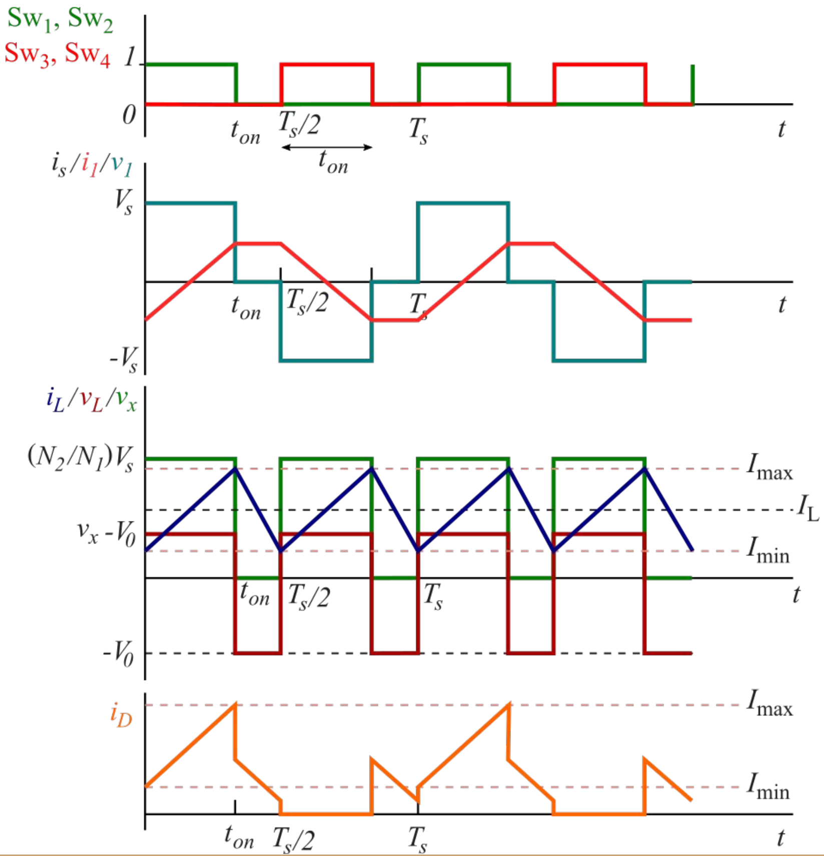
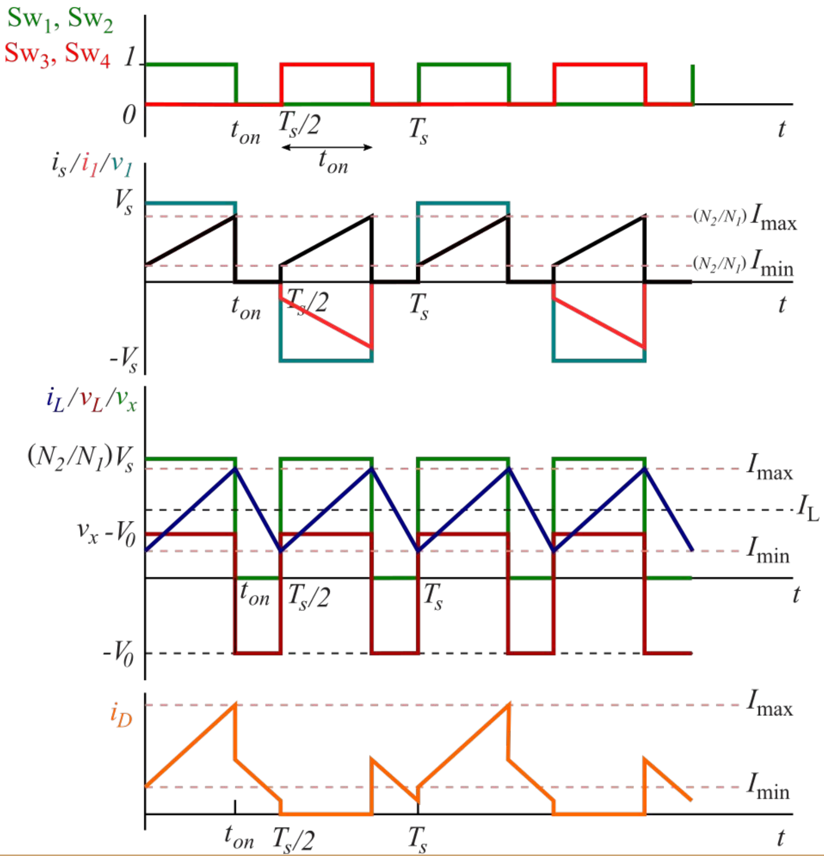




Convertidor Full-bridge: relación de conversión

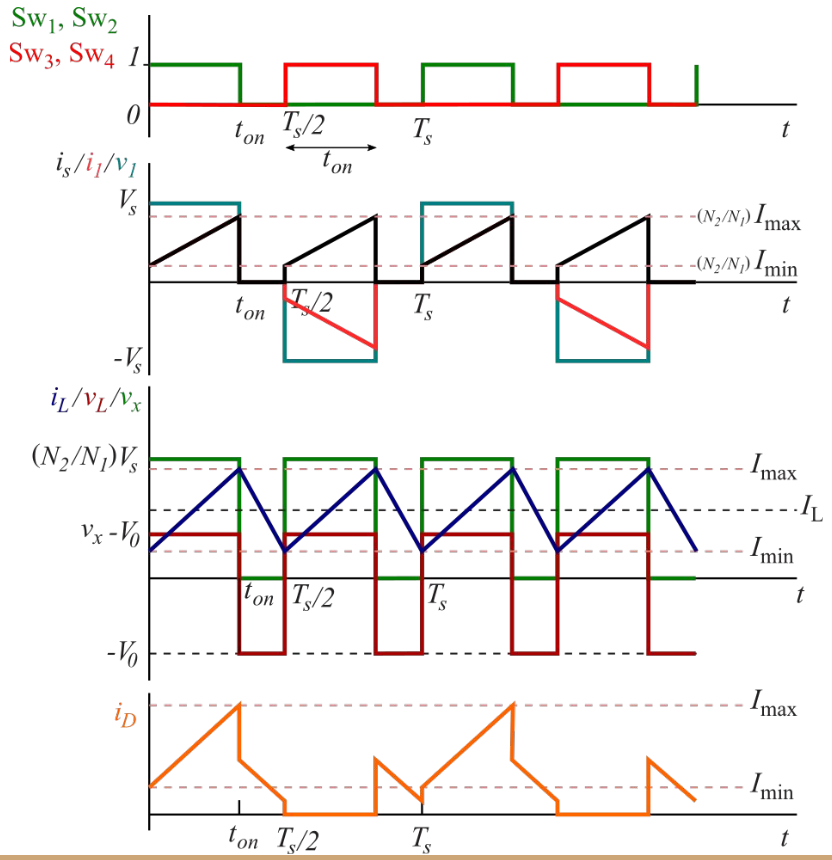
La tensión media en el inductor debe ser cero:



Convertidor Full-bridge: relación de conversión

La tensión media en el inductor debe ser cero:

$$\left(\frac{N_2}{N_1}V_s - V_0\right)t_{on} = V_0\left(\frac{T_s}{2} - t_{on}\right)$$



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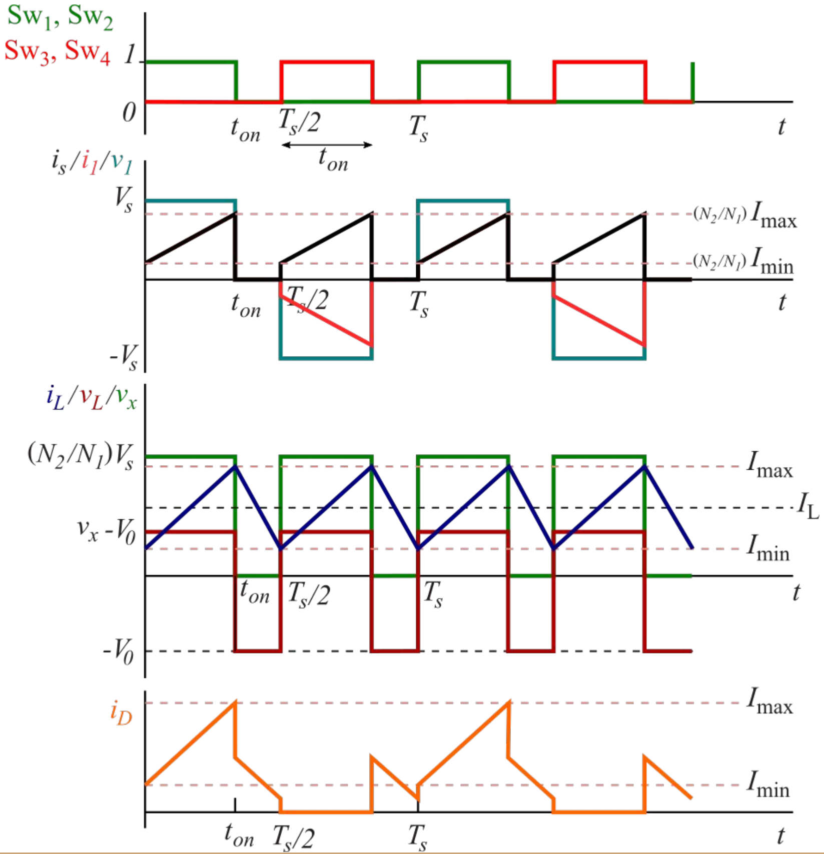

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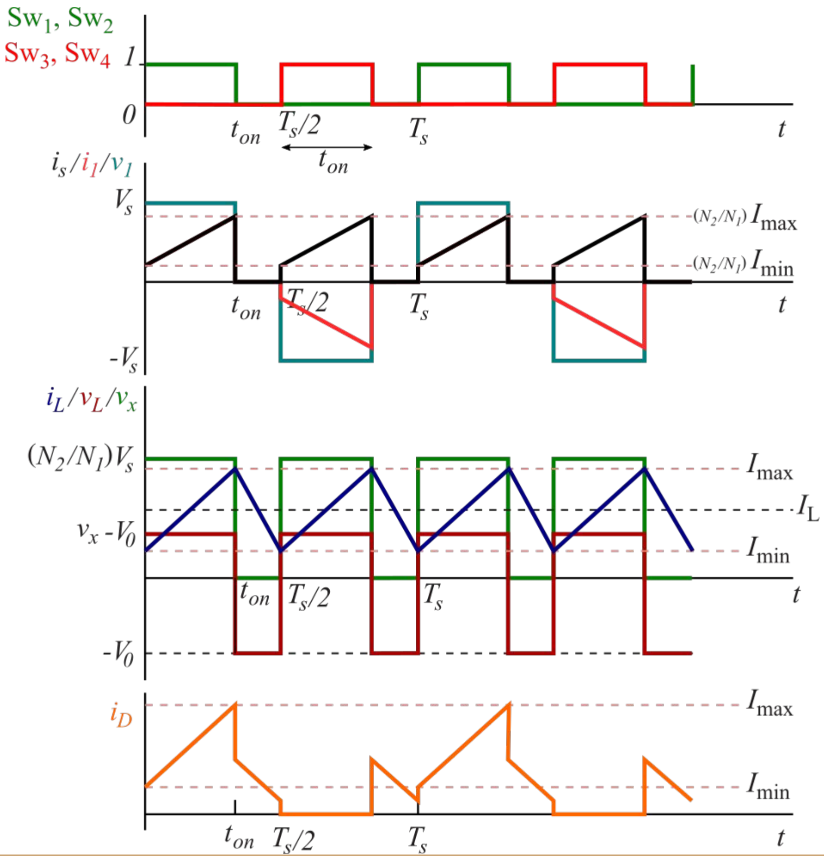
$$\frac{N_2}{N_1}V_s t_{on} - \cancel{V_0 t_{on}} = \frac{T_s}{2}V_0 - \cancel{V_0 t_{on}}$$



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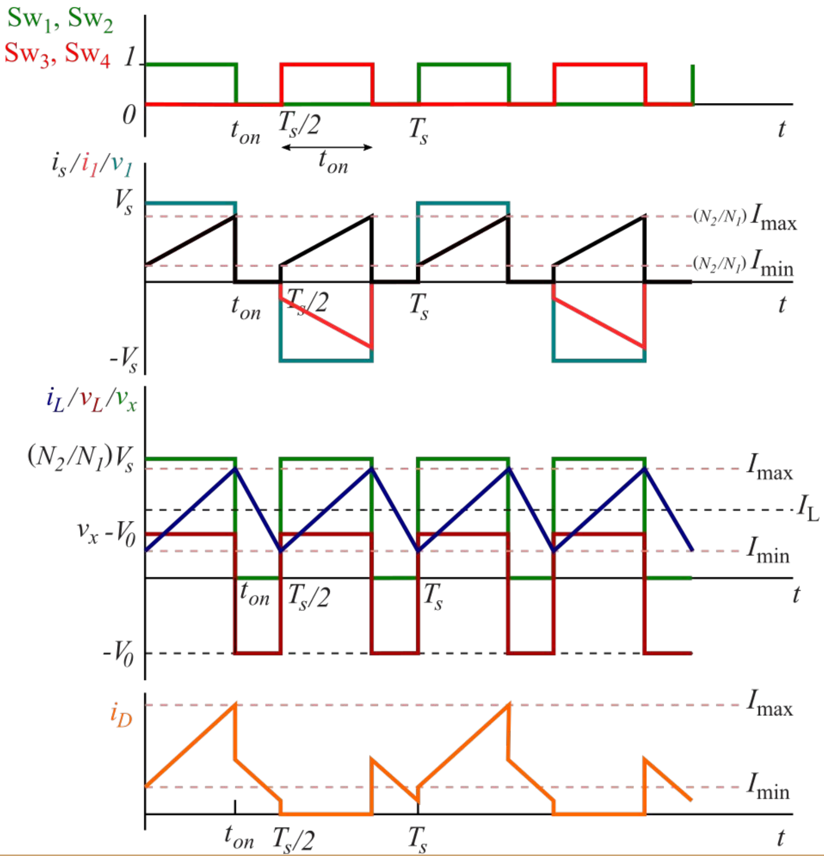
$$\frac{N_2}{N_1}V_s t_{on} - \cancel{V_0 t_{on}} = \frac{T_s}{2}V_0 - \cancel{V_0 t_{on}}$$

$$V_0 = 2\frac{N_2}{N_1}\frac{t_{on}}{T_s}V_s$$



Convertidor Full-bridge: relación de conversión

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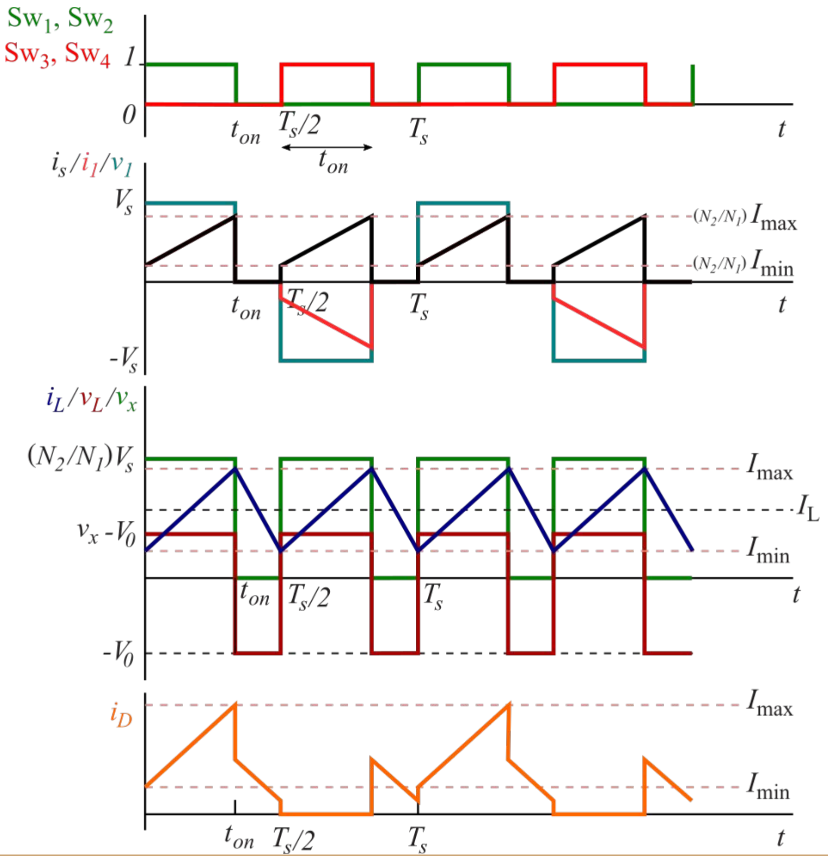
$$\boxed{V_0 = 2\frac{N_2}{N_1}DV_s}$$



Convertidor Full-bridge: relación de conversión

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$$\left(\frac{N_2}{N_1}V_s - V_0\right)t_{on} = V_0\left(\frac{T_s}{2} - t_{on}\right)$$

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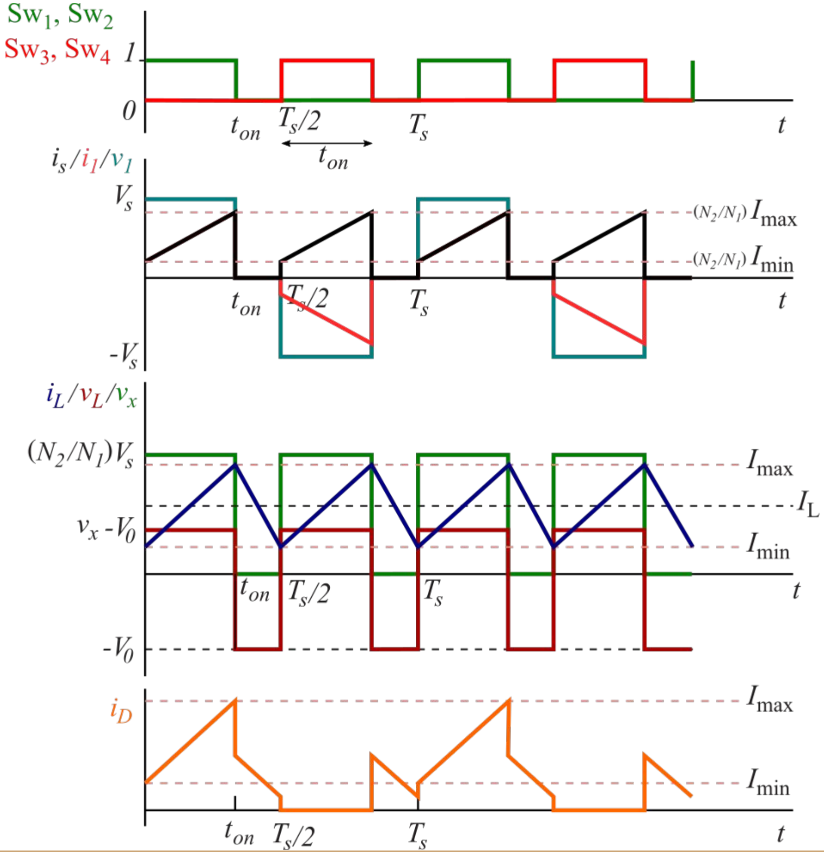
$$V_0 = 2\frac{N_2}{N_1}\frac{t_{on}}{T_s}V_s$$

$$\boxed{V_0 = 2\frac{N_2}{N_1}DV_s}$$

$$0 \leq D \leq 0{,}5$$



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- **Power Electronics. Devices, Drivers, Applications, and Passive Components** - B. W. Williams (University of Strathclyde, Glasgow. 2006).
  Descarga gratuita:
  → http://personal.strath.ac.uk/barry.williams/book.htm